

CRYPTORAVE
2026

ELKE

encrypted and lovely kage environment

UMA MARAVILHA DE AMBIENTE CRIPTOGRAFADO COM FREEBSD

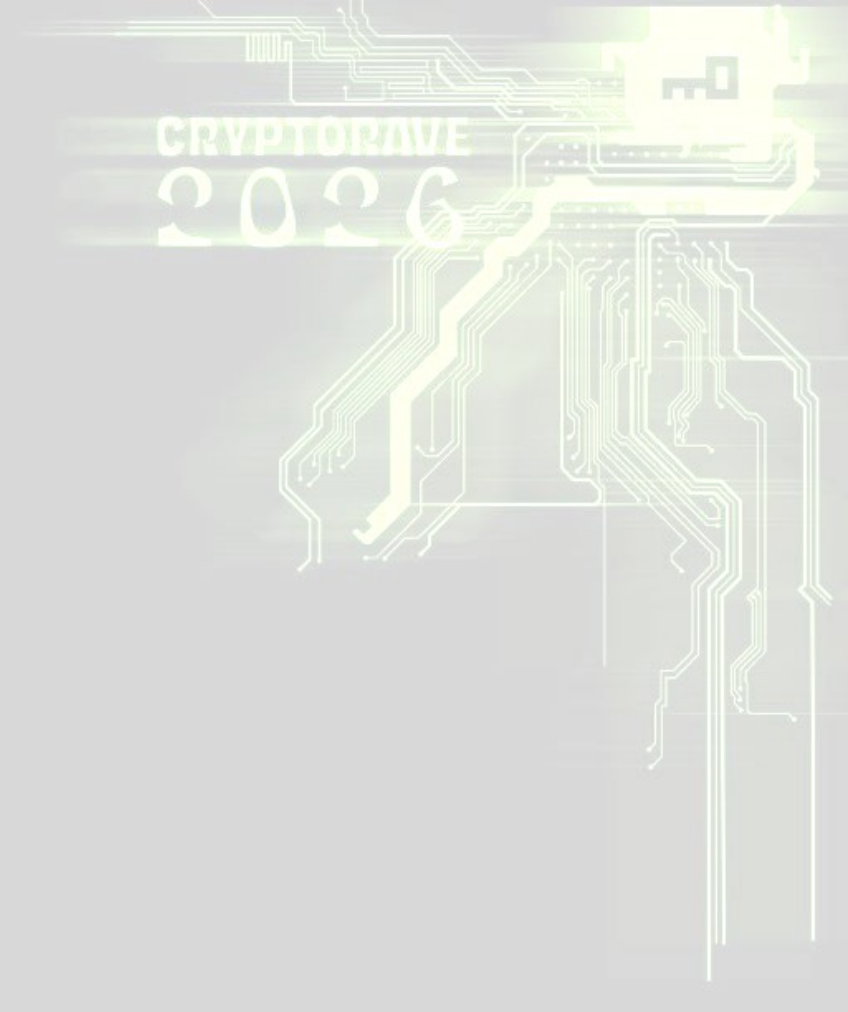


vinícius zavam

<https://keybase.io/egypcio>

<https://slideshare.net/egypcio>

- considerações iniciais
- freebsd?!1
- motivações
- prova de conceito
- evolução & refinamento
- automatização
- referências extras



CRYPTORAVE
2026



CRYPTORAVE
2026

considerações iniciais

cryptorave

- <https://2026.cryptorave.org>
- <https://we.riseup.net/cryptorave/perguntas-frequentes>
- <https://we.riseup.net/cryptorave/politica-anti-assedio>

vinícius na fila do pão

- <https://cpa.cryptorave.org/cryptorave-2026/speaker/YBWJCU>
- <https://keybase.io/egypcio>
 - 13ACCF3ED4E3B36F626FD3AE415C653413B43475 (gpg)
- <https://linkedin.com/in/viniciuszavam>
- <https://runbsd.info/people/egypcio.html>
- <https://slideshare.net/egypcio>

elke georgievna grünupp

- elke maravilha
 - <https://imdb.com/name/nm0544957>
 - https://pt.wikipedia.org/wiki/Elke_Maravilha
- encrypted & lovely kage environment
 - mdroot (base)
 - kernel
 - ntpd, sshd & tor (onion service)
 - zfs* "kage" (elke)
 - datasets criptografados



/usr/pkg/bin/flames >/dev/null

- acesso remoto
 - openssh
 - tor
- segurança física & integridade
- complexidade
- ipxe & uefi
 - secure boot
- kernel e sistema base em partição NÃO criptografada
 - dropbear initramfs & luks (linux)

se nada der certo

- paciência, leveza & convicção



mas ...

irado furioso com tudo

- <https://freebsdbrasil.com.br/empresa/irado-jorge.html>

"eu já disse e vou repetir quantas vezes você precisar, como você vai conseguir fazer isso, onde vai procurar e que documentação vai seguir de referência. pro seu bem, o que eu não vou é colar o comando que você vai simplesmente copiar, executar, agradecer porque funcionou e não terá aprendido nada. me leia com atenção e você vai descobrir por conta própria exatamente como fazer, agora entregar de mão beijada eu não vou e espero que ninguém faça isso. e, acredite em mim, é porque gosto de você"

A stylized graphic of a circuit board in shades of purple and blue. It features various traces, pads, and a prominent microchip in the upper right corner. The text 'CRYPTORAVE 2026' is overlaid on the circuitry.

CRYPTORAVE
2026

freebsd?!1

introdução & história

- sistema operacional de código aberto nos padrões unix
 - posix (portable operating system interface)
 - arm/aarch64, powerpc/ppc64, riscv64, x86/amd64
- 4.3bsd-lite, 386bsd & freebsd 1.0
 - berkeley software design, inc. & walnut creek cdrom, inc.
 - marshall kirk mckusick
 - lynne & william jolitz
 - david greenman
 - nate williams, rod grimes & jordan hubbard

licença bsd de 2 cláusulas

- adicionar os direitos e condições seguintes no código fonte
- distribuição de binários deve conter a licença em sua documentação

restrições legalmente impostas

- não diga que foi você quem desenvolveu o sistema operacional
- não processe o projeto caso seu sistema "quebre"
- não remova a licença original do código fonte ou documentação

quem usa

- apache software foundation
- apple (macos, ios)
- cisco (ironport)
- citrix (netscaler)
- dell, emc (isilon)
- juniper (junos)
- manex visual effects*
- microsoft*
- netflix (openconnect)
- nintendo*
- sony (playstation4)
- sophos (email appliance)
- whatsapp*



FreeBSD[®]

sistemas baseados em freebsd

- <https://duckduckgo.com/?q=sistemas+baseados+em+freebsd>
- ambientes desktop
 - ghostbsd
 - midnightbsd
- storage
 - nas4free
- firewalls, roteadores & universal threat management (utm)
 - bsdrp
 - opnsense
 - pfsense
 - zrouter
- hypervisor (bhyve) & virtualização (jails)
 - clonos

ambientes desktop & gerenciadores de janelas

- kde (plasma)
- gnome
- mate
- xfce
- i3
- lxde & lxqt
- awesome
- cinnamon
- sway
- gershwin
- wayfire
- openbox

freebsd na mídia

- entregando efeitos especiais de matrix em 1999 com freebsd
 - <https://www.freebsd.org/press/press-rel-1/>
- dados da copa do mundo de 2006
 - <https://www.fug.com.br/content/view/64/54/>
- segurando mais de 2 milhões de conexões do whatsapp em 2012
 - <https://blog.whatsapp.com/1-million-is-so-2011>
- vaga de trampo com objetivo de melhorar linux pra superar freebsd
 - <https://www.phoronix.com/news/MTc1NjY>
- freebsd embarcado em uma beaglebone black num robô hexapod
 - <https://www.youtube.com/watch?v=8NvpUF2BDtY#t=31m45s>
- mais rápida rede de distribuição de conteúdo usando freebsd (netflix)
 - <https://hamradio.my/2025/06/how-netflix-powers-the-worlds-fastest-content-delivery-network-with-freebsd>

algo para gamers de plantão

- https://youtube.com/results?search_query=freebsd+gaming
- <https://vermaden.wordpress.com/2020/10/14/oldschool-gaming-on-freebsd>

para quem nasceu antes dos anos 2000

- irc (efnet)
 - #bsdcode, #bsdmps, #bsdports
 - #freebsd-xorg, #freebsd-wifi

imagens de máquinas virtuais

- qcow2
- vhd
- vmdk

provedores com freebsd disponível

- amazon (aws)
- google (gcp)
- letscloud
- microsoft (azure)
- vultr

imagens para dispositivos embarcados

- beaglebone
- pine (rockpro)
- raspberry-pi

freebsd "live" com armazenamento persistente

- <https://nomadbsd.org>

instaladores gráficos

- <https://ghostbsd.org>
- <https://github.com/hellossystem>
- <https://midnightbsd.org>

cooltrainer (nicole reid)

- <https://cooltrainer.org/a-freebsd-desktop-howto>

vermaden

- <https://vermaden.wordpress.com>
 - instalação e setup de um desktop freebsd
 - zfs boot environments
 - serviços & clusters de alta disponibilidade
 - fim das tuas noites de sono :3
- <https://github.com/vermaden/verimg>
 - imagens automatizadas & customizadas

outpaddling (jason bacon)

- <https://github.com/outpaddling/desktop-installer>

cooltrainer (nicole reid)

- <https://cooltrainer.org/m-freebsd-desktop>

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- <https://github.com/outpaddling/desktop-installer>



freebsd em terras tupiniquins

- desenvolvimento brasuca
 - documentação/tradução, ports & src (base)
- bsdcon brasil
 - rio de janeiro, 2002
 - fortaleza, 2015
- grupo de usuári_s
 - lista de discussão*
 - <https://t.me/freebsdbr> & <https://web.libera.chat/#freebsd-br>
- freebsd brasil Ltda.
 - serveru appliance
- rede nacional de ensino e pesquisa, rnp
 - pontos de presença
- {agências,órgãos,institutos,universidades,instituições} federais



freebsd em terras tupiniquins

- casa hacker, com caio volpato e garga@
- <https://youtube.com/watch?v=y7S5oPYYs48>



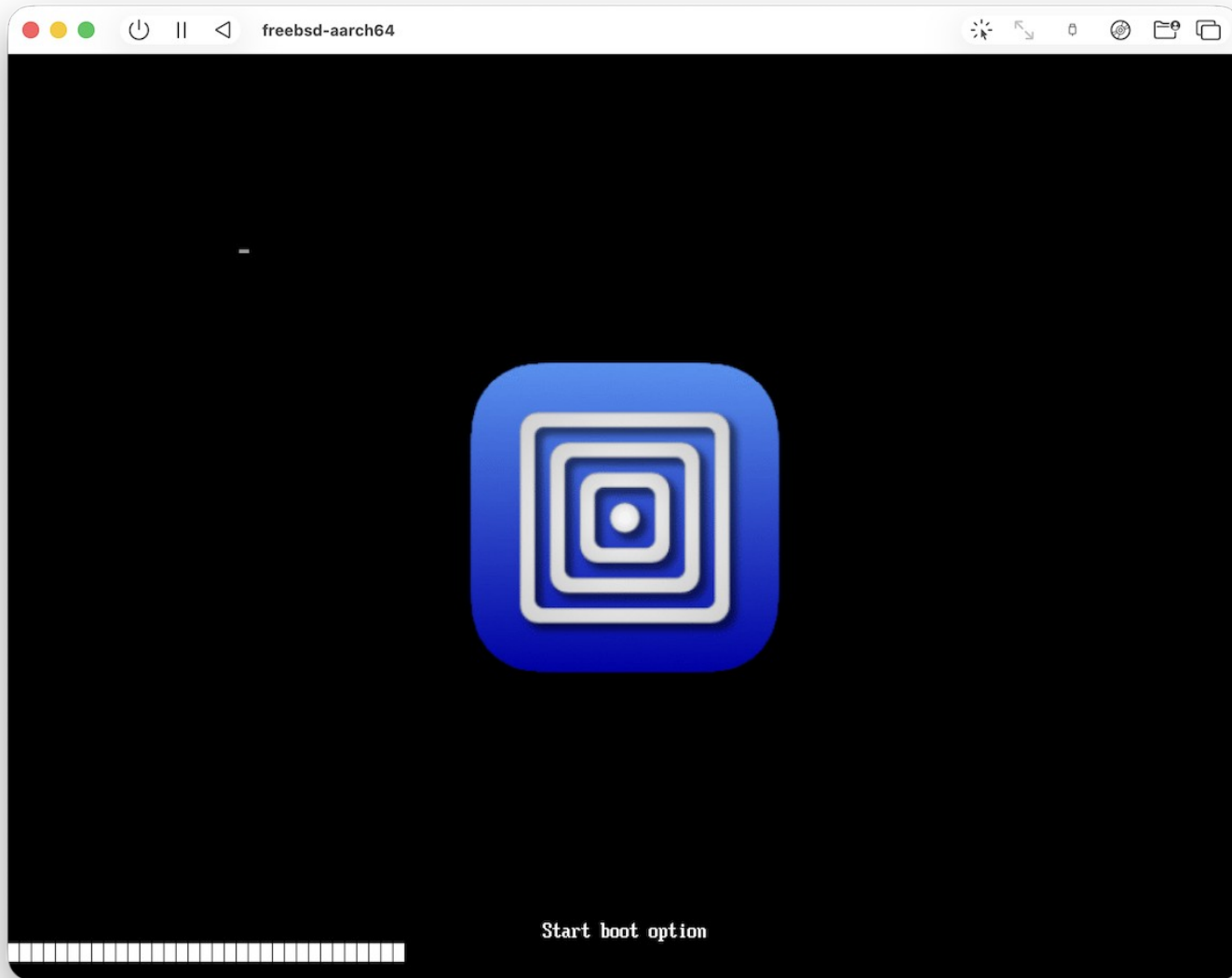
freebsd em terras tupiniquins

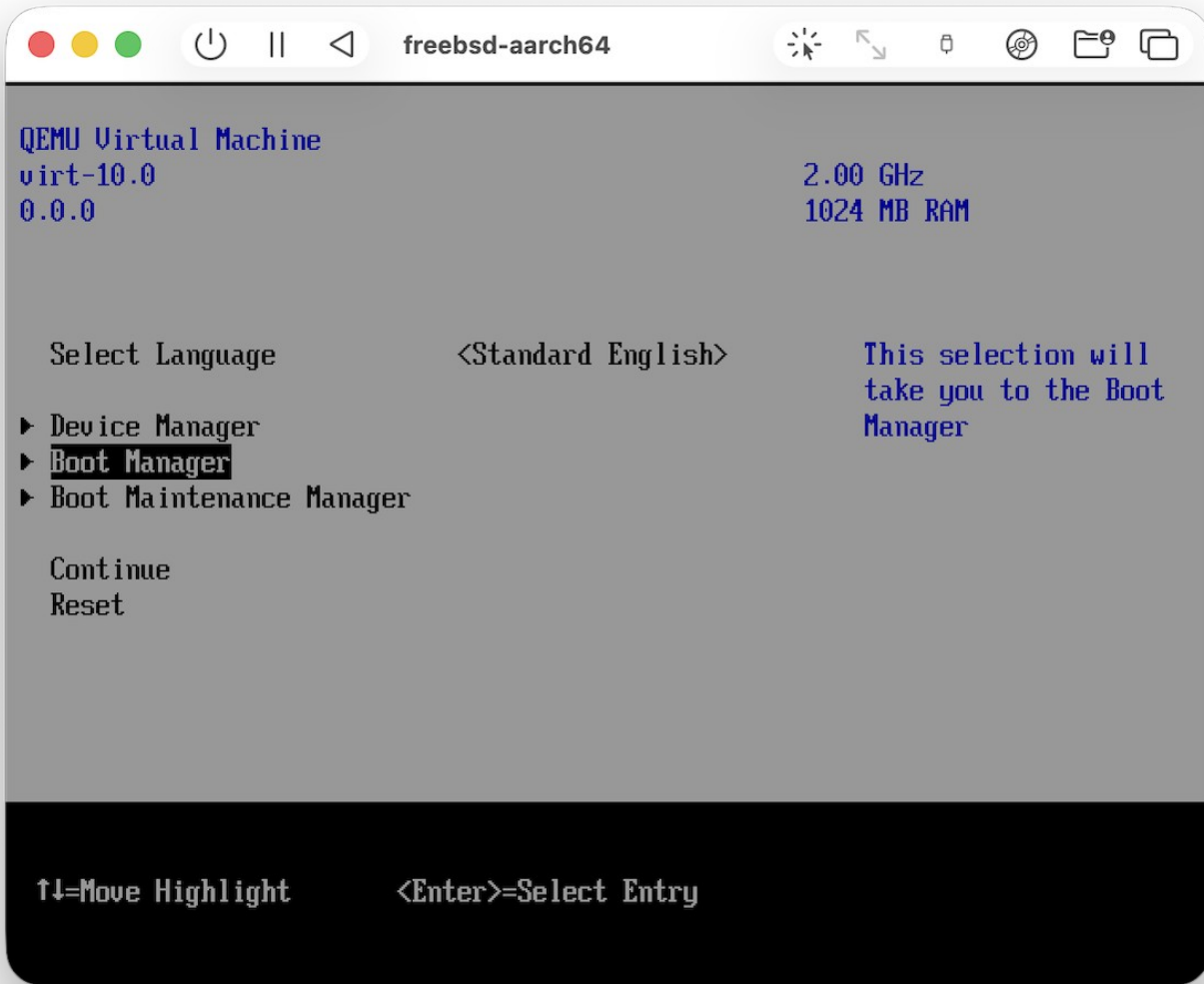
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instalação básica

- <https://docs.freebsd.org/pt-br/books/handbook/bsdinstall>
 - <https://download.freebsd.org/snapshots/ISO-IMAGES/16.0>
- hardware (virtualizado, QEMU)
 - aarch64*
 - 1 cpu
 - 1gb de ram
 - 8gb de disco
 - 1 cd-rom
 - 1 interface de rede
- setup
 - pacotes da base, offline (pkgbase)
 - sistema de arquivos zfs
 - nenhum serviço customizado







freebsd-aarch64



Boot Manager

Boot Manager Menu

UEFI QEMU QEMU USB HARDDRIVE 1-0000:00:03.0-4.1

UEFI QEMU NUMe Ctrl 2EEFCEBC-79C2-4B6A-A 1

UEFI Misc Device

EFI Internal Shell

UEFI PXEv4 (MAC:B2595F807F1C)

Device Path :

PciRoot(0x0)/Pci(0x3,0
x0)/USB(0x7,0x0)/USB(0
x0,0x0)

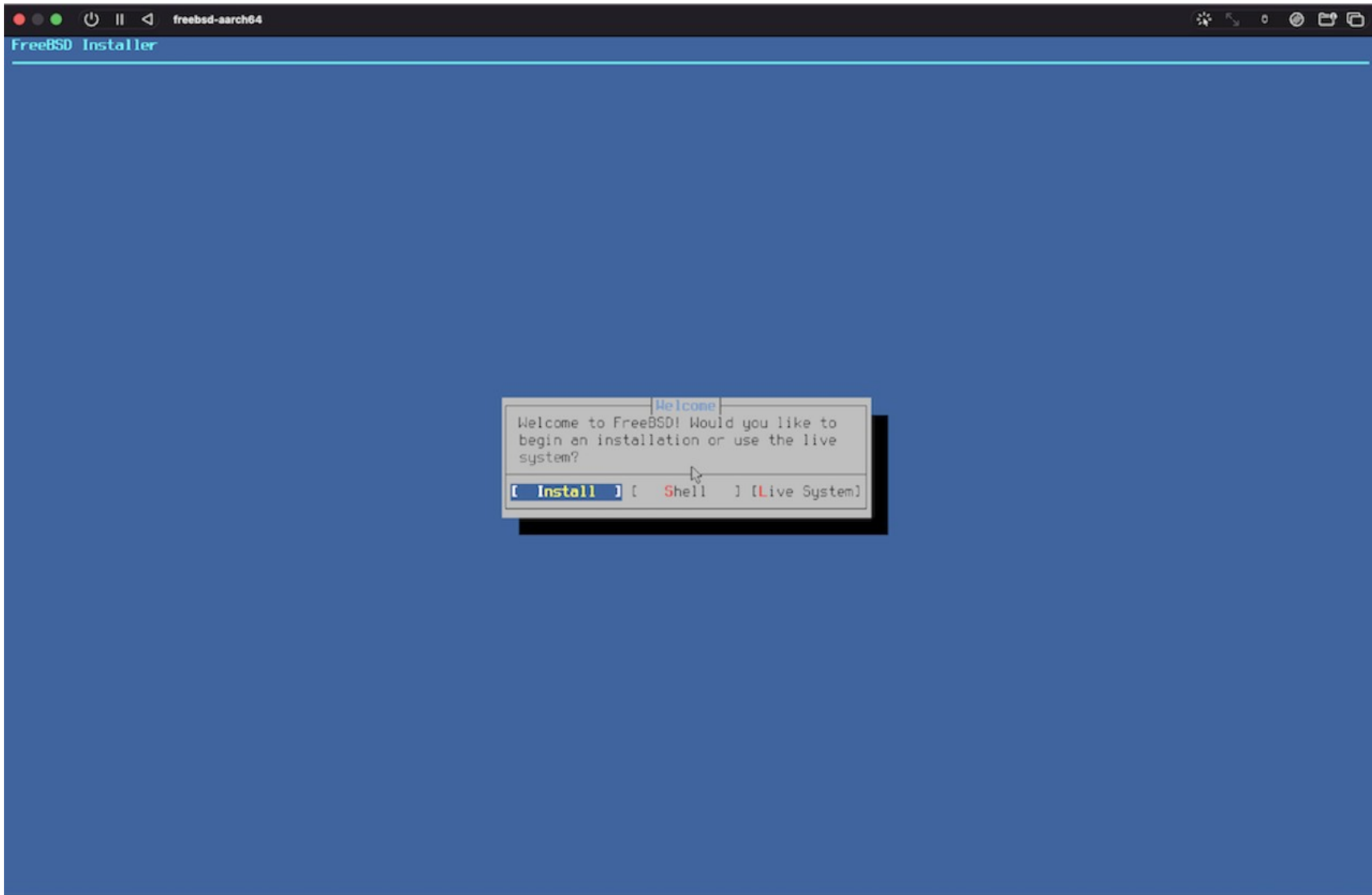
Use the <↑> and <↓> keys to choose a boot option,
the <Enter> key to select a boot option, and the
<Esc> key to exit the Boot Manager Menu.

↑↓=Move Highlight

<Enter>=Select Entry

Esc=Exit


```
freebsd-aarch64
WARNING: WITNESS option enabled, expect reduced performance.
Trying to mount root from cd9660:/dev/iso9660/16_0_CURRENT_AARCH64_CD [ro]...
nda0 at nvme0 bus 0 scbus0 target 0 lun 1
nda0: <QEMU NVMe Ctrl 10.0.2 00833EA2-45E3-4463-9>
nda0: Serial Number 00833EA2-45E3-4463-9
nda0: nvme version 1.4
nda0: 8192MB (16777216 512 byte sectors)
uhub1: 8 ports with 8 removable, self powered
uhub0: 8 ports with 8 removable, self powered
ugen0.2: <QEMU QEMU USB Tablet> at usb0
usbhid0 on uhub1
usbhid0: <QEMU QEMU USB Tablet, class 0/0, rev 2.00/0.00, addr 1> on usb0
hidbus0: <HID bus> on usbhid0
ugen0.3: <QEMU QEMU USB Mouse> at usb0
usbhid1 on uhub1
usbhid1: <QEMU QEMU USB Mouse, class 0/0, rev 2.00/0.00, addr 2> on usb0
hidbus1: <HID bus> on usbhid1
Root mount waiting for: usb0
ugen0.4: <QEMU QEMU USB Keyboard> at usb0
usbhid2 on uhub1
usbhid2: <QEMU QEMU USB Keyboard, class 0/0, rev 2.00/0.00, addr 3> on usb0
hidbus2: <HID bus> on usbhid2
hkbd0: <QEMU QEMU USB Keyboard> on hidbus2
kbd1 at hkbd0
ugen0.5: <QEMU QEMU USB Hub> at usb0
uhub2 on uhub1
uhub2: <on uhub1 USB Hub, class 9/0, rev 1.10/1.01, addr 4> on usb0
usbhid2: <QEMU QEMU USB Keyboard, class 0/0, rev 2.00/0.00, addr 3> on usb0
hidbus2: <HID bus> on usbhid2
hkbd0: <QEMU QEMU USB Keyboard> on hidbus2
kbd1 at hkbd0
ugen0.5: <QEMU QEMU USB Hub> at usb0
uhub2 on uhub1
uhub2: <QEMU QEMU USB Hub, class 9/0, rev 1.10/1.01, addr 4> on usb0
Root mount waiting for: usb0
uhub2: 8 ports with 8 removable, self powered
ugen0.6: <QEMU QEMU USB HARDDRIVE> at usb0
umass0 on uhub2
umass0: <QEMU QEMU USB HARDDRIVE, class 0/0, rev 2.00/0.00, addr 5> on usb0
umass0: SCSI over Bulk-Only; quirks = 0x0
umass0:1:0: Attached to scbus1
cd0 at umass-sim0 bus 0 scbus1 target 0 lun 0
cd0: <QEMU QEMU CD-ROM 2.5+> Removable CD-ROM SPC-3 SCSI device
cd0: Serial Number 1-0000:00:03.0-4.1
cd0: 1.000MB/s transfers
cd0: 1152MB (590121 2048 byte sectors)
cd0: quirks=0x10<10_BYTE_ONLY>
uhub_explore: illegal enable change, port 1
Dual Console: Serial Primary, Video Secondary
```



Keymap Selection

The system console driver for FreeBSD defaults to standard "US" keyboard map. Other keymaps can be chosen below.

```
>> Continue with default keymap
->- Test default keymap
( ) Armenian phonetic layout
( ) Belarusian
( ) Belgian
( ) Belgian (accent keys)
( ) Brazilian (accent keys)
( ) Brazilian (without accent keys)
( ) Bulgarian (BDS)
( ) Bulgarian (Phonetic)
( ) Canadian Bilingual
( ) Canadian Multilingual Standard
( ) Central European
( ) Central European (QWERTY)
( ) Colemak Mod-DH ergonomic alternative
( ) Colemak ergonomic alternative
( ) Croatian
( ) Czech (QWERTZ, accent keys)
( ) Danish
( ) Danish (MacBook)
( ) Danish (accent keys)
( ) Dutch (accent keys)
( ) Estonian
( ) Finnish
( ) French
( ) French (MacBook/MacBook Pro) (accent keys)
( ) French (accent keys)
( ) French Bépo
( ) French Canadian (accent keys)
( ) French Dvorak-like
( ) French Dvorak-like (accent keys)
( ) German
( ) German (accent keys)
( ) German (no accent keys)
( ) Greek (101 keys)
( ) Greek (104 keys)
( ) Greek ELOT
```

!!!

41%

[Select]

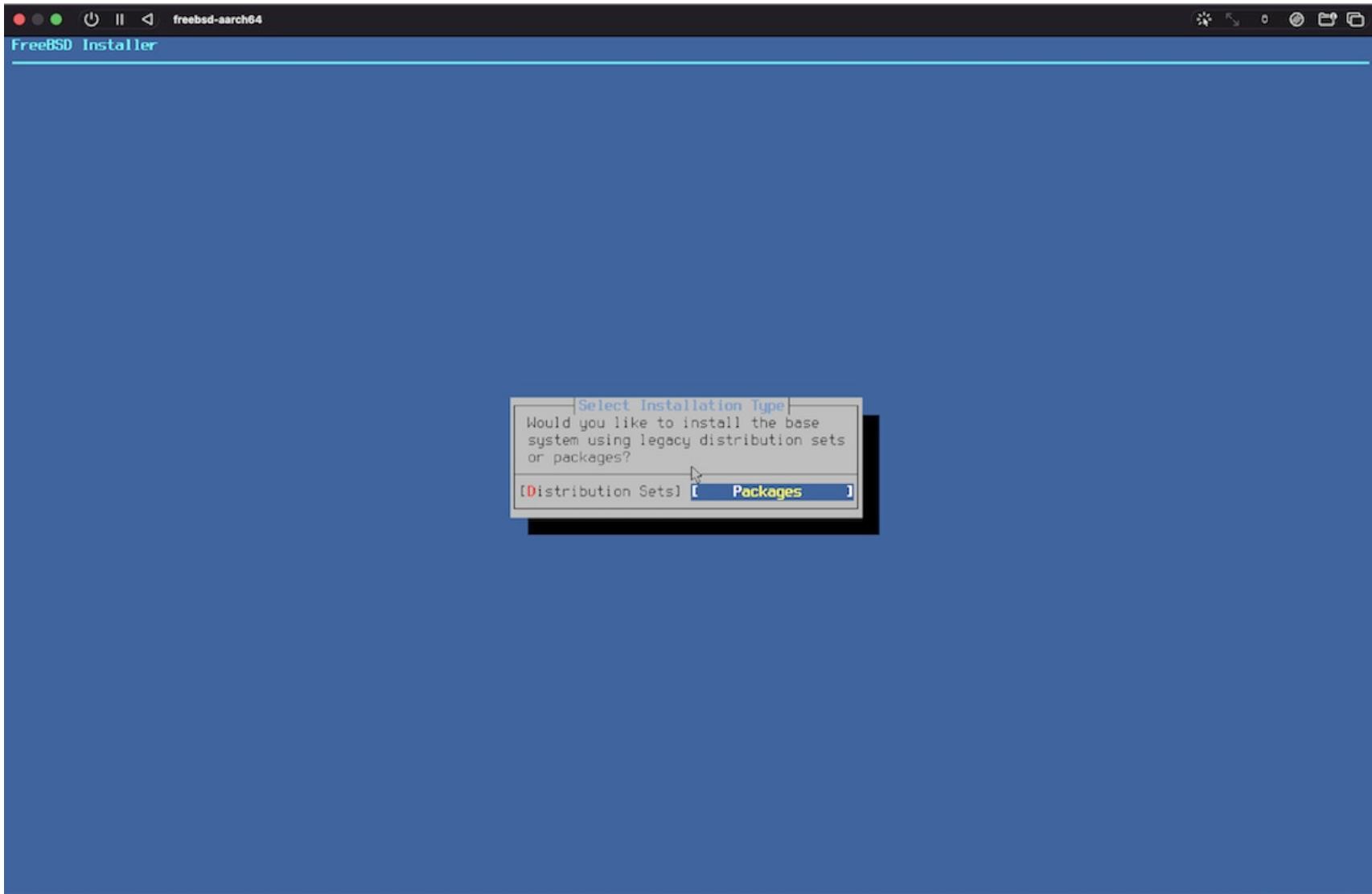
[Cancel]

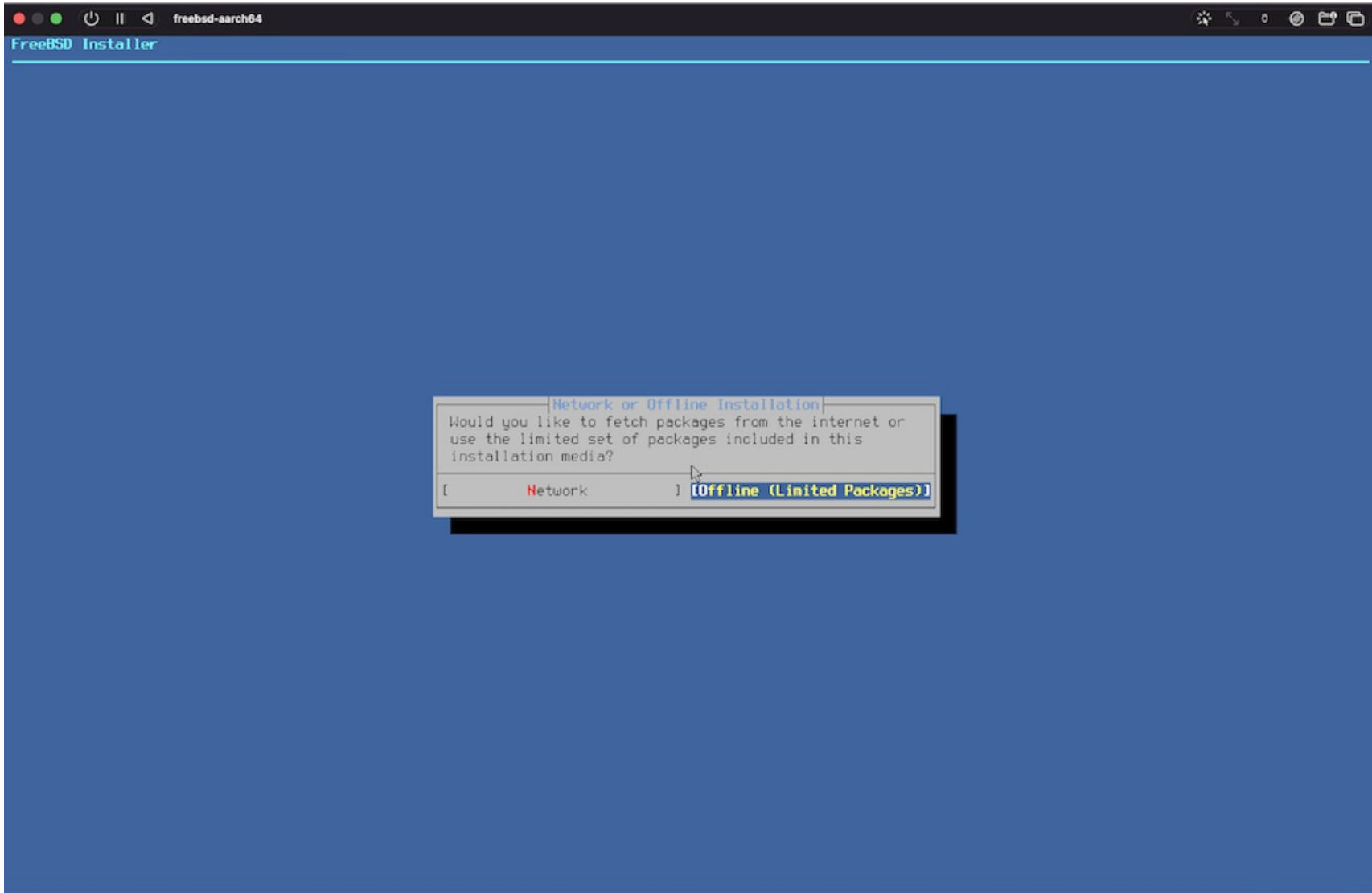
Press arrows, TAB or ENTER

Set Hostname

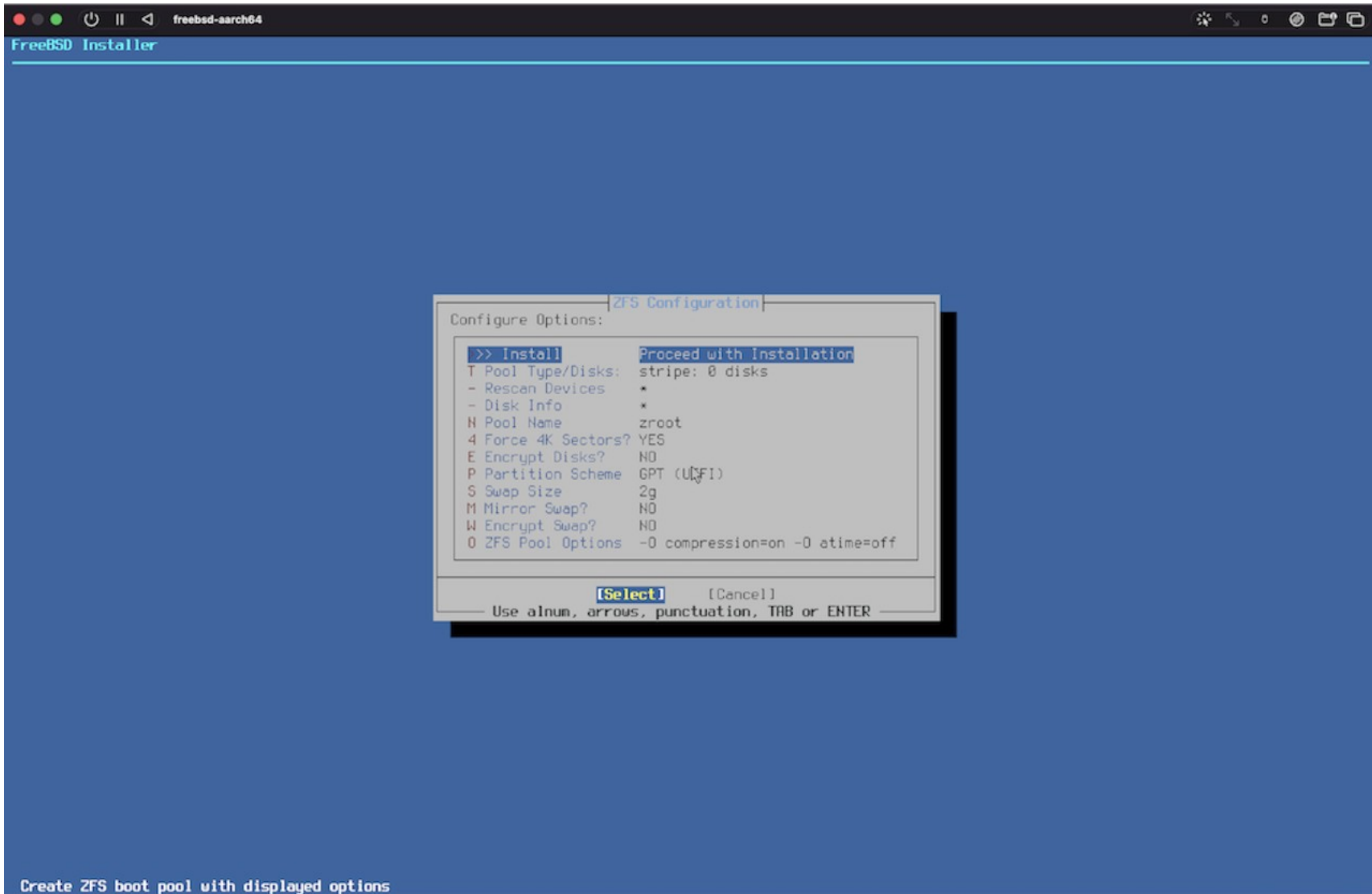
Please choose a hostname for this machine.

If you are running on a managed network, please ask your network administrator for an appropriate name.











ZFS Configuration

Please select one or more disks to create a pool:

☒ da0 QEMU NVMe Ctrl

[OK] [Back]

Use arrows, SPACE, TAB or ENTER

ZFS Configuration

Last Chance! Are you sure you want to **destroy**
the current contents of the following disks:

nda0

[YES] **[NO]**

Press arrows, TAB or ENTER

Select System Components

A minimal set of packages suitable for a multi-user system is always installed. Select additional packages you wish to install:

- ☐ base The complete base system (includes devel and optional)
- ☐ debug Debug symbols for the selected components
- ☐ devel C/C++ compilers and related utilities
- ☐ kernel-dbg Debug symbols for the kernel
- ☒ i386 32-bit compatibility libraries
- ☐ optional Optional software (excluding compilers)
- ☐ src System source tree
- ☐ tests Test suite

[OK]

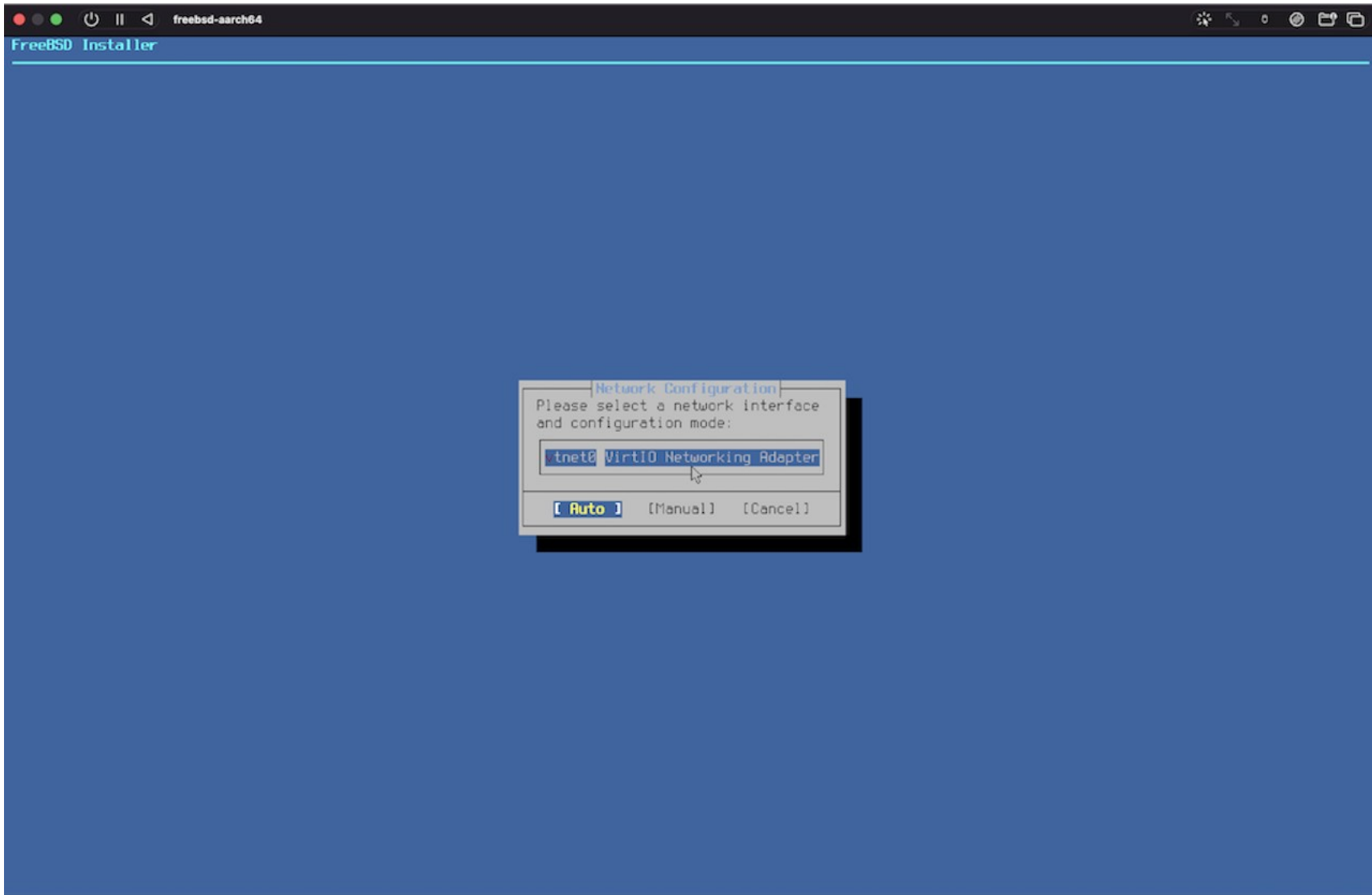
```
freebsd-aarch64
[22/71] Installing FreeBSD-libcasper-16.snap20260413132855...
[22/71] Extracting FreeBSD-libcasper-16.snap20260413132855: 100%
[23/71] Installing FreeBSD-libldns-16.snap20260413132855...
[23/71] Extracting FreeBSD-libldns-16.snap20260413132855: 100%
[24/71] Installing FreeBSD-libmagic-16.snap20260413132855...
[24/71] Extracting FreeBSD-libmagic-16.snap20260413132855: 100%
[25/71] Installing FreeBSD-libucl-16.snap20260413132855...
[25/71] Extracting FreeBSD-libucl-16.snap20260413132855: 100%
[26/71] Installing FreeBSD-libyaml-16.snap20260413132855...
[26/71] Extracting FreeBSD-libyaml-16.snap20260413132855: 100%
[27/71] Installing FreeBSD-locales-16.snap20260413132855...
[27/71] Extracting FreeBSD-locales-16.snap20260413132855: 100%
[28/71] Installing FreeBSD-mandoc-16.snap20260413132855...
[28/71] Extracting FreeBSD-mandoc-16.snap20260413132855: 100%
[29/71] Installing FreeBSD-mtree-16.snap20260413132855...
[29/71] Extracting FreeBSD-mtree-16.snap20260413132855: 100%
[30/71] Installing FreeBSD-natd-16.snap20260413132855...
[30/71] Extracting FreeBSD-natd-16.snap20260413132855: 100%
[31/71] Installing FreeBSD-ncurses-16.snap20260413132855...
[31/71] Extracting FreeBSD-ncurses-16.snap20260413132855: 100%
[32/71] Installing FreeBSD-ncurses-lib-16.snap20260413132855...
[32/71] Extracting FreeBSD-ncurses-lib-16.snap20260413132855: 100%
[33/71] Installing FreeBSD-newsyslog-16.snap20260413132855...
[33/71] Extracting FreeBSD-newsyslog-16.snap20260413132855: 100%
[34/71] Installing FreeBSD-nuageinit-16.snap20260413132855...
[34/71] Extracting FreeBSD-nuageinit-16.snap20260413132855: 100%
[35/71] Installing FreeBSD-openssl-16.snap20260413132855...
[35/71] Extracting FreeBSD-openssl-16.snap20260413132855: 100%
[36/71] Installing FreeBSD-certctl-16.snap20260413132855...
[36/71] Extracting FreeBSD-certctl-16.snap20260413132855: 100%
[37/71] Installing FreeBSD-caroot-16.snap20260413132855...
[37/71] Extracting FreeBSD-caroot-16.snap20260413132855: 100%
[38/71] Installing FreeBSD-openssl-lib-16.snap20260413132855...
[38/71] Extracting FreeBSD-openssl-lib-16.snap20260413132855: 100%
[39/71] Installing FreeBSD-pam-16.snap20260413132855...
[39/71] Extracting FreeBSD-pam-16.snap20260413132855: 100%
[40/71] Installing FreeBSD-pam-lib-16.snap20260413132855...
[40/71] Extracting FreeBSD-pam-lib-16.snap20260413132855: 100%
[41/71] Installing FreeBSD-periodic-16.snap20260413132855...
[41/71] Extracting FreeBSD-periodic-16.snap20260413132855: 100%
[42/71] Installing FreeBSD-pkg-bootstrap-16.snap20260413132855...
[42/71] Extracting FreeBSD-pkg-bootstrap-16.snap20260413132855: 100%
[43/71] Installing FreeBSD-powerd-16.snap20260413132855...
[43/71] Extracting FreeBSD-powerd-16.snap20260413132855: 100%
[44/71] Installing FreeBSD-ppp-16.snap20260413132855...
[44/71] Extracting FreeBSD-ppp-16.snap20260413132855: 100%
[45/71] Installing FreeBSD-rdma-lib-16.snap20260413132855...
[45/71] Extracting FreeBSD-rdma-lib-16.snap20260413132855: 100%
[46/71] Installing FreeBSD-rescue-16.snap20260413132855...
[46/71] Extracting FreeBSD-rescue-16.snap20260413132855: 0%
```

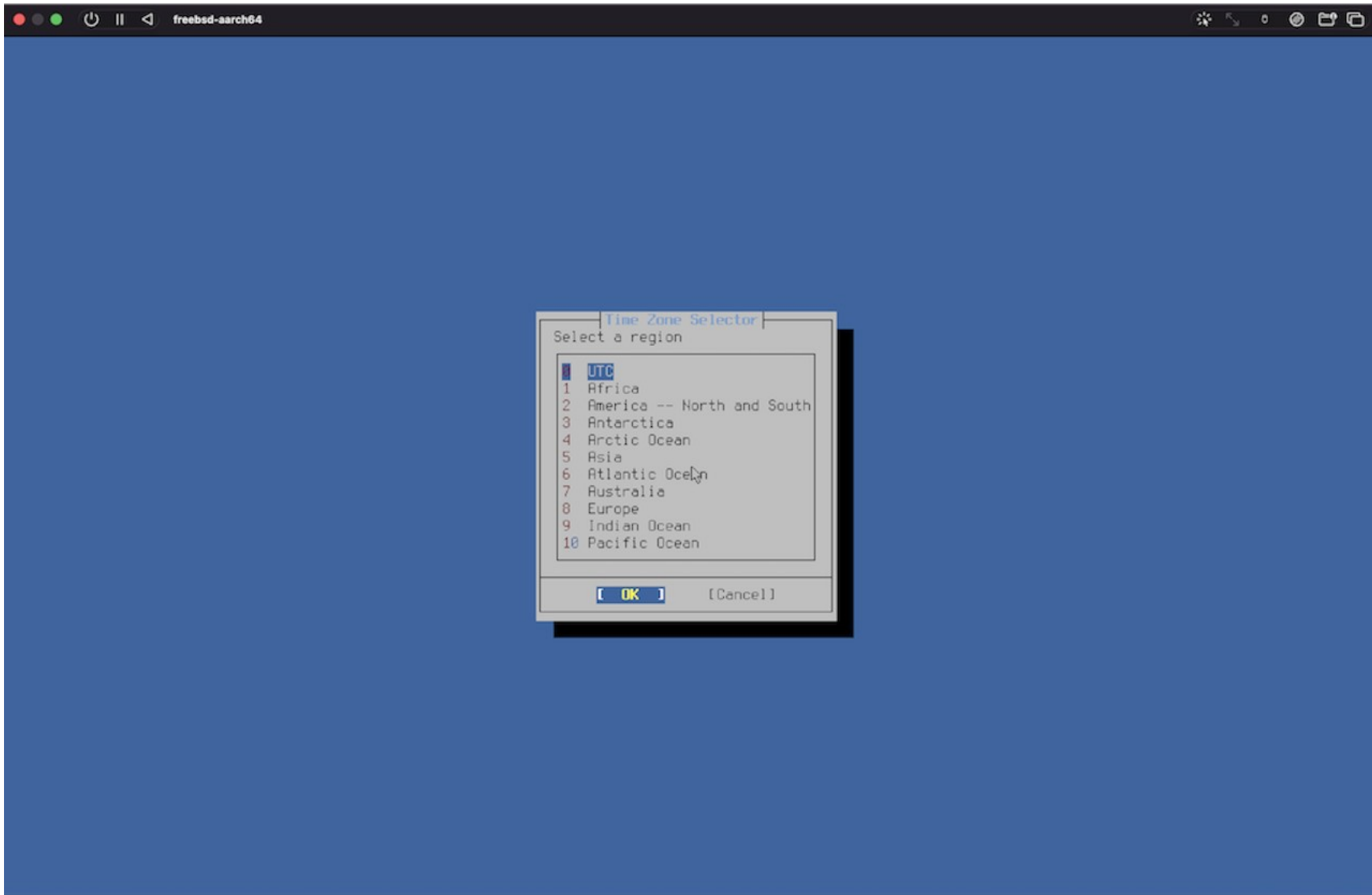
Set root password

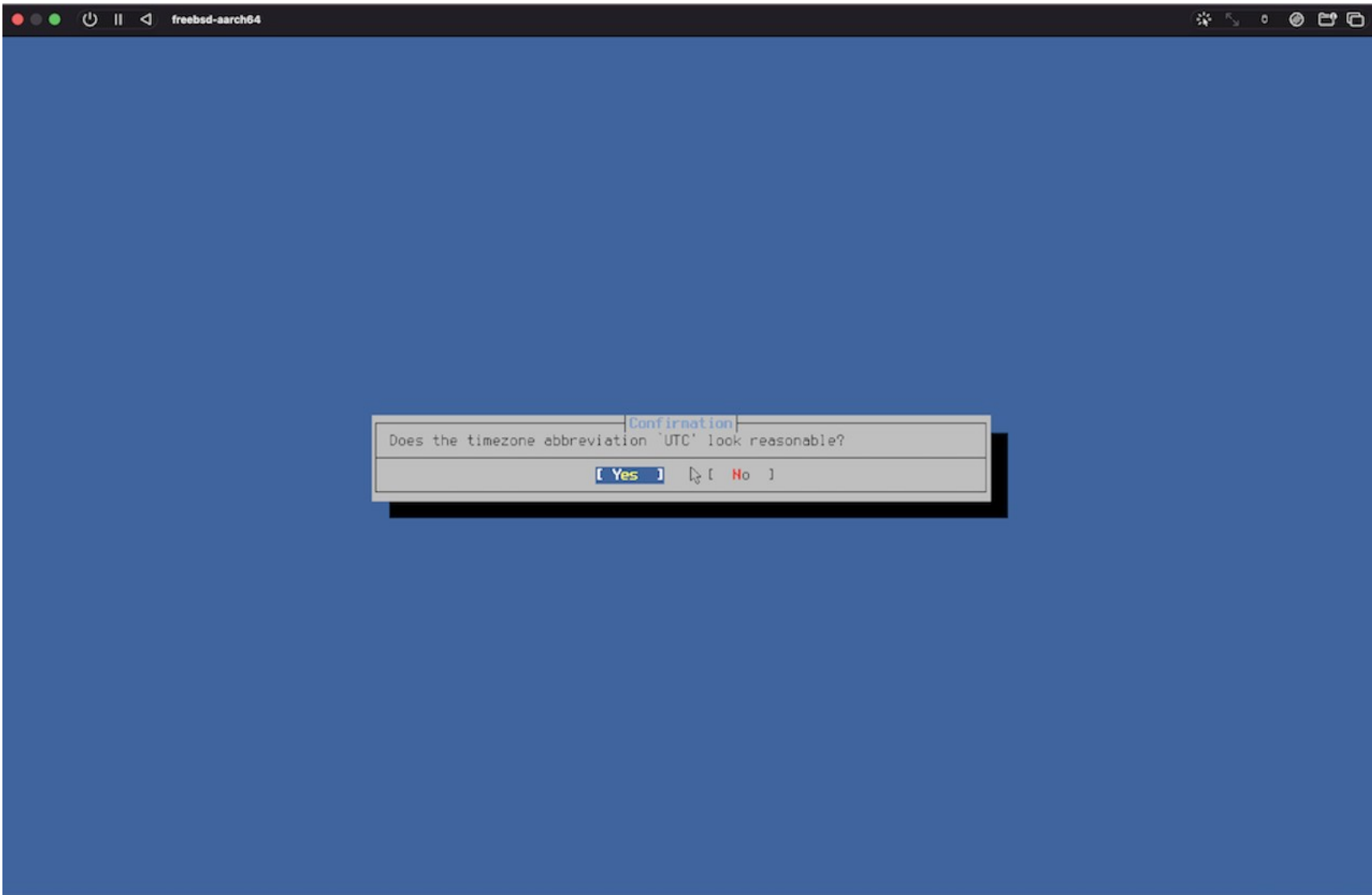
Please select a password for the system management account (root)

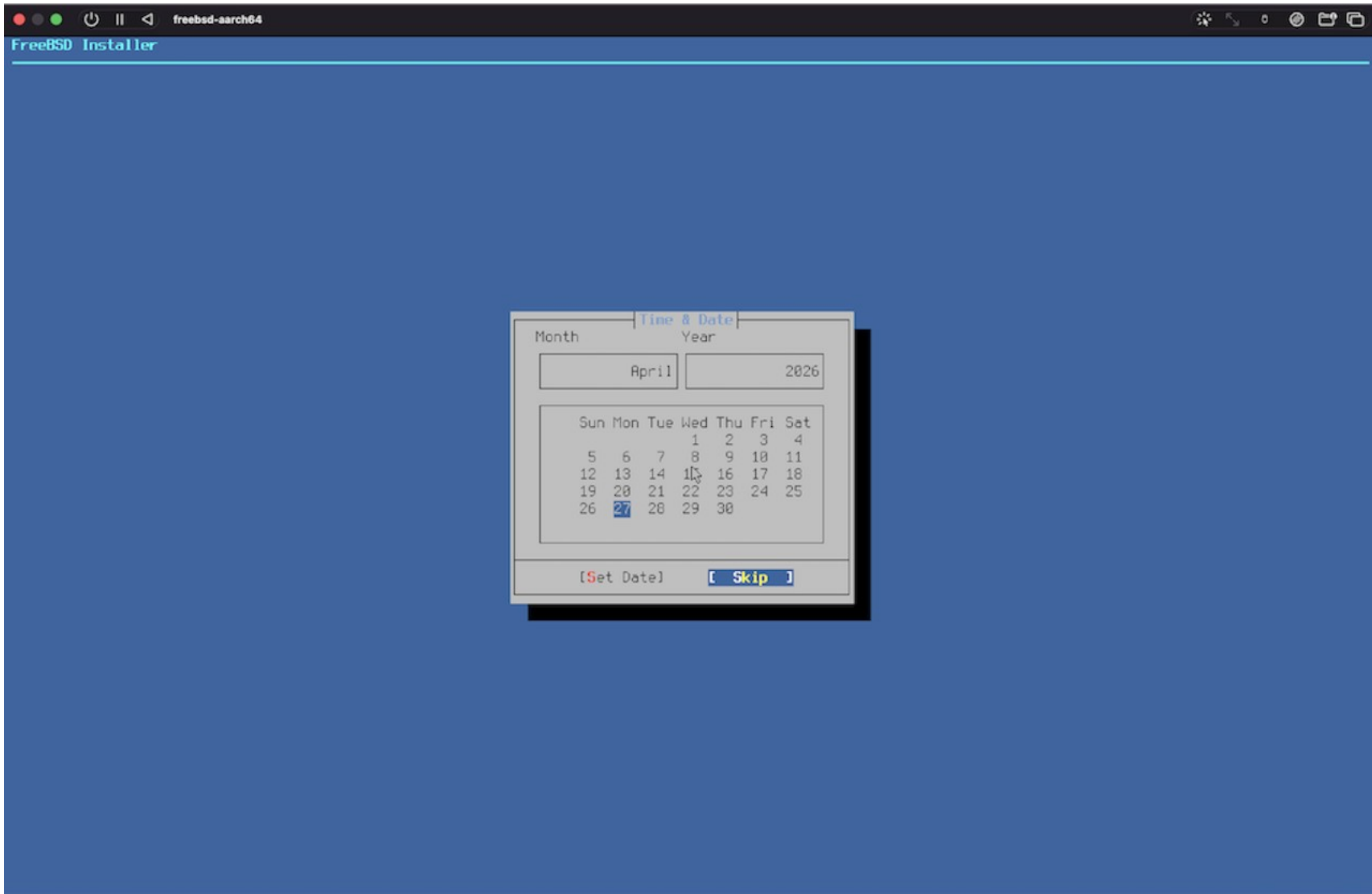
Password	***
Repeat password	***

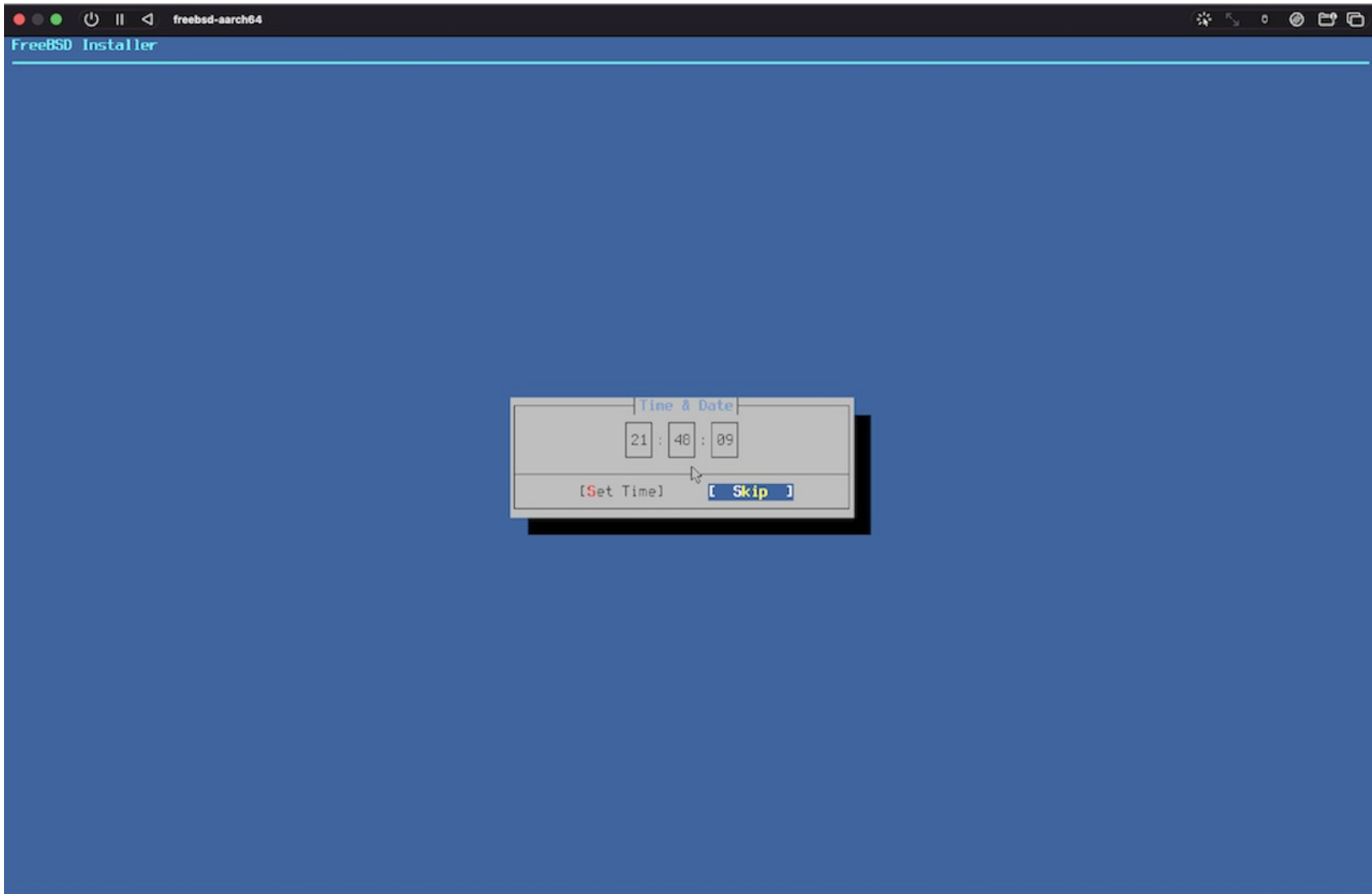
[OK] [Skip]











System Configuration

Choose the services you would like to be started at boot:

- ☒ powerd Adjust CPU frequency dynamically if supported
- ☒ dumpdev Enable kernel crash dumps to /var/crash

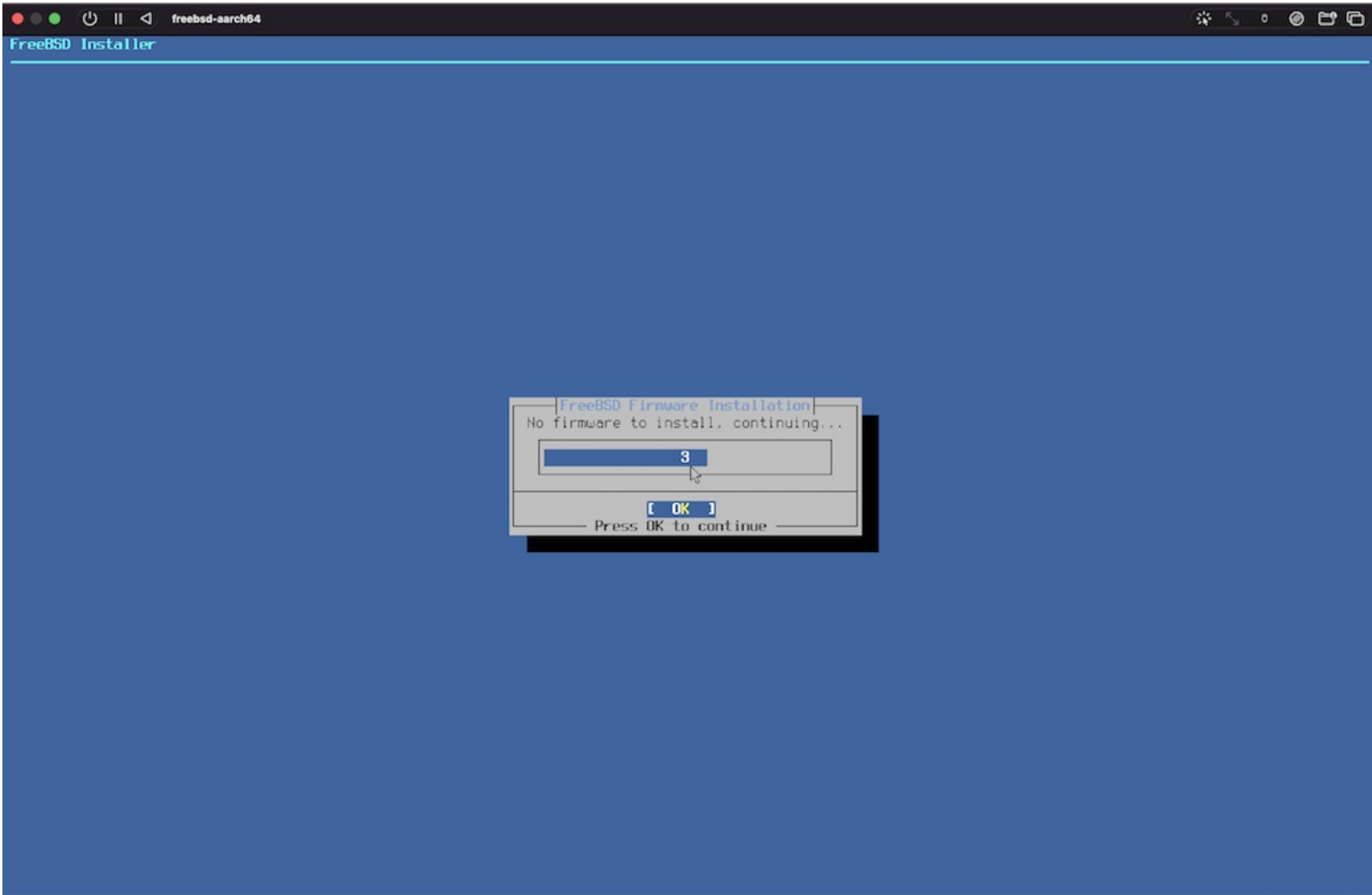
[OK]

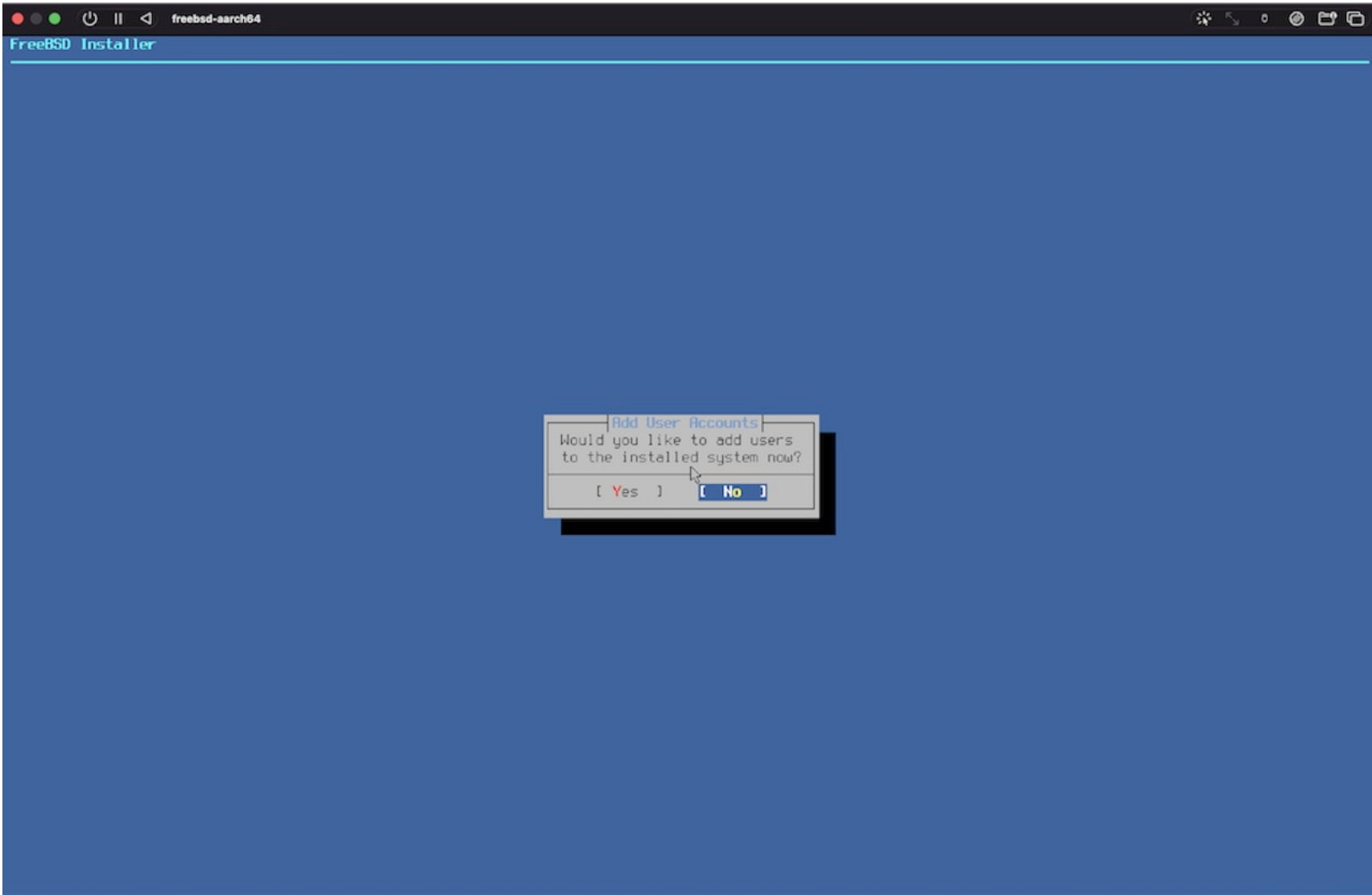
System Hardening

Choose system security hardening options:

- | | | |
|---------------------------------------|-----------------|--|
| ☒ 1 | hide_uids | Hide processes running as other users |
| ☐ 2 | hide_gids | Hide processes running as other groups |
| ☐ 3 | hide_jail | Hide processes running in jails |
| ☐ 4 | read_msgbuf | Disable reading kernel message buffer for unprivileged users |
| ☐ 5 | proc_debug | Disable process debugging facilities for unprivileged users |
| ☐ 6 | random_pid | Randomize the PID of newly created processes |
| ☐ 7 | clear_tmp | Clean the /tmp filesystem on system startup |
| ☐ 8 | disable_syslogd | Disable opening Syslogd network socket (disables remote logging) |
| ☐ 9 | secure_console | Enable console password prompt |
| ☐ 10 | disable_ddtrace | Disallow DTrace destructive-mode |

[OK]





Final Configuration

Setup of your FreeBSD system is nearly complete. You can now modify your configuration choices. After this screen, you will have an opportunity to make more complex changes using a shell.

Finish	Apply configuration and exit installer
Add User	Add a user to the system
Root Password	Change root password
Hostname	Set system hostname
Network	Networking configuration
Services	Set daemons to run on startup
System Hardening	Set security options
Time Zone	Set system timezone
Firmware	Install Firmware (requires network)
Handbook	Install FreeBSD Handbook (requires network)

[Select]

Manual Configuration

The installation is now finished. Before exiting the installer, would you like to open a shell in the new system to make any final manual modifications?

[Yes]

[No]

Complete

Installation of FreeBSD complete! Would
you like to reboot into the installed
system now?

[**Reboot**] [Shutdown] [Live System]

```
#17 0xffff000000d1258 at fork_trampoline+0x18
lock order reversal: (sleepable after non-sleepable)
 1st 0xffffa00001509d00 vtnet0-rx0 (vtnet0-rx0, sleep mutex) @ /usr/src/sys/dev/virtio/network/if_vtnet.c:2250
 2nd 0xffff0000011b92f0 in6_multi_sx (in6_multi_sx, sx) @ /usr/src/sys/netinet6/in6_mcast.c:1216
lock order vtnet0-rx0 -> in6_multi_sx attempted at:
#0 0xffff000000055f708 at witness_checkorder+0xb70
#1 0xffff00000004f0670 at _sx_xlock+0x70
#2 0xffff00000006e40e8 at in6_joingroup+0x40
#3 0xffff00000006db920 at in6_update_ifa+0xc6c
#4 0xffff0000000709550 at in6_ifadd+0x2b0
#5 0xffff0000000705070 at nd6_ra_input+0xf30
#6 0xffff00000006d5a54 at icmp6_input+0x0c8
#7 0xffff00000006f02bc at ip6_input+0xe54
#8 0xffff000000064e5f8 at netisr_dispatch_src+0xd8
#9 0xffff000000062f900 at ether_demux+0x174
#10 0xffff0000000630ed4 at ether_nh_input+0x374
#11 0xffff000000064e5f8 at netisr_dispatch_src+0xd8
#12 0xffff000000062fdc8 at ether_input+0xdc
#13 0xffff00000003401f0 at vtnet_rxq_eof+0x750
#14 0xffff00000003479d4 at vtnet_rx_vq_process+0xb0
#15 0xffff000000033f9a0 at vtpci_intx_intr+0xe8
#16 0xffff0000000497fa0 at ithread_loop+0x29c
#17 0xffff00000004940bc at fork_exit+0x70
```

FreeBSD/arm64 (freebsd-aarch64) (ttyv0)

login: root

Password:

FreeBSD 16.0-CURRENT (GENERIC) #0 main-n285005-e9fc0c538264: Mon Apr 13 12:48:35 UTC 2026

Welcome to FreeBSD!

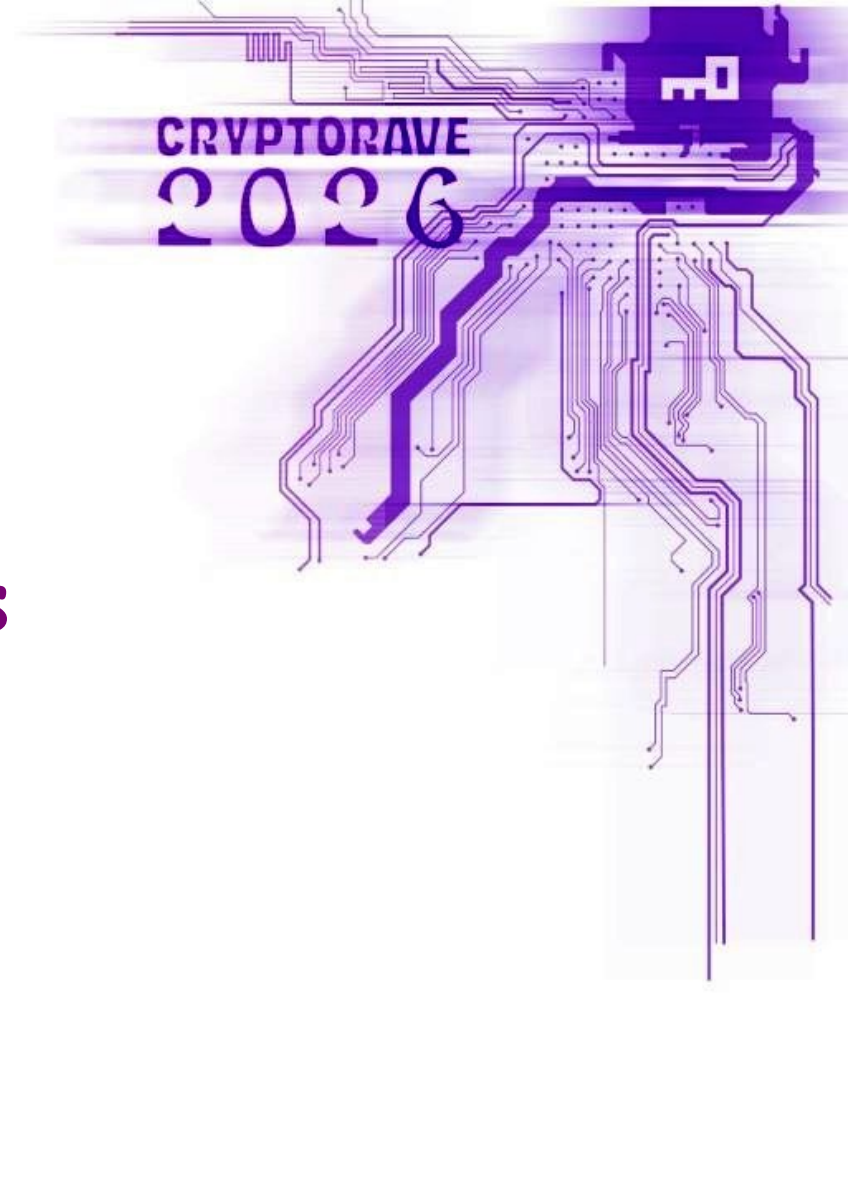
Release Notes, Errata: <https://www.FreeBSD.org/releases/>
 Security Advisories: <https://www.FreeBSD.org/security/>
 FreeBSD Handbook: <https://www.FreeBSD.org/handbook/>
 FreeBSD FAQ: <https://www.FreeBSD.org/faq/>
 Questions List: <https://www.FreeBSD.org/lists/questions/>
 FreeBSD Forums: <https://forums.FreeBSD.org/>

Documents installed with the system are in the /usr/local/share/doc/freebsd/
 directory, or can be installed later with: pkg install en-freebsd-doc
 For other languages, replace "en" with a language code like de or fr.

Show the version of FreeBSD installed: freebsd-version ; uname -a
 Please include that output and any error messages when posting questions.
 Introduction to manual pages: man man
 FreeBSD directory layout: man hier

To change this login announcement, see motd(5).

root@freebsd-aarch64:~ #

A stylized graphic of a circuit board in shades of purple and blue, featuring various traces, pads, and a small component resembling a microchip or connector in the upper right corner.

CRYPTORAVE
2026

motivações

é claro que isso roda... **netbsd**

- ambientes linux com dropbear
 - <https://duckduckgo.com/?q=dropbear+initramfs+luks>
- conjunto de funcionalidades no "estado da arte"
 - pesquisa e desenvolvimento
- "tomar/usar da própria medicina" (dogfooding)

meus dados, minhas regras

- autonomia
- backups diversos
- lgpd/gdpr
- setup de laboratório forense

projetos relacionados

- <https://github.com/clinta/geliUnlocker>
- <https://github.com/oxfox22/freebsd-remote-crypto-single-drive>
- <https://github.com/Sec42/freebsd-remote-crypto>
- <https://github.com/emtiu/freebsd-outerbase>

automatize todas as coisas

- eurobsdcon 2021

- <https://slideshare.net/slideshow/eurobsdcon-2021-autoinstalling-bsd-systems/250243872>
- <https://www.youtube.com/watch?v=7F3UwfNB2JA>

- eurobsdcon 2023

- <https://slideshare.net/slideshow/eurobsdcon-2023-autoinstalling-bsd-systems-cases-using-pfsense-truenas-and-more/261357863>
- <https://www.youtube.com/watch?v=ByvSwhCl8k8>

cd /pub && more \$drink

- amig_s, família, pets, plantas, ...
- praia, serra, vale, fazenda, ...
- esportes, hobbies, ...



A stylized graphic of a circuit board in shades of purple and blue, located in the top right corner. It features various traces, pads, and a small component that looks like a microcontroller or chip.

CRYPTORAVE
2026

prova de conceito

código fonte do instalador oficial

- lexi winter (ivy@), glen barber & jordan hubbard
 - <https://cgит.freebsd.org/src/tree/release>
 - release.sh
 - \${target}/make-memstick.sh
 - \${target}/mkisoimages.sh
- devin teske, rebecca cran, allan jude & nathan whitehorn
 - <https://cgит.freebsd.org/src/tree/usr.sbin>
 - bsdconfig/
 - bsdinstall/

mfsbsd como referência na criação da "imagem base"

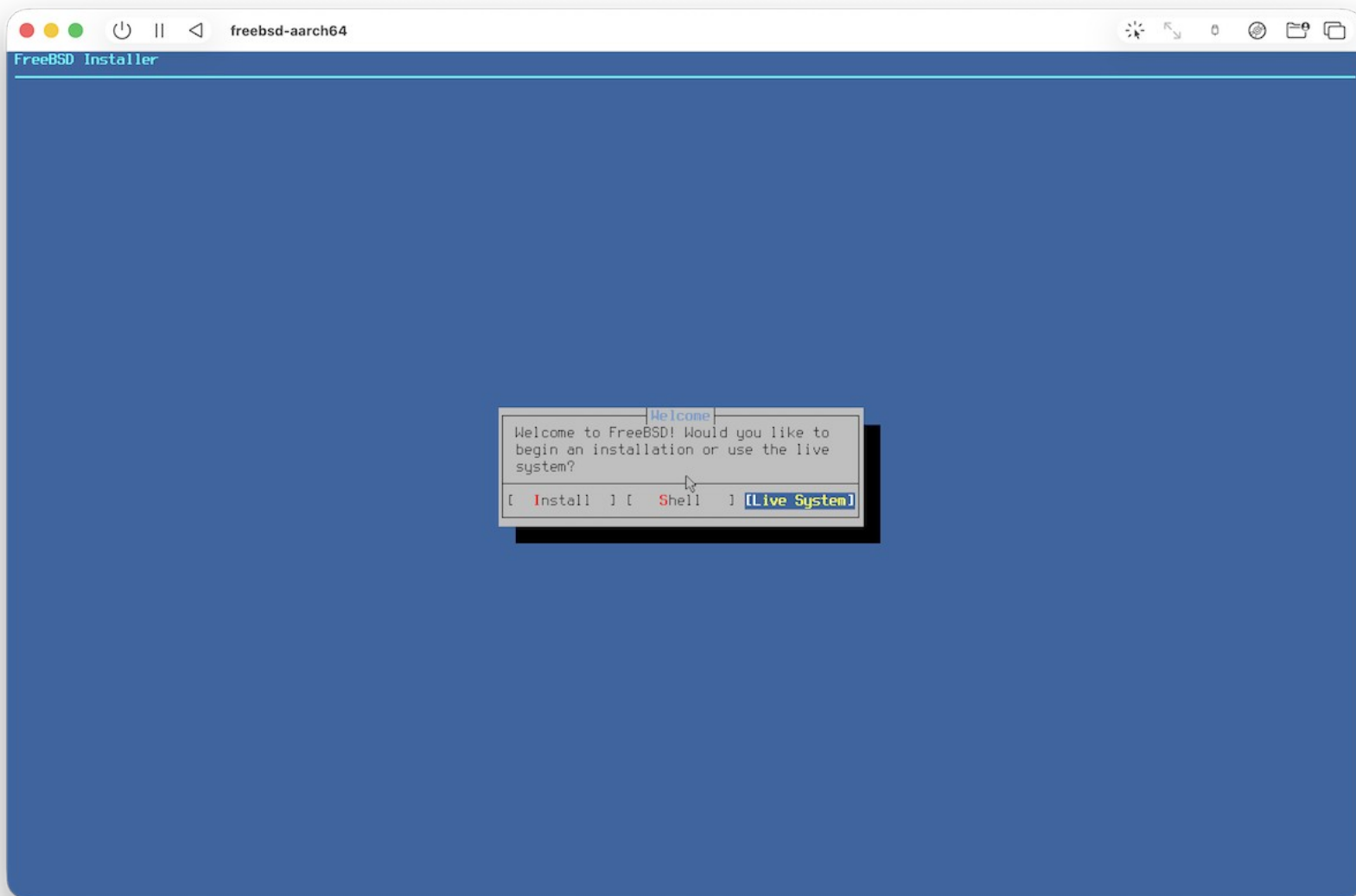
- martin matuška
 - <https://github.com/mmatуска/mfsbsd>

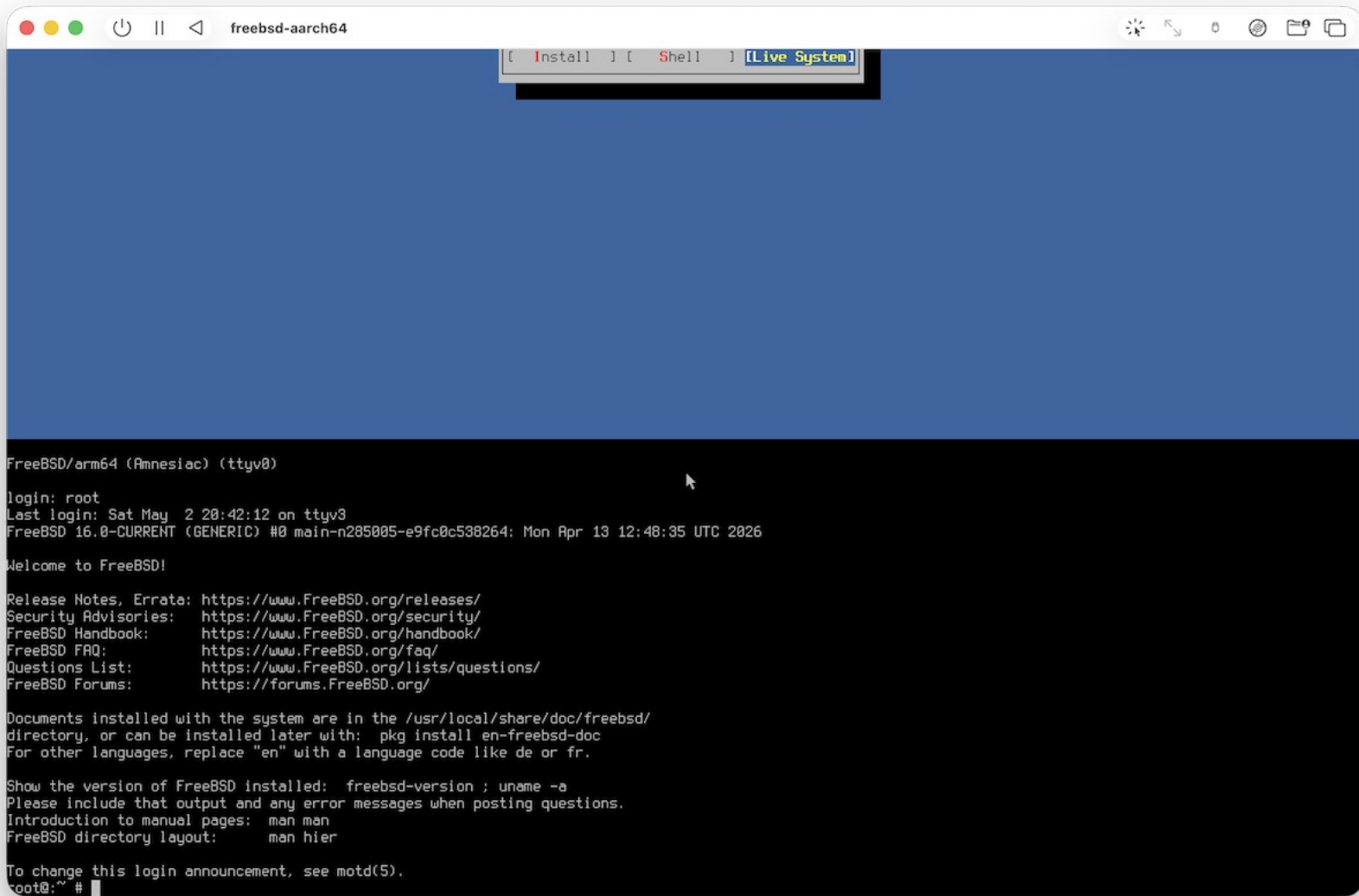
instalação customizada

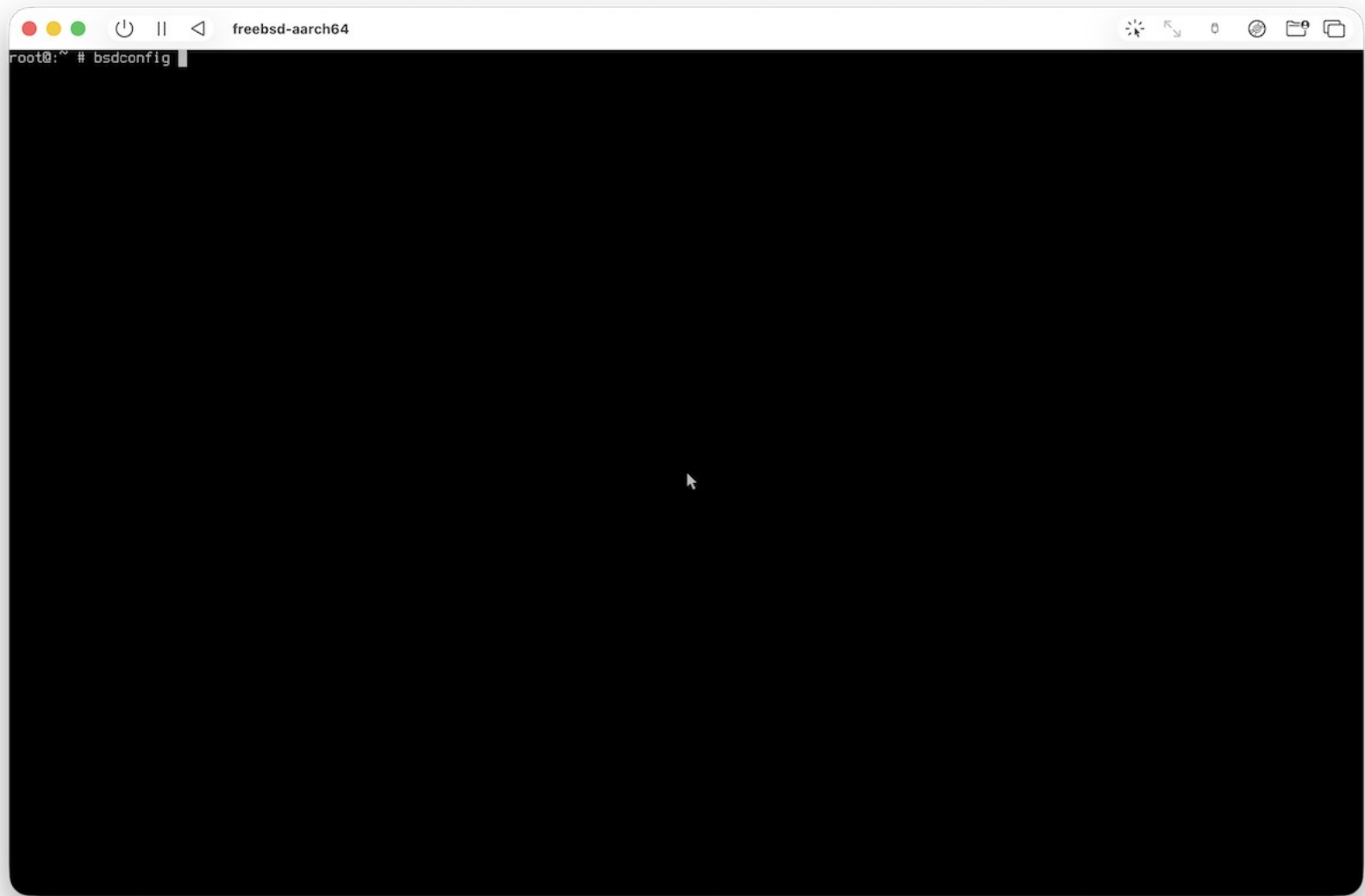
- partição "base" (com kernel)
 - sem gerenciador de pacotes
 - elementos mínimos do sistema base
- partição "elke" (sem kernel)
 - <https://docs.freebsd.org/pt-br/books/handbook/zfs/>
 - cenário 1: sem criptografia
 - cenário 2: com criptografia

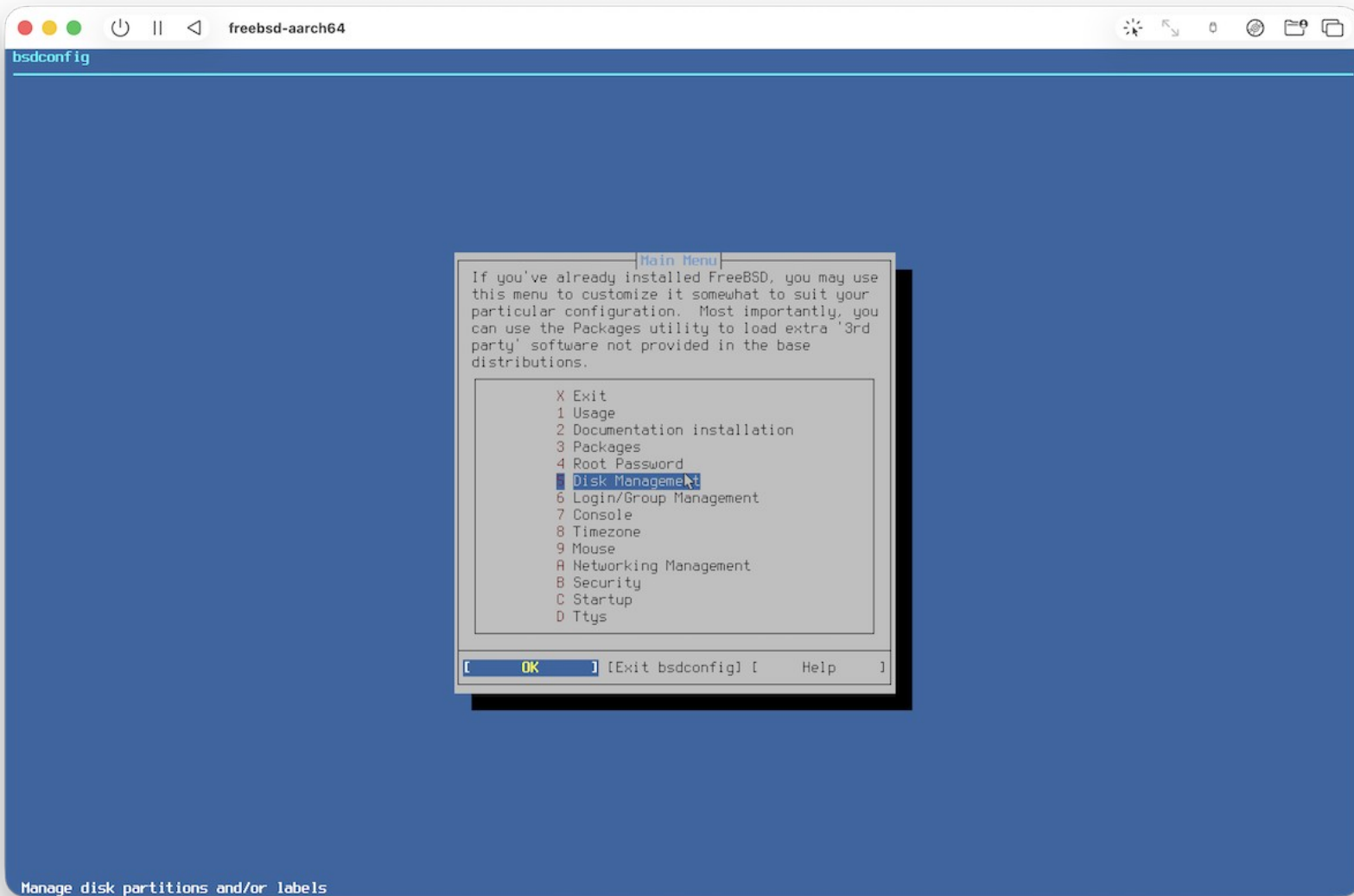
instalação customizada via "live system"

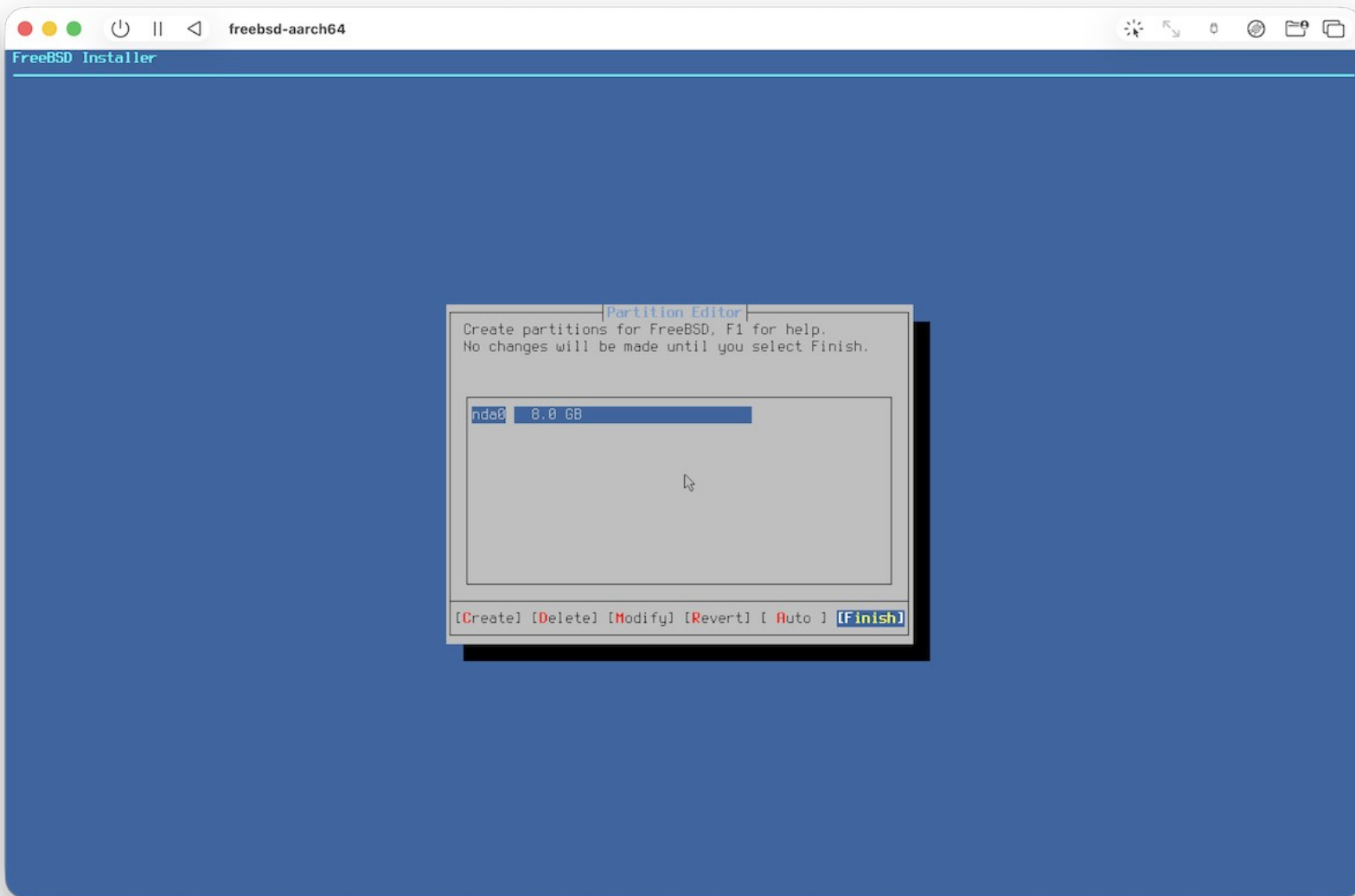
- particionamento de disco
 - bsdconfig diskmgmt
 - gpart & newfs/newfs_msdos
- bootloader (uefi)
 - efibootmgr
- elementos do sistema
 - kernel & freebsd "sets"
 - zfs (pool & datasets)
- configuração do sistema
 - hostname, networking, sshd, ...
- modificações customizadas
 - pacotes adicionais

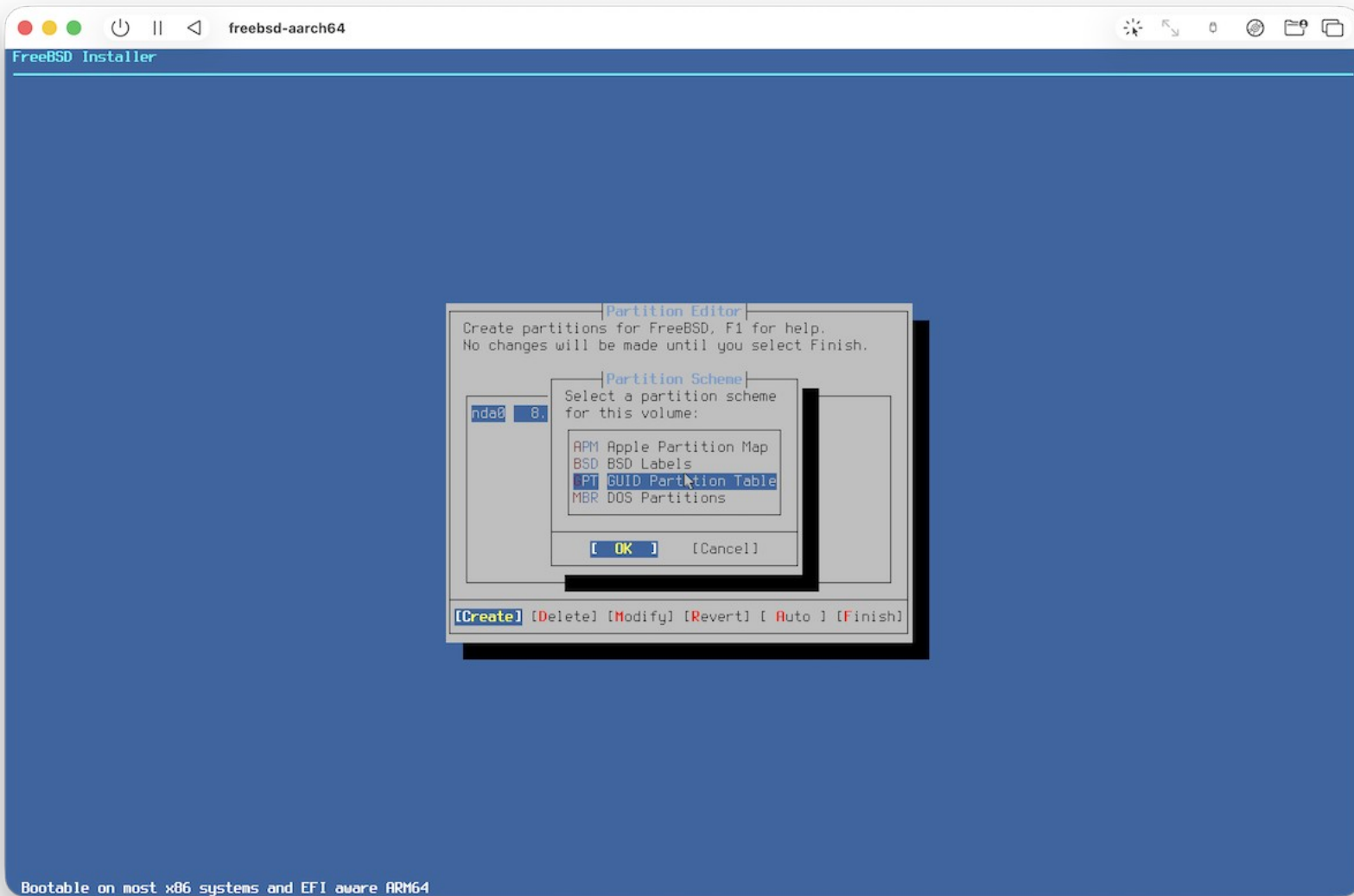


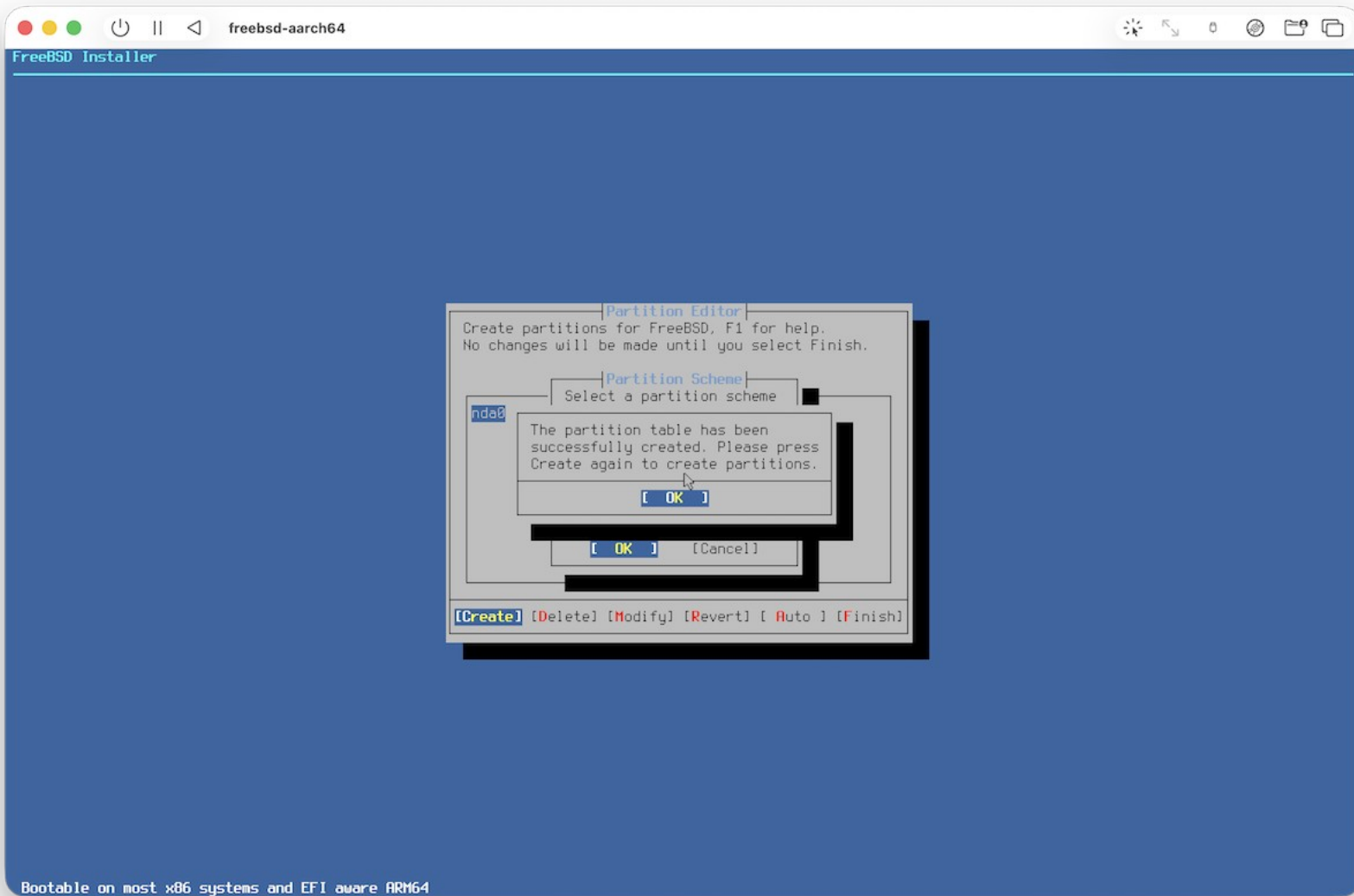










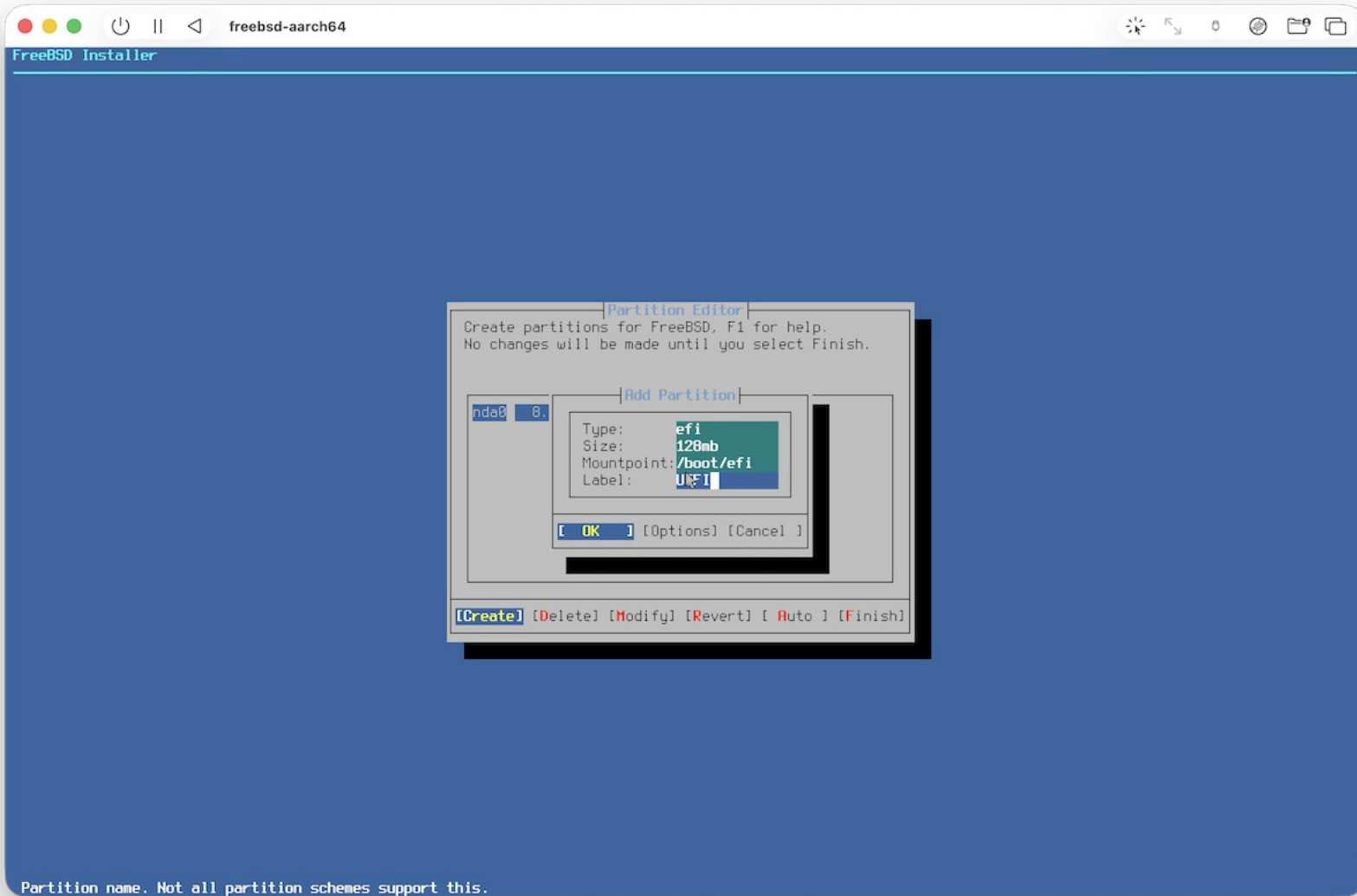


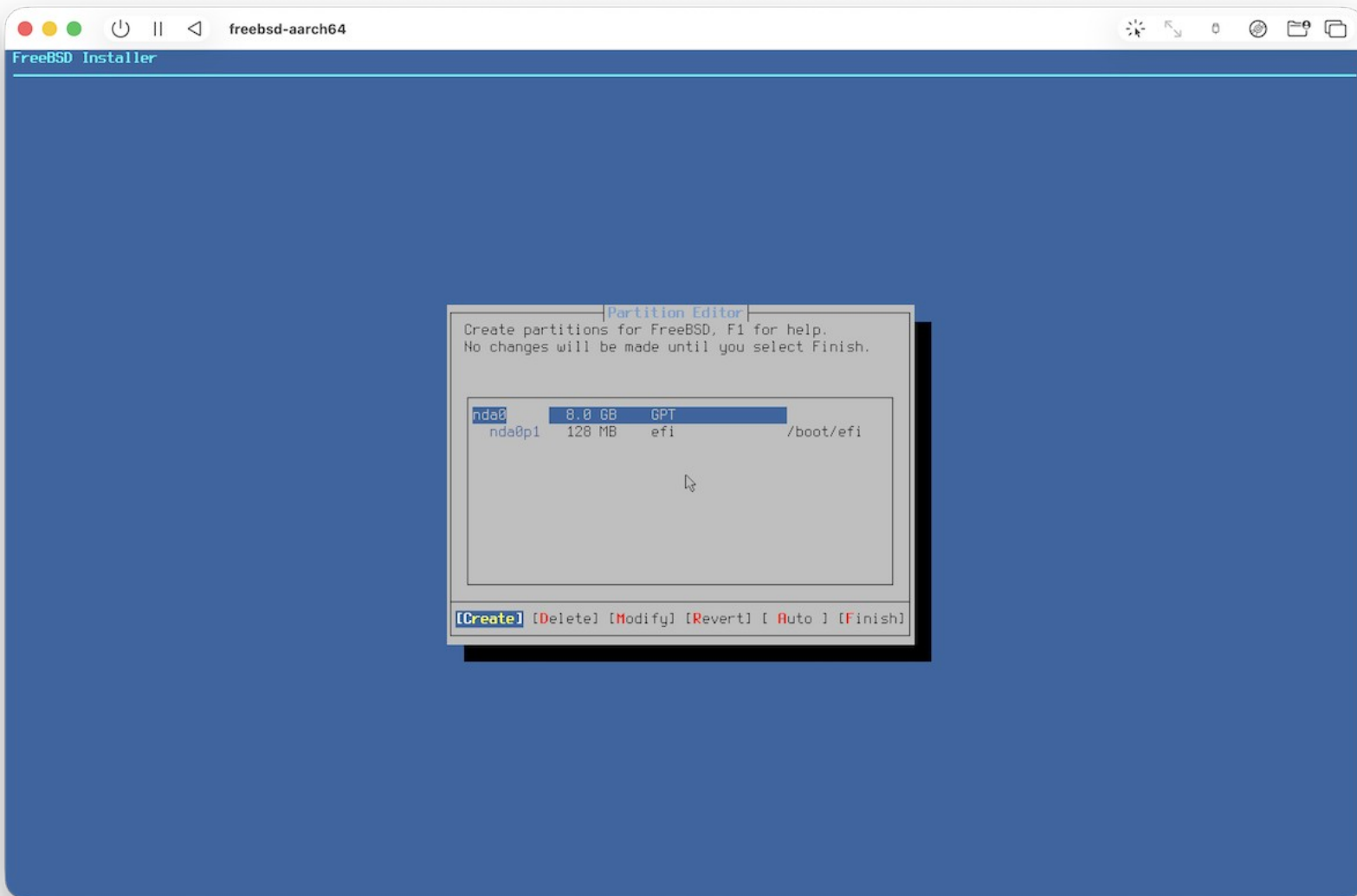
componentes da instalação customizada (base)

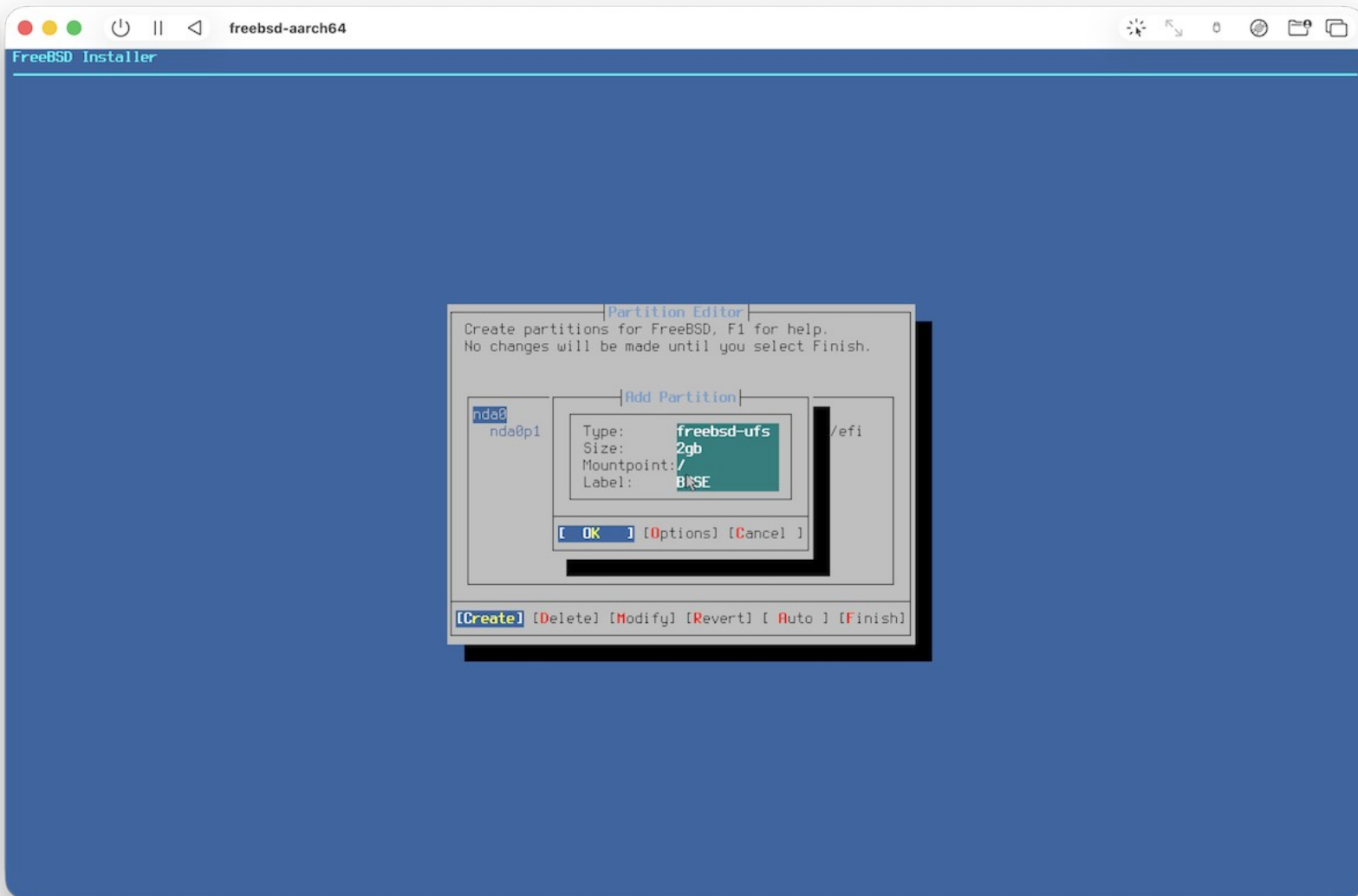
- /dev/gpt/uefi
 - bootloader (loader.efi)
- /dev/gpt/base
 - sistema de arquivos raíz (ufs)
 - kernel (FreeBSD-kernel-generic)
 - base mínima (FreeBSD-set-minimal)

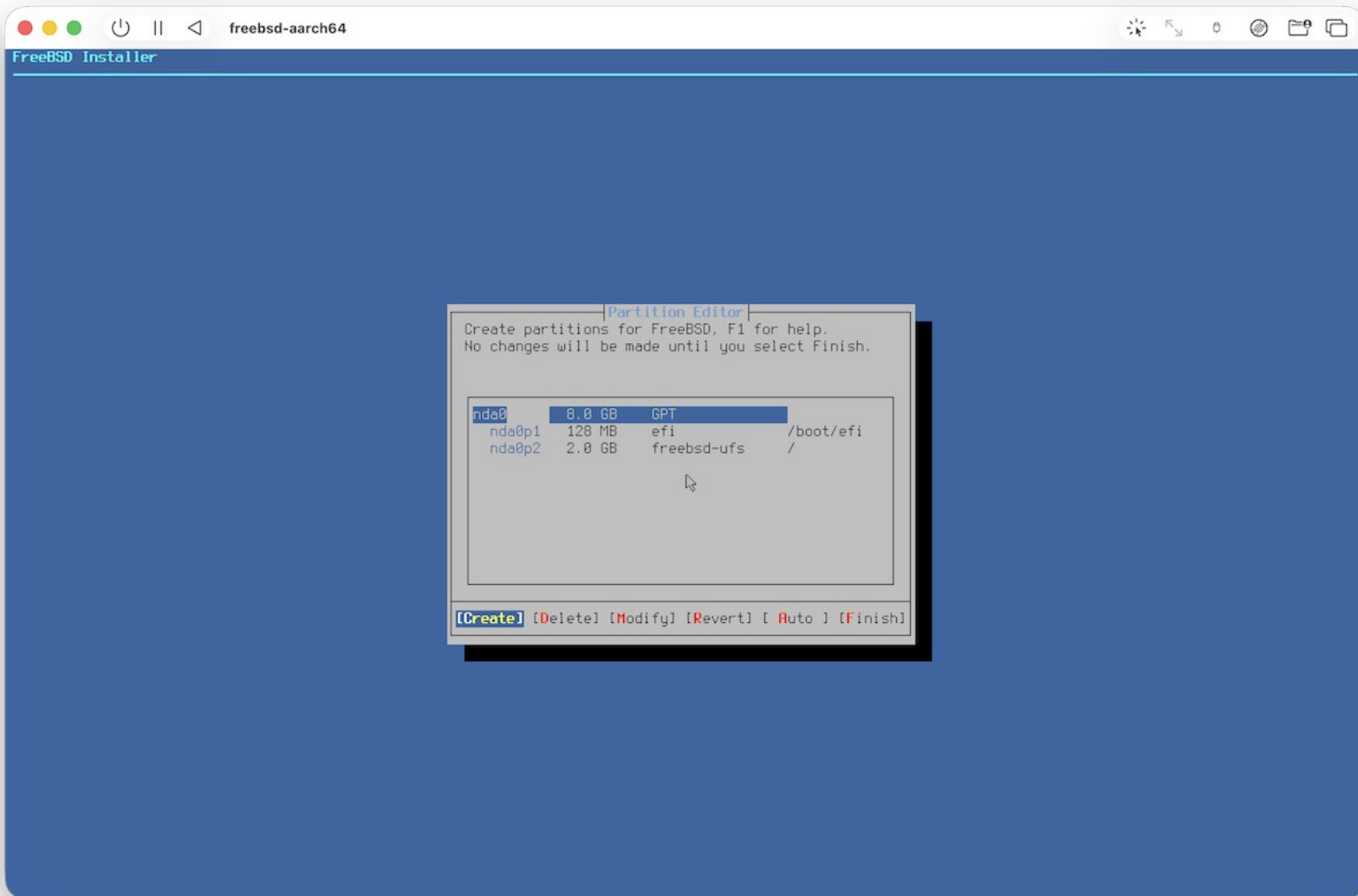
componentes da instalação customizada (elke), cenário 1

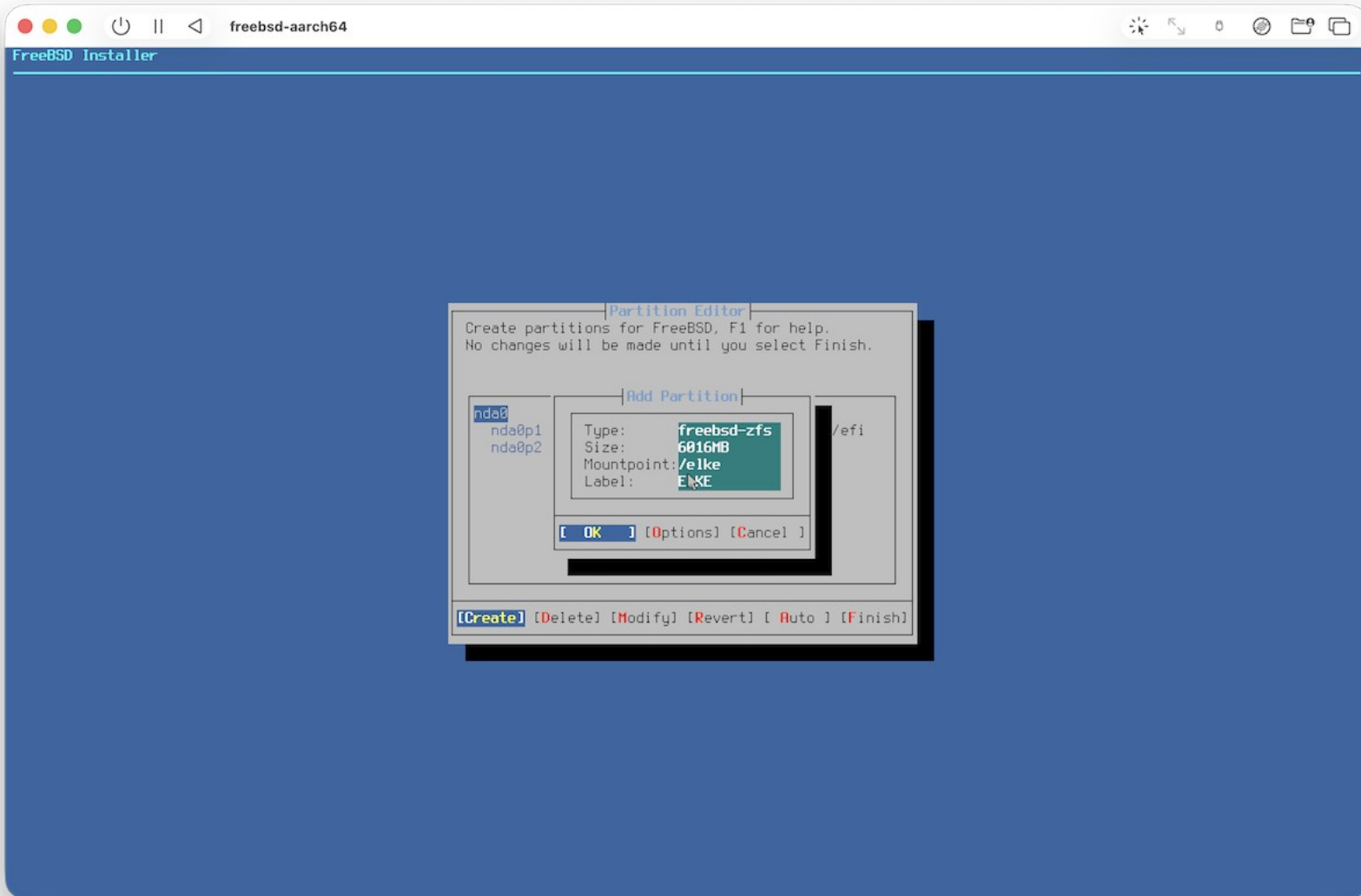
- zpool (zfs)
 - dataset raíz
- sistema operacional
 - gerenciador de pacotes (pkg)
 - base (FreeBSD-set-base)

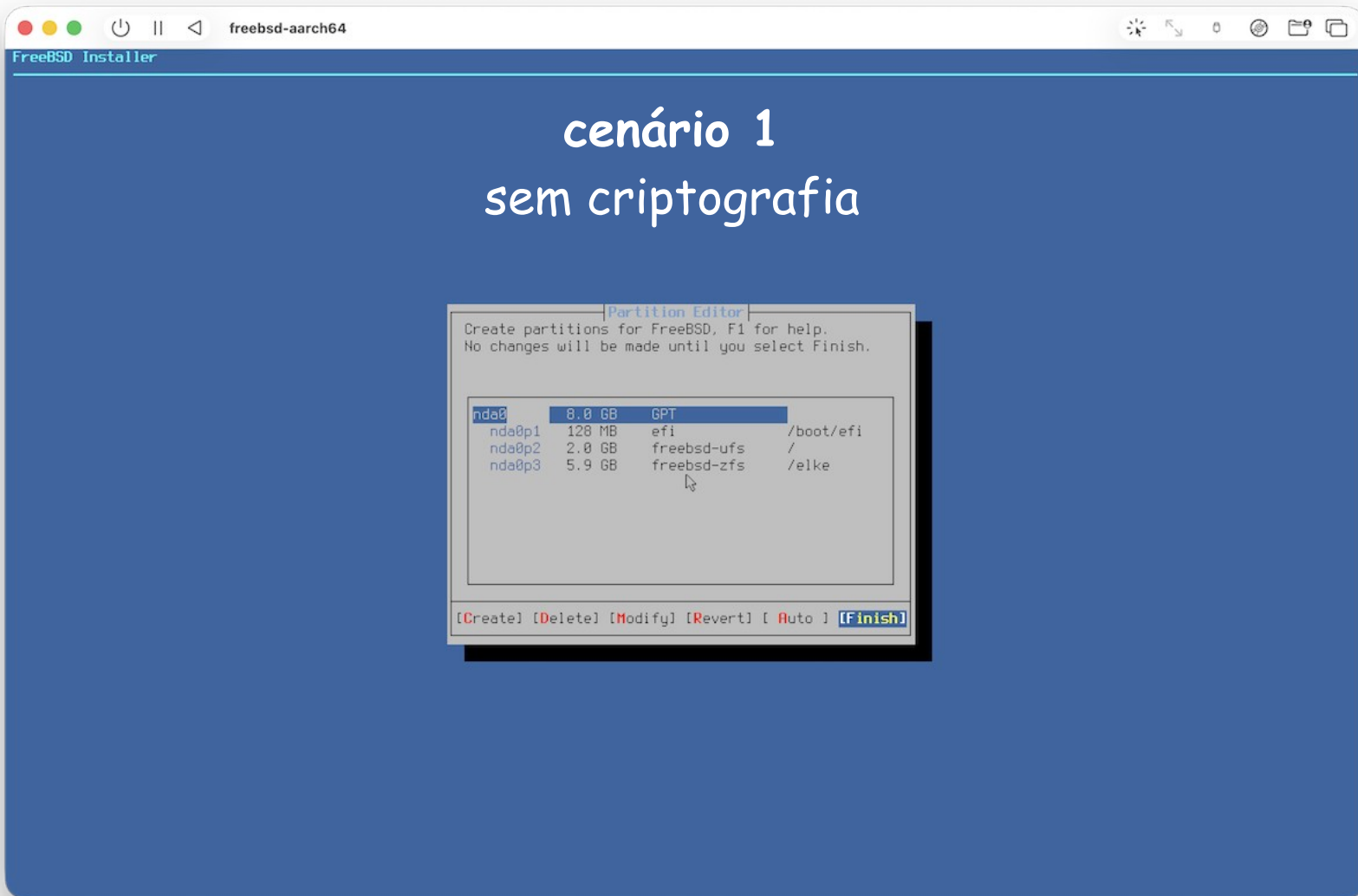


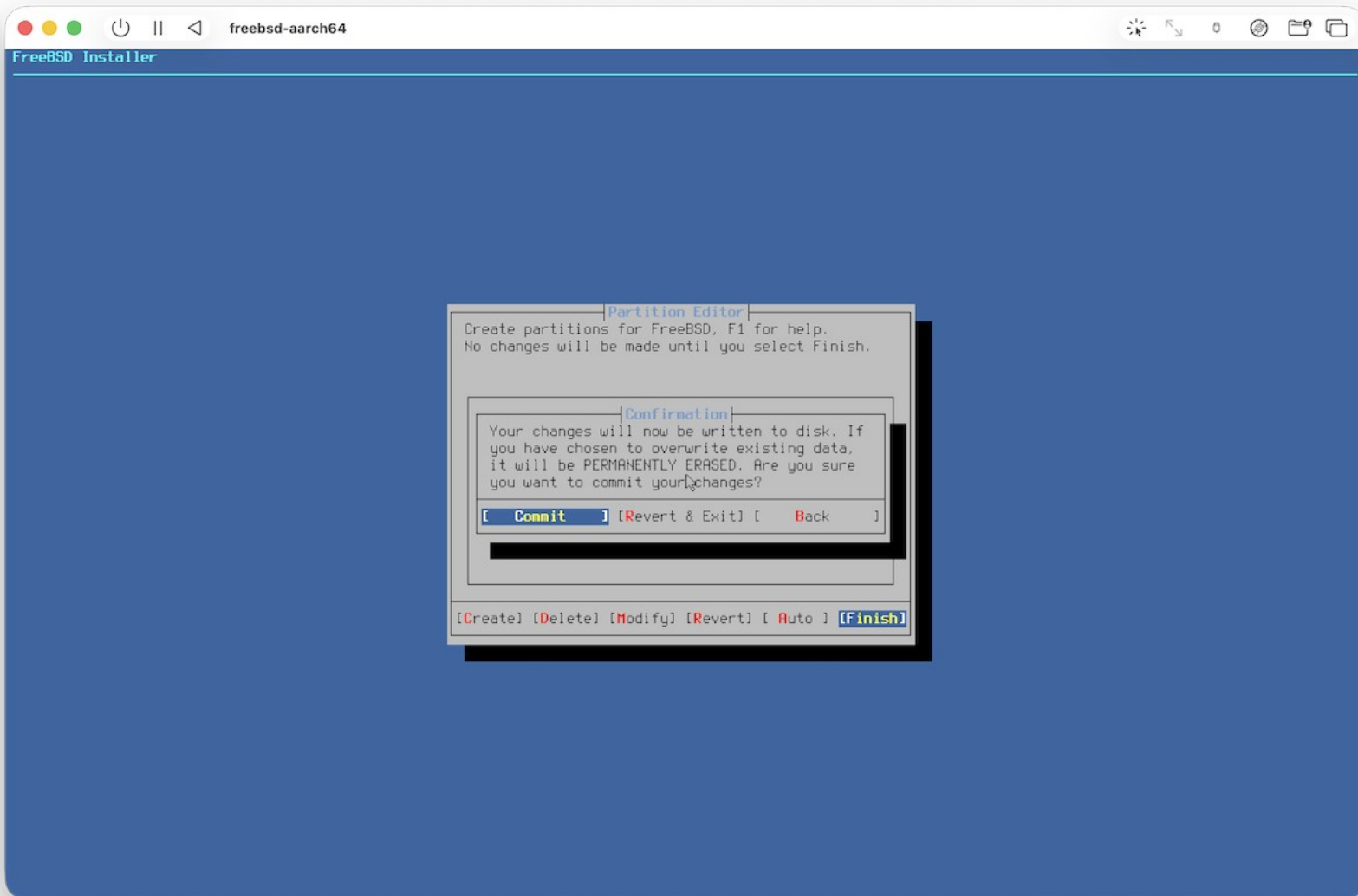


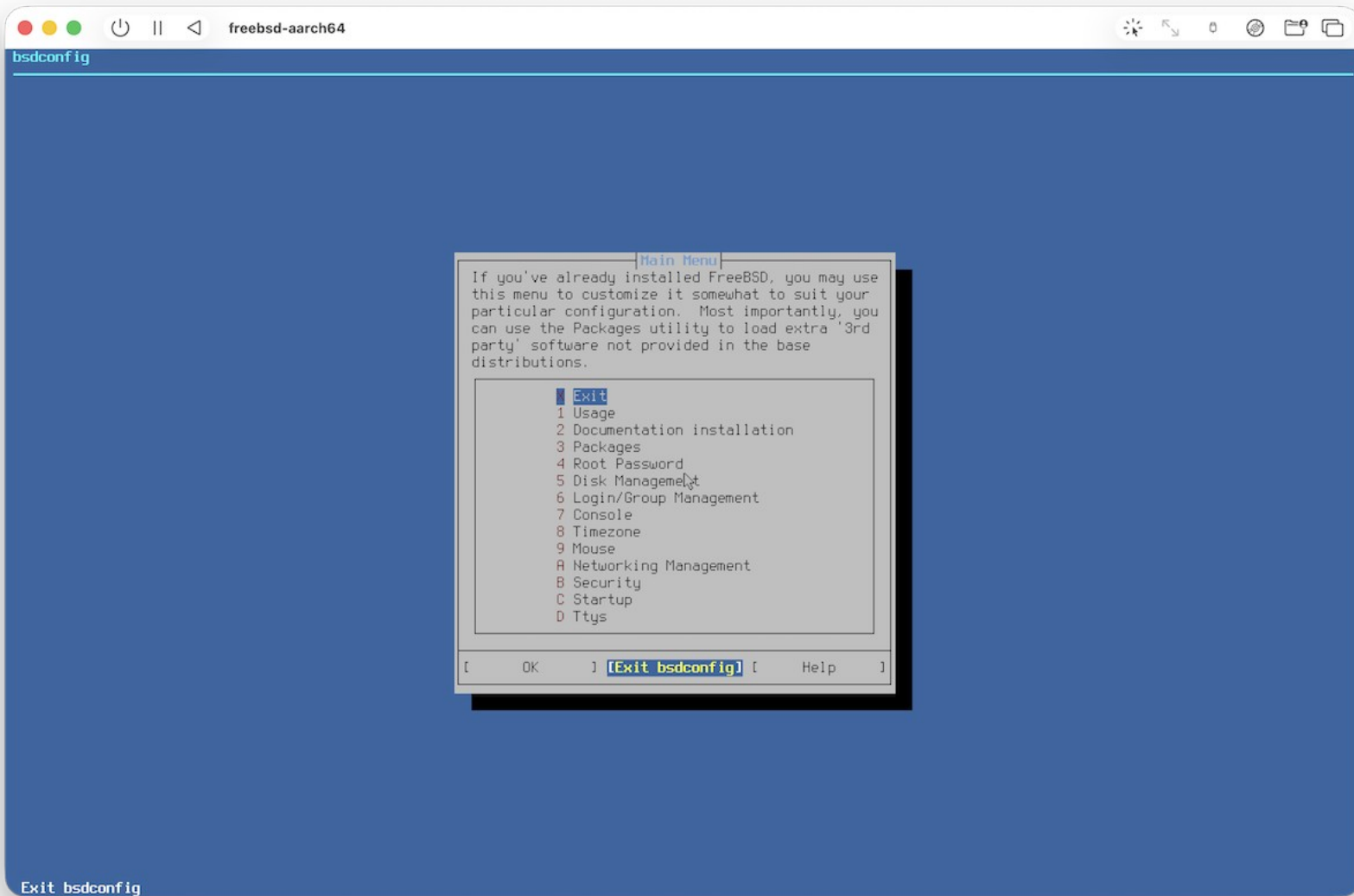


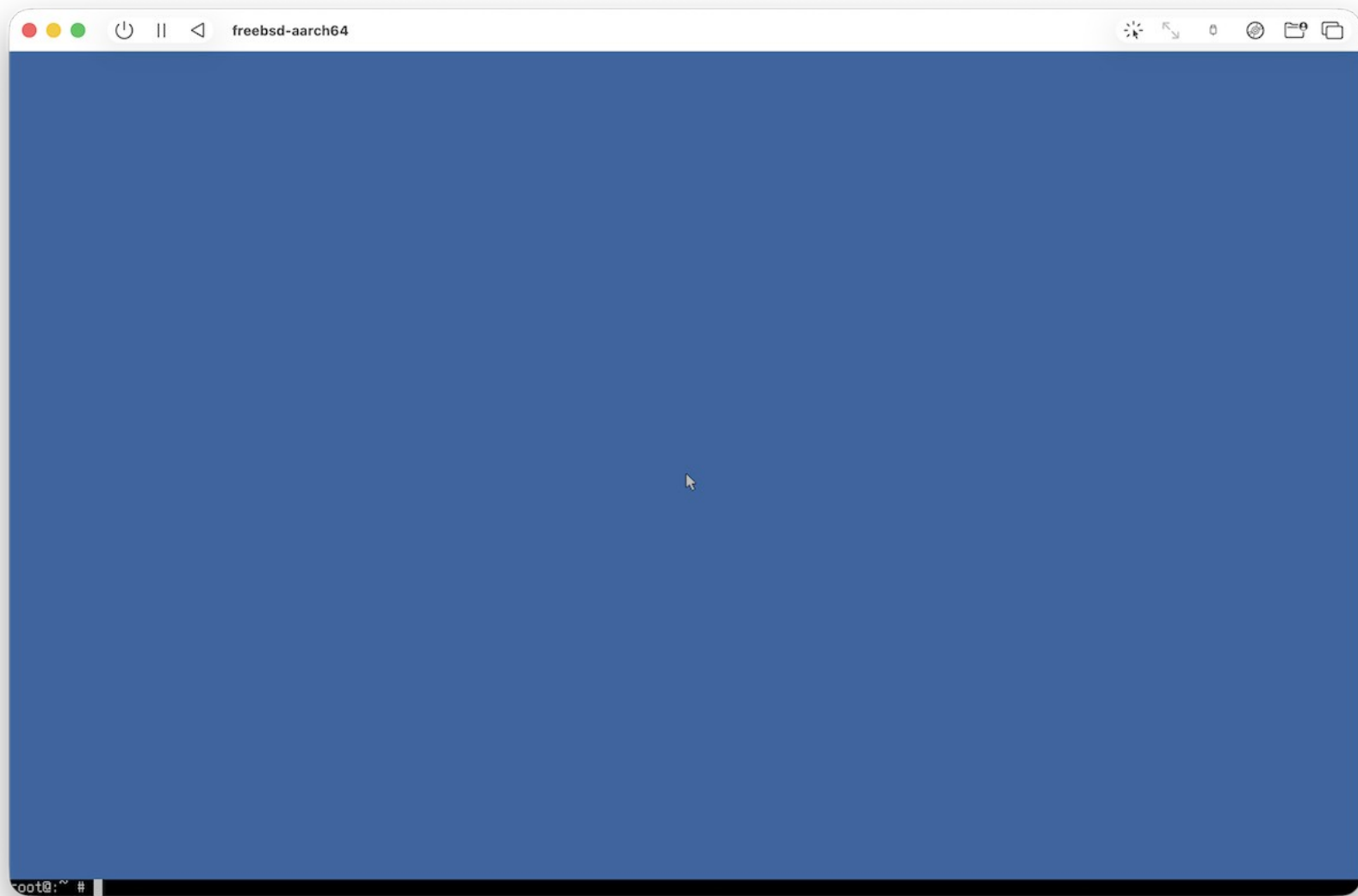




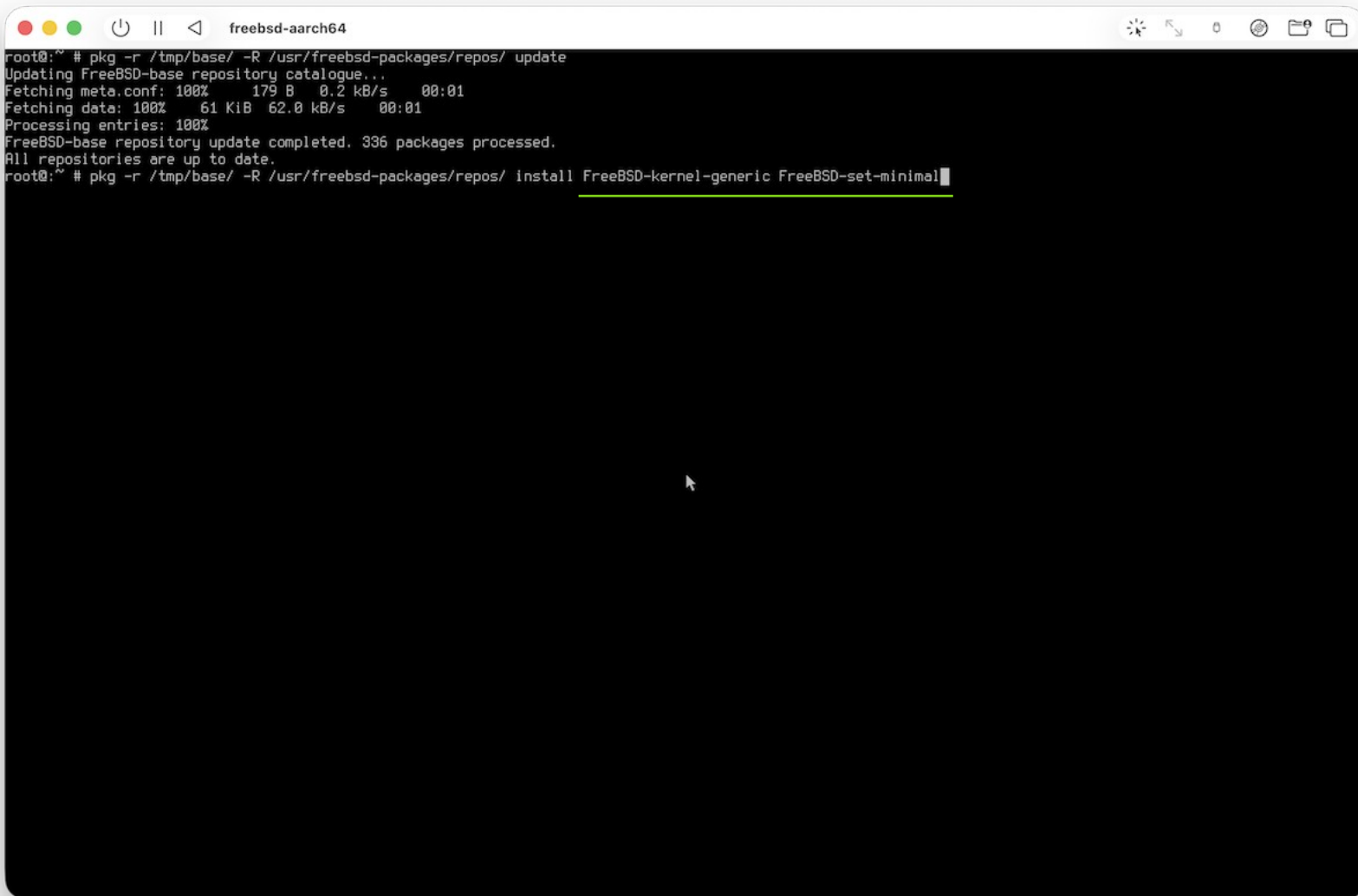








```
freebsd-aarch64
root@:~ # df -Th
Filesystem                                Type      Size  Used Avail Capacity  Mounted on
/dev/iso9660/16_0_CURRENT_AARCH64_CD     cd9660    1.1G  1.1G    0B   100%      /
devfs                                     devfs      1.0K    0B   1.0K    0%     /dev
tmpfs                                     tmpfs      32M   132K   32M    0%     /var
tmpfs                                     tmpfs      20M    16K   20M    0%     /tmp
root@:~ # mkdir /tmp/base
root@:~ # mount /dev/gpt/BASE /tmp/base
root@:~ # mkdir -p /tmp/base/boot/efi
root@:~ # mount_msdosfs /dev/gpt/UEFI /tmp/base/boot/efi
root@:~ # df -Th
Filesystem                                Type      Size  Used Avail Capacity  Mounted on
/dev/iso9660/16_0_CURRENT_AARCH64_CD     cd9660    1.1G  1.1G    0B   100%      /
devfs                                     devfs      1.0K    0B   1.0K    0%     /dev
tmpfs                                     tmpfs      32M   132K   32M    0%     /var
tmpfs                                     tmpfs      20M    16K   20M    0%     /tmp
/dev/gpt/BASE                             ufs       1.9G   16M   1.8G    1%     /tmp/base
/dev/gpt/UEFI                             msdosfs   126M   512B  126M    0%     /tmp/base/boot/efi
root@:~ #
```

```
root@:~ # pkg -r /tmp/base/ -R /usr/freebsd-packages/repos/ update
Updating FreeBSD-base repository catalogue...
Fetching meta.conf: 100% 179 B 0.2 kB/s 00:01
Fetching data: 100% 61 KiB 62.0 kB/s 00:01
Processing entries: 100%
FreeBSD-base repository update completed. 336 packages processed.
All repositories are up to date.
root@:~ # pkg -r /tmp/base/ -R /usr/freebsd-packages/repos/ install FreeBSD-kernel-generic FreeBSD-set-minimal
```



FreeBSD-ssh (pkgbase)

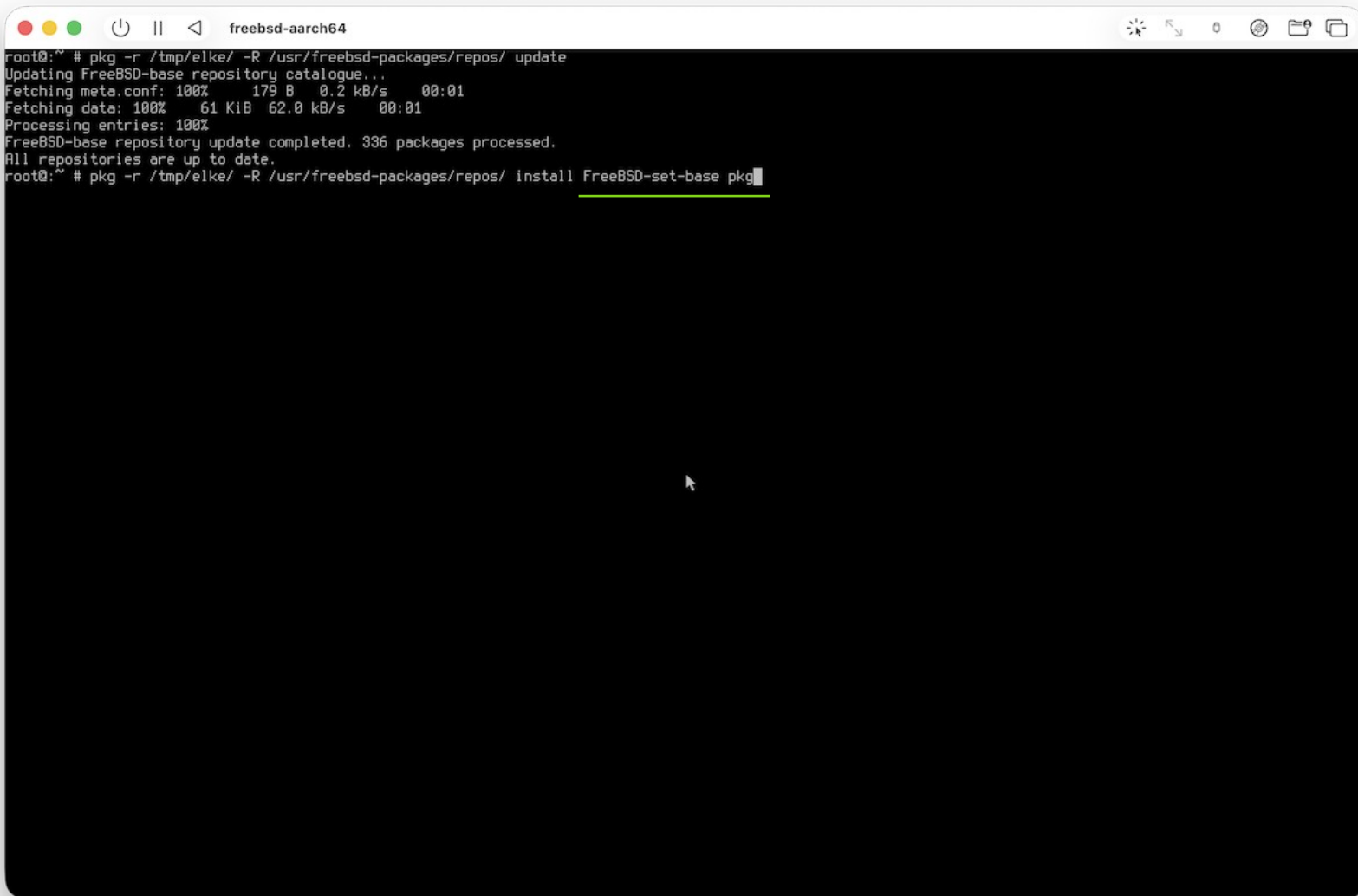
```
freebsd-aarch64
FreeBSD-libcasper: 16.snap20260413132855
FreeBSD-libldns: 16.snap20260413132855
FreeBSD-libmagic: 16.snap20260413132855
FreeBSD-libucl: 16.snap20260413132855
FreeBSD-libyaml: 16.snap20260413132855
FreeBSD-locales: 16.snap20260413132855
FreeBSD-mandoc: 16.snap20260413132855
FreeBSD-mtree: 16.snap20260413132855
FreeBSD-natd: 16.snap20260413132855
FreeBSD-ncurses: 16.snap20260413132855
FreeBSD-ncurses-lib: 16.snap20260413132855
FreeBSD-newsyslog: 16.snap20260413132855
FreeBSD-nuageinit: 16.snap20260413132855
FreeBSD-openssl: 16.snap20260413132855
FreeBSD-openssl-lib: 16.snap20260413132855
FreeBSD-pam: 16.snap20260413132855
FreeBSD-pam-lib: 16.snap20260413132855
FreeBSD-periodic: 16.snap20260413132855
FreeBSD-pkg-bootstrap: 16.snap20260413132855
FreeBSD-powerd: 16.snap20260413132855
FreeBSD-ppp: 16.snap20260413132855
FreeBSD-rc: 16.snap20260413132855
FreeBSD-rdma-lib: 16.snap20260413132855
FreeBSD-rescue: 16.snap20260413132855
FreeBSD-resolvconf: 16.snap20260413132855
FreeBSD-runtime: 16.snap20260413132855
FreeBSD-set-minimal: 16.snap20260413132855
FreeBSD-syslogd: 16.snap20260413132855
FreeBSD-tcpd: 16.snap20260413132855
FreeBSD-ufs: 16.snap20260413132855
FreeBSD-ufs-lib: 16.snap20260413132855
FreeBSD-utilities: 16.snap20260413132855
FreeBSD-vi: 16.snap20260413132855
FreeBSD-vt-data: 16.snap20260413132855
FreeBSD-wpa: 16.snap20260413132855
FreeBSD-xz: 16.snap20260413132855
FreeBSD-xz-lib: 16.snap20260413132855
FreeBSD-yp-lib: 16.snap20260413132855
FreeBSD-zfs: 16.snap20260413132855
FreeBSD-zfs-lib: 16.snap20260413132855
FreeBSD-zlib: 16.snap20260413132855
FreeBSD-zoneinfo: 16.snap20260413132855
FreeBSD-zstd: 16.snap20260413132855
FreeBSD-zstd-lib: 16.snap20260413132855

Number of packages to be installed: 70

The process will require 247 MiB more space.

Proceed with this action? [y/N]: y
```

```
freebsd-aarch64
[46/70] Extracting FreeBSD-rescue-16.snap20260413132855: 100%
[47/70] Installing FreeBSD-resolvconf-16.snap20260413132855...
[47/70] Extracting FreeBSD-resolvconf-16.snap20260413132855: 100%
[48/70] Installing FreeBSD-dhclient-16.snap20260413132855...
[48/70] Extracting FreeBSD-dhclient-16.snap20260413132855: 100%
[49/70] Installing FreeBSD-runtime-16.snap20260413132855...
[49/70] Extracting FreeBSD-runtime-16.snap20260413132855: 100%
[50/70] Installing FreeBSD-at-16.snap20260413132855...
[50/70] Extracting FreeBSD-at-16.snap20260413132855: 100%
[51/70] Installing FreeBSD-devd-16.snap20260413132855...
[51/70] Extracting FreeBSD-devd-16.snap20260413132855: 100%
[52/70] Installing FreeBSD-rc-16.snap20260413132855...
[52/70] Extracting FreeBSD-rc-16.snap20260413132855: 100%
[53/70] Installing FreeBSD-syslogd-16.snap20260413132855...
[53/70] Extracting FreeBSD-syslogd-16.snap20260413132855: 100%
[54/70] Installing FreeBSD-tcpd-16.snap20260413132855...
[54/70] Extracting FreeBSD-tcpd-16.snap20260413132855: 100%
[55/70] Installing FreeBSD-ufs-16.snap20260413132855...
[55/70] Extracting FreeBSD-ufs-16.snap20260413132855: 100%
[56/70] Installing FreeBSD-ufs-lib-16.snap20260413132855...
[56/70] Extracting FreeBSD-ufs-lib-16.snap20260413132855: 100%
[57/70] Installing FreeBSD-utilities-16.snap20260413132855...
[57/70] Extracting FreeBSD-utilities-16.snap20260413132855: 100%
[58/70] Installing FreeBSD-vi-16.snap20260413132855...
[58/70] Extracting FreeBSD-vi-16.snap20260413132855: 100%
[59/70] Installing FreeBSD-vt-data-16.snap20260413132855...
[59/70] Extracting FreeBSD-vt-data-16.snap20260413132855: 100%
[60/70] Installing FreeBSD-wpa-16.snap20260413132855...
[60/70] Extracting FreeBSD-wpa-16.snap20260413132855: 100%
[61/70] Installing FreeBSD-xz-16.snap20260413132855...
[61/70] Extracting FreeBSD-xz-16.snap20260413132855: 100%
[62/70] Installing FreeBSD-xz-lib-16.snap20260413132855...
[62/70] Extracting FreeBSD-xz-lib-16.snap20260413132855: 100%
[63/70] Installing FreeBSD-yp-lib-16.snap20260413132855...
[63/70] Extracting FreeBSD-yp-lib-16.snap20260413132855: 100%
[64/70] Installing FreeBSD-zfs-16.snap20260413132855...
[64/70] Extracting FreeBSD-zfs-16.snap20260413132855: 100%
[65/70] Installing FreeBSD-zfs-lib-16.snap20260413132855...
[65/70] Extracting FreeBSD-zfs-lib-16.snap20260413132855: 100%
[66/70] Installing FreeBSD-zlib-16.snap20260413132855...
[66/70] Extracting FreeBSD-zlib-16.snap20260413132855: 100%
[67/70] Installing FreeBSD-zoneinfo-16.snap20260413132855...
[67/70] Extracting FreeBSD-zoneinfo-16.snap20260413132855: 100%
[68/70] Installing FreeBSD-zstd-16.snap20260413132855...
[68/70] Extracting FreeBSD-zstd-16.snap20260413132855: 100%
[69/70] Installing FreeBSD-zstd-lib-16.snap20260413132855...
[69/70] Extracting FreeBSD-zstd-lib-16.snap20260413132855: 100%
[70/70] Installing FreeBSD-set-minimal-16.snap20260413132855...
==> Running trigger: mandoc.ucl
root@:~ #
```



```
root@:~ # pkg -r /tmp/elke/ -R /usr/freebsd-packages/repos/ update
Updating FreeBSD-base repository catalogue...
Fetching meta.conf: 100% 179 B 0.2 kB/s 00:01
Fetching data: 100% 61 KiB 62.0 kB/s 00:01
Processing entries: 100%
FreeBSD-base repository update completed. 336 packages processed.
All repositories are up to date.
root@:~ # pkg -r /tmp/elke/ -R /usr/freebsd-packages/repos/ install FreeBSD-set-base pkg
```

```
freebsd-aarch64
FreeBSD-runtime: 16.snap20260413132855
FreeBSD-runtime-dev: 16.snap20260413132855
FreeBSD-sendmail: 16.snap20260413132855
FreeBSD-set-base: 16.snap20260413132855
FreeBSD-set-devel: 16.snap20260413132855
FreeBSD-set-minimal: 16.snap20260413132855
FreeBSD-set-optional: 16.snap20260413132855
FreeBSD-smbutils: 16.snap20260413132855
FreeBSD-smbutils-dev: 16.snap20260413132855
FreeBSD-sound: 16.snap20260413132855
FreeBSD-sound-dev: 16.snap20260413132855
FreeBSD-ssh: 16.snap20260413132855
FreeBSD-ssh-dev: 16.snap20260413132855
FreeBSD-syscons-data: 16.snap20260413132855
FreeBSD-syslogd: 16.snap20260413132855
FreeBSD-tcpd: 16.snap20260413132855
FreeBSD-tcpd-dev: 16.snap20260413132855
FreeBSD-telnet: 16.snap20260413132855
FreeBSD-toolchain: 16.snap20260413132855
FreeBSD-toolchain-dev: 16.snap20260413132855
FreeBSD-ufs: 16.snap20260413132855
FreeBSD-ufs-dev: 16.snap20260413132855
FreeBSD-ufs-lib: 16.snap20260413132855
FreeBSD-utilities: 16.snap20260413132855
FreeBSD-utilities-dev: 16.snap20260413132855
FreeBSD-vi: 16.snap20260413132855
FreeBSD-vt-data: 16.snap20260413132855
FreeBSD-wpa: 16.snap20260413132855
FreeBSD-wx: 16.snap20260413132855
FreeBSD-wx-dev: 16.snap20260413132855
FreeBSD-wx-lib: 16.snap20260413132855
FreeBSD-yp: 16.snap20260413132855
FreeBSD-yp-dev: 16.snap20260413132855
FreeBSD-yp-lib: 16.snap20260413132855
FreeBSD-zfs: 16.snap20260413132855
FreeBSD-zfs-dev: 16.snap20260413132855
FreeBSD-zfs-lib: 16.snap20260413132855
FreeBSD-zlib: 16.snap20260413132855
FreeBSD-zlib-dev: 16.snap20260413132855
FreeBSD-zoneinfo: 16.snap20260413132855
FreeBSD-zstd: 16.snap20260413132855
FreeBSD-zstd-dev: 16.snap20260413132855
FreeBSD-zstd-lib: 16.snap20260413132855
pkg: 2.6.2_1

Number of packages to be installed: 216

The process will require 691 MiB more space.

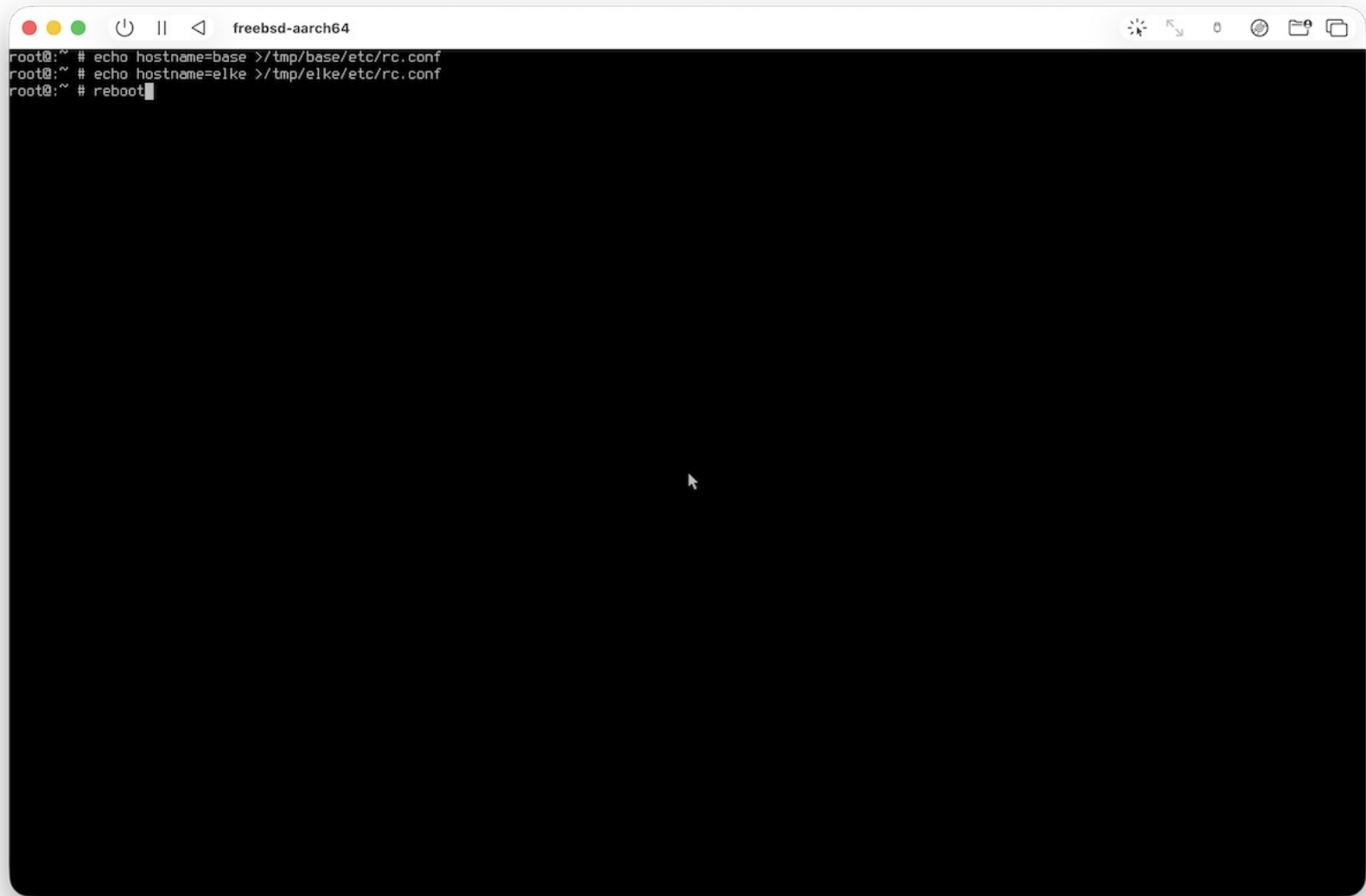
Proceed with this action? [y/N]: y
```

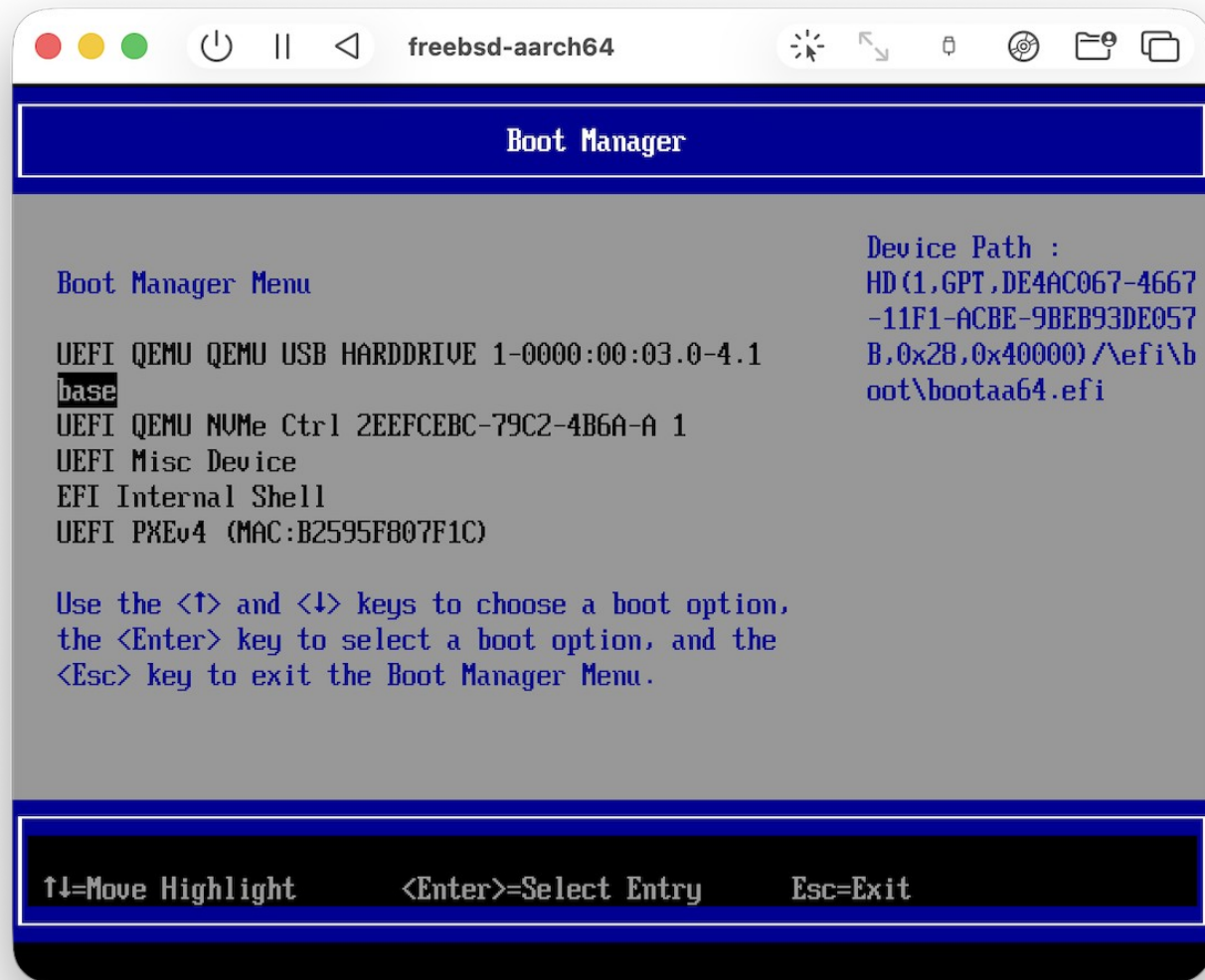


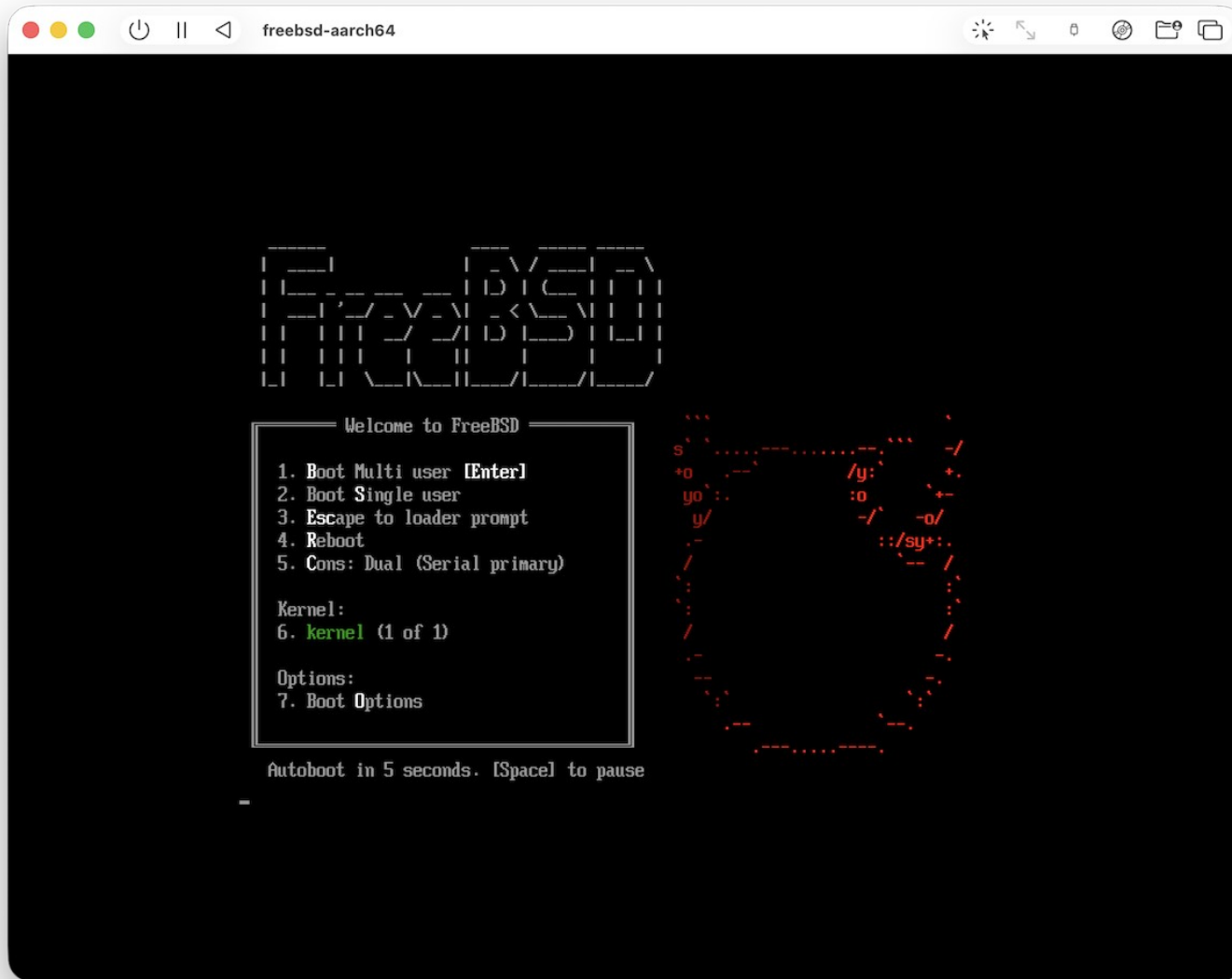
```
freebsd-aarch64

[191/216] Installing FreeBSD-utilities-16.snap20260413132855...
[191/216] Extracting FreeBSD-utilities-16.snap20260413132855: 100%
[192/216] Installing FreeBSD-rcmds-16.snap20260413132855...
[192/216] Extracting FreeBSD-rcmds-16.snap20260413132855: 100%
[193/216] Installing FreeBSD-utilities-dev-16.snap20260413132855...
[193/216] Extracting FreeBSD-utilities-dev-16.snap20260413132855: 100%
[194/216] Installing FreeBSD-vi-16.snap20260413132855...
[194/216] Extracting FreeBSD-vi-16.snap20260413132855: 100%
[195/216] Installing FreeBSD-vt-data-16.snap20260413132855...
[195/216] Extracting FreeBSD-vt-data-16.snap20260413132855: 100%
[196/216] Installing FreeBSD-wpa-16.snap20260413132855...
[196/216] Extracting FreeBSD-wpa-16.snap20260413132855: 100%
[197/216] Installing FreeBSD-xz-16.snap20260413132855...
[197/216] Extracting FreeBSD-xz-16.snap20260413132855: 100%
[198/216] Installing FreeBSD-xz-dev-16.snap20260413132855...
[198/216] Extracting FreeBSD-xz-dev-16.snap20260413132855: 100%
[199/216] Installing FreeBSD-xz-lib-16.snap20260413132855...
[199/216] Extracting FreeBSD-xz-lib-16.snap20260413132855: 100%
[200/216] Installing FreeBSD-yp-16.snap20260413132855...
[200/216] Extracting FreeBSD-yp-16.snap20260413132855: 100%
[201/216] Installing FreeBSD-yp-dev-16.snap20260413132855...
[201/216] Extracting FreeBSD-yp-dev-16.snap20260413132855: 100%
[202/216] Installing FreeBSD-yp-lib-16.snap20260413132855...
[202/216] Extracting FreeBSD-yp-lib-16.snap20260413132855: 100%
[203/216] Installing FreeBSD-zfs-16.snap20260413132855...
[203/216] Extracting FreeBSD-zfs-16.snap20260413132855: 100%
[204/216] Installing FreeBSD-zfs-dev-16.snap20260413132855...
[204/216] Extracting FreeBSD-zfs-dev-16.snap20260413132855: 100%
[205/216] Installing FreeBSD-zfs-lib-16.snap20260413132855...
[205/216] Extracting FreeBSD-zfs-lib-16.snap20260413132855: 100%
[206/216] Installing FreeBSD-zlib-16.snap20260413132855...
[206/216] Extracting FreeBSD-zlib-16.snap20260413132855: 100%
[207/216] Installing FreeBSD-zlib-dev-16.snap20260413132855...
[207/216] Extracting FreeBSD-zlib-dev-16.snap20260413132855: 100%
[208/216] Installing FreeBSD-zoneinfo-16.snap20260413132855...
[208/216] Extracting FreeBSD-zoneinfo-16.snap20260413132855: 100%
[209/216] Installing FreeBSD-zstd-16.snap20260413132855...
[209/216] Extracting FreeBSD-zstd-16.snap20260413132855: 100%
[210/216] Installing FreeBSD-zstd-dev-16.snap20260413132855...
[210/216] Extracting FreeBSD-zstd-dev-16.snap20260413132855: 100%
[211/216] Installing FreeBSD-set-devel-16.snap20260413132855...
[212/216] Installing FreeBSD-zstd-lib-16.snap20260413132855...
[212/216] Extracting FreeBSD-zstd-lib-16.snap20260413132855: 100%
[213/216] Installing FreeBSD-set-minimal-16.snap20260413132855...
[214/216] Installing FreeBSD-set-optional-16.snap20260413132855...
[215/216] Installing FreeBSD-set-base-16.snap20260413132855...
[216/216] Installing pkg-2.6.2_1...
[216/216] Extracting pkg-2.6.2_1: 100%
==> Running trigger: mandoc.ucl
root@:~ #
```

```
freebsd-aarch64
root@:~ # df -Th
Filesystem                                Type      Size  Used Avail Capacity  Mounted on
/dev/iso9660/16_0_CURRENT_AARCH64_CD     cd9660    1.1G  1.1G   0B    100%      /
devfs                                     devfs      1.0K   0B    1.0K   0%      /dev
tmpfs                                     tmpfs      32M   164K   32M    1%      /var
tmpfs                                     tmpfs      20M    12K   20M    0%      /tmp
/dev/gpt/BASE                             ufs        1.9G  288M   1.5G   16%      /tmp/base
/dev/gpt/UEFI                             msdosfs    126M   512B  126M    0%      /tmp/base/boot/efi
elke/rootfs                               zfs        5.3G  348M   5.0G    6%      /tmp/elke
root@:~ # mkdir -p /tmp/base/boot/efi/efi/boot
root@:~ # cp /boot/loader.efi /tmp/base/boot/efi/efi/boot/bootaa64.efi
root@:~ # efibootmgr -a -c -l /tmp/base/boot/efi/efi/boot/bootaa64.efi -L base
Boot to FW : false
BootCurrent: 0001
Timeout     : 3 seconds
BootOrder   : 0005, 0001, 0003, 0000, 0002, 0004
  Boot0005* base
+Boot0001* UEFI QEMU QEMU USB HARDDRIVE 1-0000:00:03.0-4.1
  Boot0003* UEFI QEMU NVMe Ctrl 2EEFCEBC-79C2-4B6A-A 1
  Boot0000* UIApp
  Boot0002* UEFI Misc Device
  Boot0004* EFI Internal Shell
root@:~ #
```





```
freebsd-aarch64
ugen0.2: <QEMU QEMU USB Tablet> at usb0
usbhid0 on uhub1
usbhid0: <QEMU QEMU USB Tablet, class 0/0, rev 2.00/0.00, addr 1> on usb0
hidbus0: <HID bus> on usbhid0
ugen0.3: <QEMU QEMU USB Mouse> at usb0
usbhid1 on uhub1
usbhid1: <QEMU QEMU USB Mouse, class 0/0, rev 2.00/0.00, addr 2> on usb0
hidbus1: <HID bus> on usbhid1
Root mount waiting for: usb0
ugen0.4: <QEMU QEMU USB Keyboard> at usb0
usbhid2 on uhub1
usbhid2: <QEMU QEMU USB Keyboard, class 0/0, rev 2.00/0.00, addr 3> on usb0
hidbus2: <HID bus> on usbhid2
hkbd0: <QEMU QEMU USB Keyboard> on hidbus2
kbd1 at hkbd0
ugen0.5: <QEMU QEMU USB Hub> at usb0
uhub2 on uhub1
uhub2: <QEMU QEMU USB Hub, class 9/0, rev 1.10/1.01, addr 4> on usb0
Root mount waiting for: usb0
uhub2: 8 ports with 8 removable, self powered
ugen0.6: <QEMU QEMU USB HARDDRIVE> at usb0
umass0 on uhub2
umass0: <QEMU QEMU USB HARDDRIVE, class 0/0, rev 2.00/0.00, addr 5> on usb0
umass0: SCSI over Bulk-Only; quirks = 0x0
umass0:1:0: Attached to scbus1
cd0 at umass-sim0 bus 0 scbus1 target 0 lun 0
cd0: <QEMU QEMU CD-ROM 2.5+> Removable CD-ROM SPC-3 SCSI device
cd0: Serial Number 1-0000:00:03.0-4.1
cd0: 1.000MB/s transfers
cd0: Attempt to query device size failed: NOT READY, Medium not present
cd0: quirks=0x10<10_BYTE_ONLY>

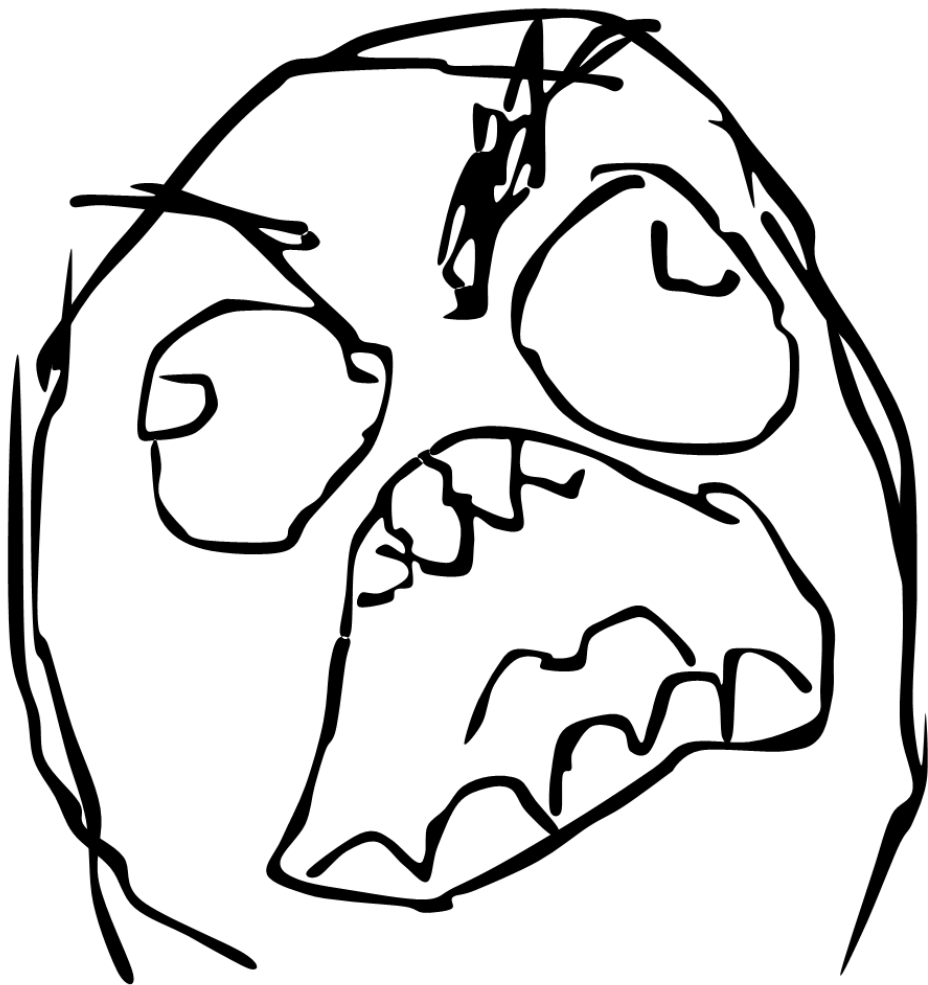
Loader variables:

Manual root filesystem specification:
<fstype>:<device> [options]
    Mount <device> using filesystem <fstype>
    and with the specified (optional) option list.

eg. ufs:/dev/ds0s1a
    zfs:zroot/ROOT/default
    cd9660:/dev/cd0 ro
    (which is equivalent to: mount -t cd9660 -o ro /dev/cd0 /)

?          List valid disk boot devices
.          Yield 1 second (for background tasks)
<empty line> Abort manual input

mountroot> uhub_explore: illegal enable change, port 1
```



ffffff

fffuuuu

uuuuuuuu

uuuu-

```
freebsd-aarch64

and with the specified (optional) option list.

eg. ufs:/dev/da0s1a
    zfs:zroot/R00T/default
    cd9660:/dev/cd0 ro
    (which is equivalent to: mount -t cd9660 -o ro /dev/cd0 /)

?          List valid disk boot devices
.          Yield 1 second (for background tasks)
<empty line> Abort manual input

mountroot> uhub_explore: illegal enable change, port 1
ufs:/dev/gpt/BASE
Trying to mount root from ufs:/dev/gpt/BASE [...]
Dual Console: Serial Primary, Video Secondary
hms0: <QEMU QEMU USB Tablet> on hidbus0
hms0: 3 buttons and [XYW] coordinates ID=0
hms1: <QEMU QEMU USB Mouse> on hidbus1
hms1: 5 buttons and [XYW] coordinates ID=0
vtcon0: <VirtIO Console Adapter> on virtio_pci2
vtrnd0: <VirtIO Entropy Adapter> on virtio_pci3
random: registering fast source VirtIO Entropy Adapter
lo0: link state changed to UP
vtnet0: link state changed to UP

FreeBSD/arm64 (BASE) (ttyv0)

login: root
FreeBSD 16.0-CURRENT (GENERIC) #0 main-n285005-e9fc0c538264: Mon Apr 13 12:48:35 UTC 2026

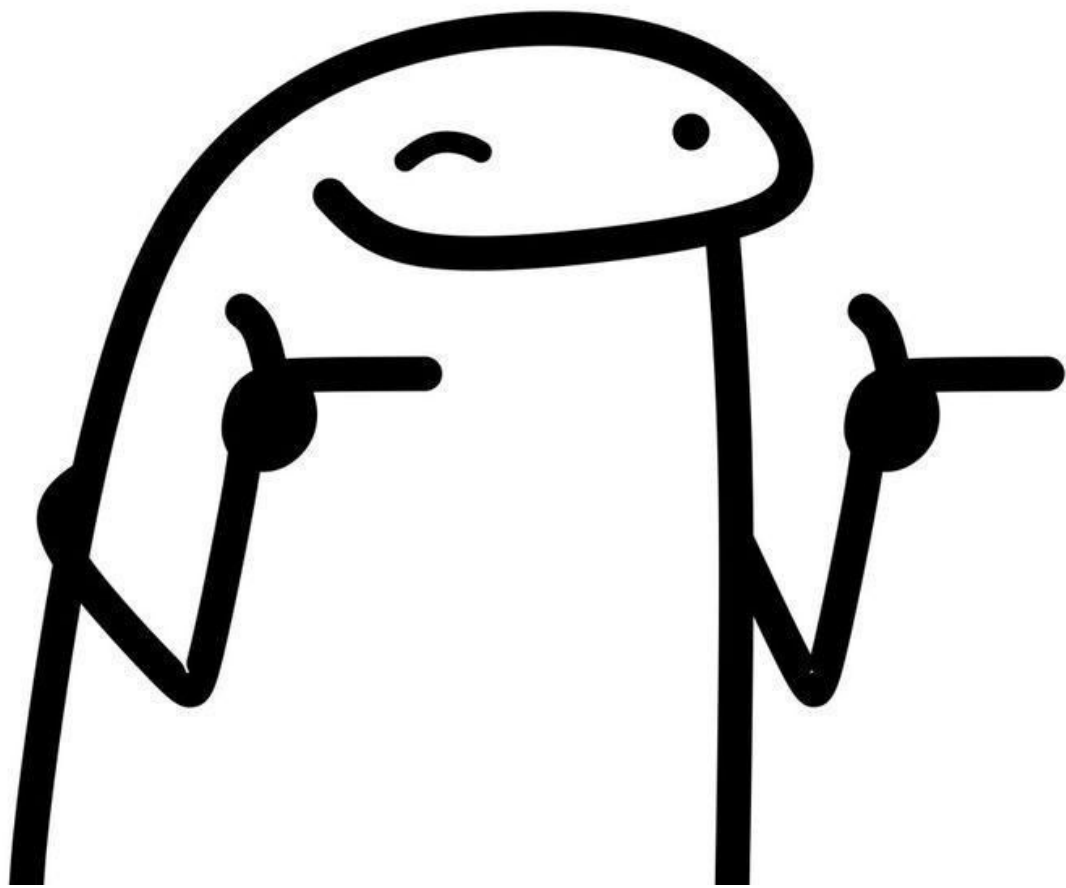
Welcome to FreeBSD!

Release Notes, Errata: https://www.FreeBSD.org/releases/
Security Advisories:  https://www.FreeBSD.org/security/
FreeBSD Handbook:     https://www.FreeBSD.org/handbook/
FreeBSD FAQ:          https://www.FreeBSD.org/faq/
Questions List:        https://www.FreeBSD.org/lists/questions/
FreeBSD Forums:       https://forums.FreeBSD.org/

Documents installed with the system are in the /usr/local/share/doc/freebsd/
directory, or can be installed later with: pkg install en-freebsd-doc
For other languages, replace "en" with a language code like de or fr.

Show the version of FreeBSD installed: freebsd-version ; uname -a
Please include that output and any error messages when posting questions.
Introduction to manual pages: man man
FreeBSD directory layout:    man hier

To change this login announcement, see motd(5).
root@BASE:~ #
```



```
freebsd-aarch64
root@BASE:~ # kldstat
Id Refs Address          Size Name
  1   19 0xffff000000000000 15126d8 kernel
  2    1 0xffff000096200000   22000 hms.ko
  3    1 0xffff000096220000   24000 hidmap.ko
  4    1 0xffff000096246000   25000 virtio_console.ko
  5    1 0xffff00009626b000   23000 virtio_random.ko
root@BASE:~ # kldload zfs
ZFS filesystem version: 5
ZFS storage pool version: features support (5000)
root@BASE:~ # kenv vfs.root.mountfrom="zfs:elke/ROOT/main"
vfs.root.mountfrom="zfs:elke/ROOT/main"
root@BASE:~ # reboot -r
Trying to mount root from zfs:elke/ROOT/main [!...

FreeBSD/arm64 (ELKE) (ttyv0)

login: root
FreeBSD 16.0-CURRENT (GENERIC) #0 main-n285005-e9fc0c538264: Mon Apr 13 12:48:35 UTC 2026

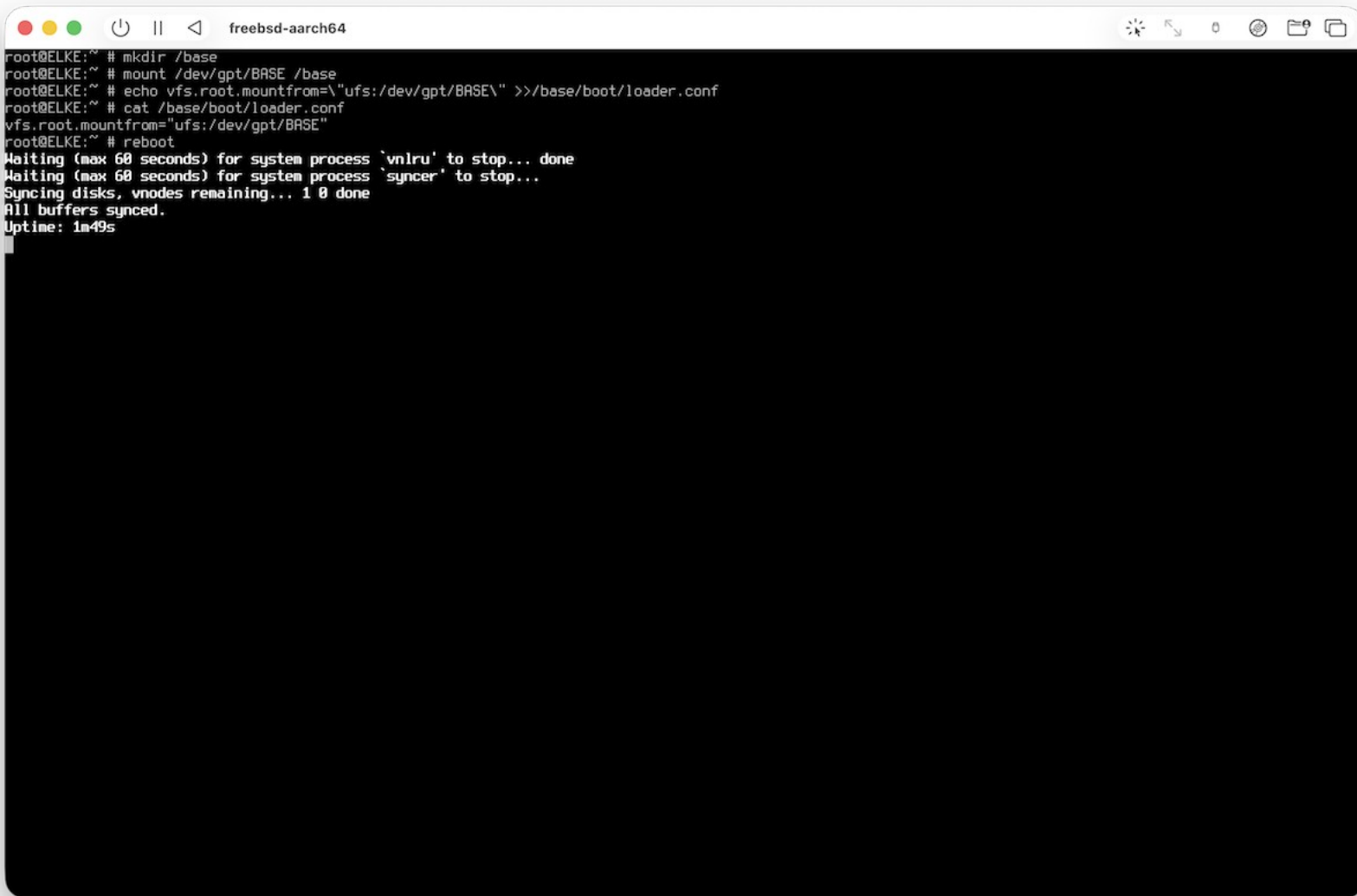
Welcome to FreeBSD!

Release Notes, Errata: https://www.FreeBSD.org/releases/
Security Advisories:  https://www.FreeBSD.org/security/
FreeBSD Handbook:     https://www.FreeBSD.org/handbook/
FreeBSD FAQ:          https://www.FreeBSD.org/faq/
Questions List:       https://www.FreeBSD.org/lists/questions/
FreeBSD Forums:       https://forums.FreeBSD.org/

Documents installed with the system are in the /usr/local/share/doc/freebsd/
directory, or can be installed later with:  pkg install en-freebsd-doc
For other languages, replace "en" with a language code like de or fr.

Show the version of FreeBSD installed:  freebsd-version ; uname -a
Please include that output and any error messages when posting questions.
Introduction to manual pages:  man man
FreeBSD directory layout:      man hier

To change this login announcement, see motd(5).
root@ELKE:~ #
```

```
root@ELKE:~ # mkdir /base
root@ELKE:~ # mount /dev/gpt/BASE /base
root@ELKE:~ # echo vfs.root.mountfrom="ufs:/dev/gpt/BASE\" >>/base/boot/loader.conf
root@ELKE:~ # cat /base/boot/loader.conf
vfs.root.mountfrom="ufs:/dev/gpt/BASE"
root@ELKE:~ # reboot
Waiting (max 60 seconds) for system process `vnlr' to stop... done
Waiting (max 60 seconds) for system process `syncer' to stop...
Syncing disks, vnodes remaining... 1 0 done
All buffers synced.
Uptime: 1m49s
```

```
freebsd-aarch64
nda0: nvme version 1.4
nda0: 8192MB (16777216 512 byte sectors)
uhub1: 8 ports with 8 removable, self powered
uhub0: 8 ports with 8 removable, self powered
ugen0.2: <QEMU QEMU USB Tablet> at usb0
usbhid0 on uhub1
usbhid0: <QEMU QEMU USB Tablet, class 0/0, rev 2.00/0.00, addr 1> on usb0
hidbus0: <HID bus> on usbhid0
ugen0.3: <QEMU QEMU USB Mouse> at usb0
usbhid1 on uhub1
usbhid1: <QEMU QEMU USB Mouse, class 0/0, rev 2.00/0.00, addr 2> on usb0
hidbus1: <HID bus> on usbhid1
Root mount waiting for: usb0
ugen0.4: <QEMU QEMU USB Keyboard> at usb0
usbhid2 on uhub1
usbhid2: <QEMU QEMU USB Keyboard, class 0/0, rev 2.00/0.00, addr 3> on usb0
hidbus2: <HID bus> on usbhid2
hkbd0: <QEMU QEMU USB Keyboard> on hidbus2
kbd1 at hkbd0
ugen0.5: <QEMU QEMU USB Hub> at usb0
uhub2 on uhub1
uhub2: <QEMU QEMU USB Hub, class 9/0, rev 1.10/1.01, addr 4> on usb0
Root mount waiting for: usb0
uhub2: 8 ports with 8 removable, self powered
ugen0.6: <QEMU QEMU USB HARDDRIVE> at usb0
umass0 on uhub2
umass0: <QEMU QEMU USB HARDDRIVE, class 0/0, rev 2.00/0.00, addr 5> on usb0
umass0: SCSI over Bulk-Only; quirks = 0x0
umass0:1:0: Attached to scbus1
cd0 at umass-sim0 bus 0 scbus1 target 0 lun 0
cd0: <QEMU QEMU CD-ROM 2.5+> Removable CD-ROM SPC-3 SCSI device
cd0: Serial Number 1-0000:00:03.0-4.1
cd0: 1.000MB/s transfers
cd0: Attempt to query device size failed: NOT READY, Medium not present
cd0: quirks=0x10<10_BYTE_ONLY>
Dual Console: Serial Primary, Video Secondary
uhub_explore: illegal enable change, port 1
hms0: <QEMU QEMU USB Tablet> on hidbus0
hms0: 3 buttons and [XYW] coordinates ID=0
hms1: <QEMU QEMU USB Mouse> on hidbus1
hms1: 5 buttons and [XYW] coordinates ID=0
vtcon0: <VirtIO Console Adapter> on virtio_pci2
vtrnd0: <VirtIO Entropy Adapter> on virtio_pci3
random: registering fast source VirtIO Entropy Adapter
lo0: link state changed to UP
vtnet0: link state changed to UP

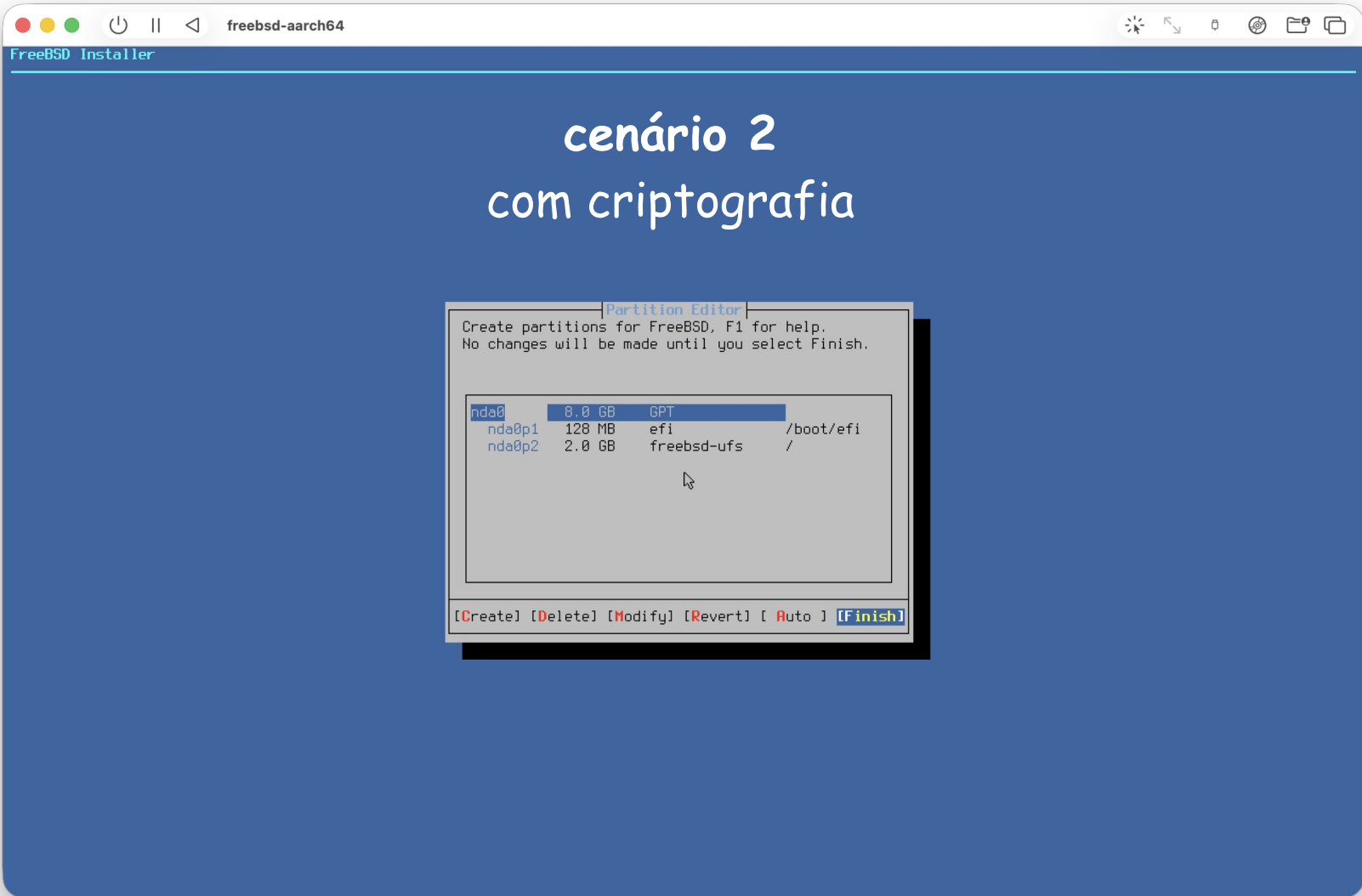
FreeBSD/arm64 (BASE) (ttyv0)
login: 
```

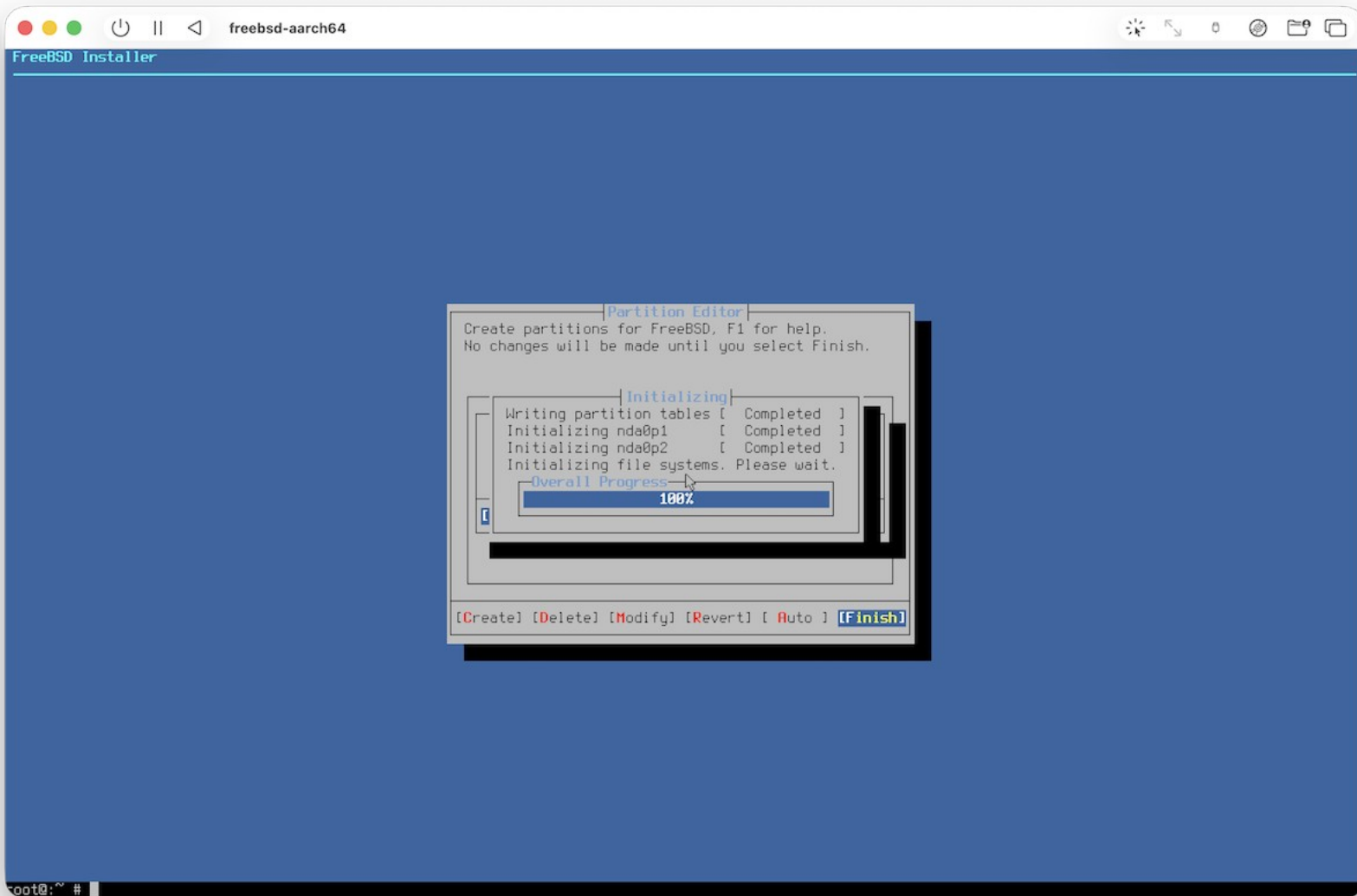
componentes da instalação customizada (elke), cenário 2

- geli
 - criptografia de disco (partição)
 - utilizado pelo instalador oficial
- zfs pool
 - dataset raíz
 - reside na partição criptografada
- sistema operacional
 - gerenciador de pacotes (pkg)
 - base (FreeBSD-set-base)

Directed by
ROBERT B. WEIDE

<https://youtube.com/watch?v=qzbtclsJXw>





```
freebsd-aarch64
root@:~ # gpart show -p nda0
=>      40  16777136   nda0  GPT  (8G)
        40    262144   nda0p1  efi  (128M)
        262184  4194384   nda0p2  freebsd-ufs  (2G)
        4456488 12320688   - free -  (6G)

root@:~ # gpart add -t freebsd-zfs nda0
nda0p3 added
root@:~ # gpart show -p nda0
=>      40  16777136   nda0  GPT  (8G)
        40    262144   nda0p1  efi  (128M)
        262184  4194384   nda0p2  freebsd-ufs  (2G)
        4456488 12320688   nda0p3  freebsd-zfs  (6G)

root@:~ # kldload geom_eli
root@:~ # geli init nda0p3
Enter new passphrase:
Reenter new passphrase:

Metadata backup for provider nda0p3 can be found in /var/backups/nda0p3.eli
and can be restored with the following command:

    # geli restore /var/backups/nda0p3.eli nda0p3

root@:~ # geli attach nda0p3
Enter passphrase:
GEOM_ELI: Device nda0p3.eli created.
GEOM_ELI: Encryption: AES-XTS 128
GEOM_ELI:   Crypto: accelerated software
root@:~ #
```

```
freebsd-aarch64
root@:~ # kldload geom_eli
root@:~ # geli status
root@:~ # geli list
root@:~ # geli init nda0p3
Enter new passphrase:
Reenter new passphrase:

Metadata backup for provider nda0p3 can be found in /var/backups/nda0p3.eli
and can be restored with the following command:

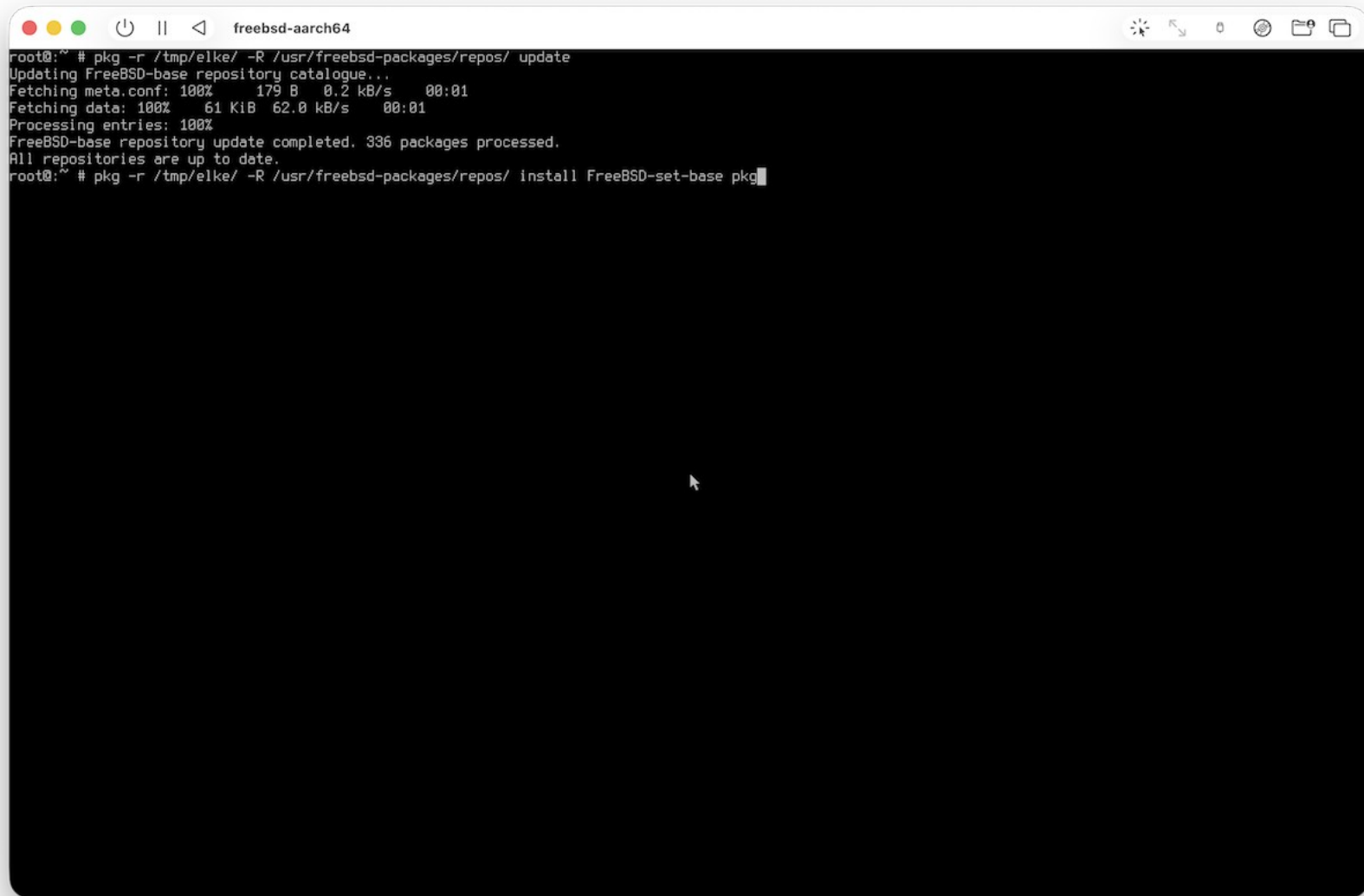
    # geli restore /var/backups/nda0p3.eli nda0p3

root@:~ # geli status
root@:~ # geli list
root@:~ # geli attach nda0p3
Enter passphrase:
GEOM_ELI: Device nda0p3.eli created.
GEOM_ELI: Encryption: AES-XTS 128
GEOM_ELI:   Crypto: accelerated software
root@:~ # geli status
      Name Status Components
nda0p3.eli ACTIVE nda0p3
root@:~ # geli list
Geom name: nda0p3.eli
State: ACTIVE
EncryptionAlgorithm: AES-XTS
KeyLength: 128
Crypto: accelerated software
Version: 7
UsedKey: 0
Flags: AUTORESIZE
KeysAllocated: 12
KeysTotal: 12
Providers:
1. Name: nda0p3.eli
   Mediasize: 6308191744 (5.9G)
   Sectorsize: 512
   Mode: r0w0e0
Consumers:
1. Name: nda0p3
   Mediasize: 6308192256 (5.9G)
   Sectorsize: 512
   StripeSize: 0
   Stripeoffset: 2281721856
   Mode: r1w1e1

root@:~ #
root@:~ #
root@:~ #
root@:~ #
```



```
freebsd-aarch64
root@:~ # kidload zfs
ZFS filesystem version: 5
ZFS storage pool version: features support (5000)
root@:~ # zpool list
no pools available
root@:~ # zfs list
no datasets available
root@:~ # zpool create -f -R /tmp/elke elke nda0p3.eli
root@:~ # zfs create -o mountpoint=/ elke/rootfs
root@:~ # zpool list
NAME      SIZE  ALLOC   FREE CKPOINT  EXPANDSZ   FRAG    CAP  DEDUP    HEALTH  ALTROOT
elke  5.50G  236K   5.50G      -          -         0%    0%   1.00x    ONLINE  /tmp/elke
root@:~ # zfs list
NAME                USED  AVAIL  REFER  MOUNTPOINT
elke                 158K  5.33G   24K    /tmp/elke/elke
elke/rootfs          24K  5.33G   24K    /tmp/elke
root@:~ #
```



```
root@:~ # pkg -r /tmp/elke/ -R /usr/freebsd-packages/repos/ update
Updating FreeBSD-base repository catalogue...
Fetching meta.conf: 100% 179 B 0.2 kB/s 00:01
Fetching data: 100% 61 KiB 62.0 kB/s 00:01
Processing entries: 100%
FreeBSD-base repository update completed. 336 packages processed.
All repositories are up to date.
root@:~ # pkg -r /tmp/elke/ -R /usr/freebsd-packages/repos/ install FreeBSD-set-base pkg
```

```
freebsd-aarch64
FreeBSD-runtime: 16.snap20260413132855
FreeBSD-runtime-dev: 16.snap20260413132855
FreeBSD-sendmail: 16.snap20260413132855
FreeBSD-set-base: 16.snap20260413132855
FreeBSD-set-devel: 16.snap20260413132855
FreeBSD-set-minimal: 16.snap20260413132855
FreeBSD-set-optional: 16.snap20260413132855
FreeBSD-smbutils: 16.snap20260413132855
FreeBSD-smbutils-dev: 16.snap20260413132855
FreeBSD-sound: 16.snap20260413132855
FreeBSD-sound-dev: 16.snap20260413132855
FreeBSD-ssh: 16.snap20260413132855
FreeBSD-ssh-dev: 16.snap20260413132855
FreeBSD-syscons-data: 16.snap20260413132855
FreeBSD-syslogd: 16.snap20260413132855
FreeBSD-tcpd: 16.snap20260413132855
FreeBSD-tcpd-dev: 16.snap20260413132855
FreeBSD-telnet: 16.snap20260413132855
FreeBSD-toolchain: 16.snap20260413132855
FreeBSD-toolchain-dev: 16.snap20260413132855
FreeBSD-ufs: 16.snap20260413132855
FreeBSD-ufs-dev: 16.snap20260413132855
FreeBSD-ufs-lib: 16.snap20260413132855
FreeBSD-utilities: 16.snap20260413132855
FreeBSD-utilities-dev: 16.snap20260413132855
FreeBSD-vi: 16.snap20260413132855
FreeBSD-vt-data: 16.snap20260413132855
FreeBSD-wpa: 16.snap20260413132855
FreeBSD-wx: 16.snap20260413132855
FreeBSD-wx-dev: 16.snap20260413132855
FreeBSD-wx-lib: 16.snap20260413132855
FreeBSD-yp: 16.snap20260413132855
FreeBSD-yp-dev: 16.snap20260413132855
FreeBSD-yp-lib: 16.snap20260413132855
FreeBSD-zfs: 16.snap20260413132855
FreeBSD-zfs-dev: 16.snap20260413132855
FreeBSD-zfs-lib: 16.snap20260413132855
FreeBSD-zlib: 16.snap20260413132855
FreeBSD-zlib-dev: 16.snap20260413132855
FreeBSD-zoneinfo: 16.snap20260413132855
FreeBSD-zstd: 16.snap20260413132855
FreeBSD-zstd-dev: 16.snap20260413132855
FreeBSD-zstd-lib: 16.snap20260413132855
pkg: 2.6.2_1

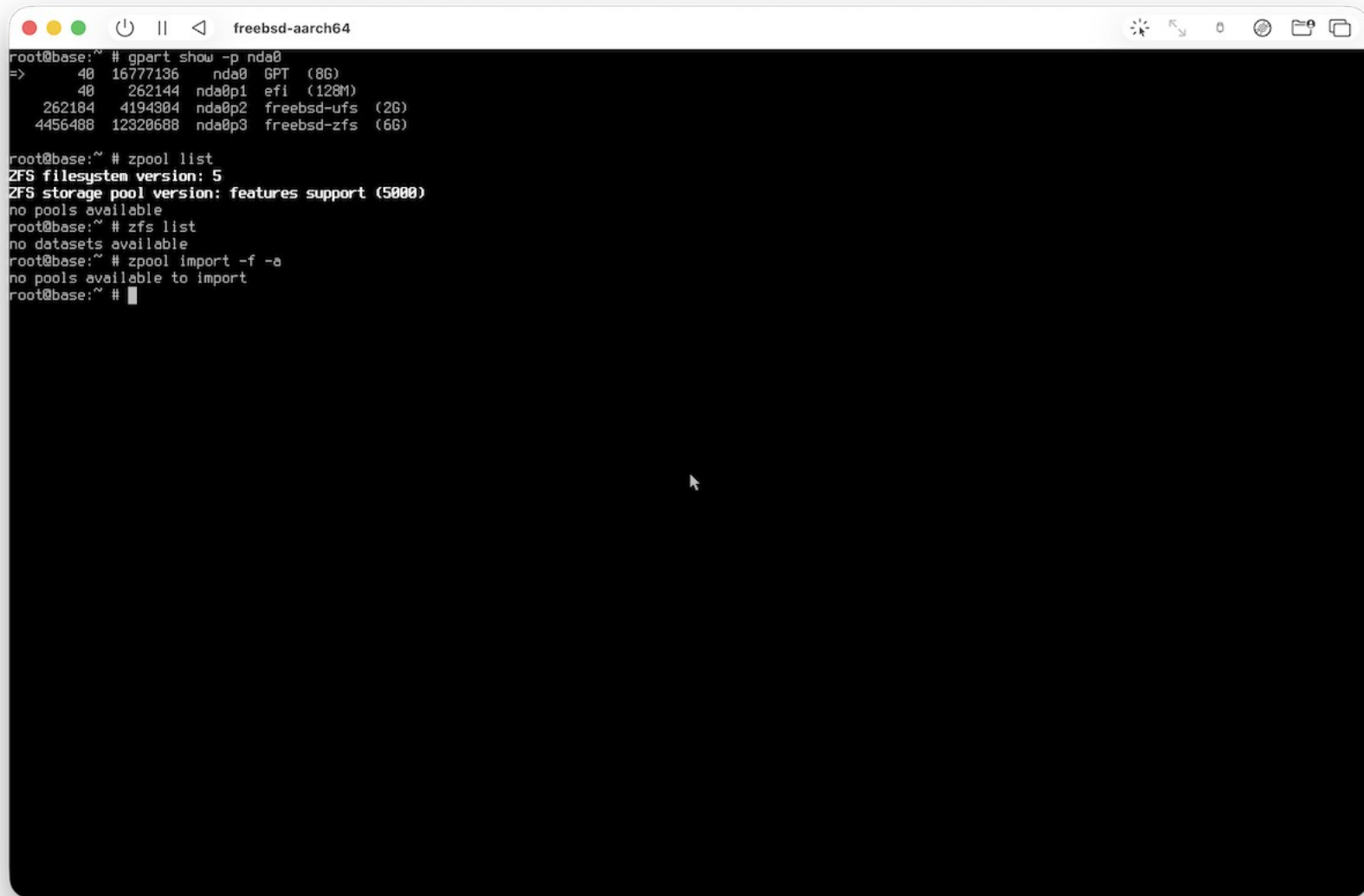
Number of packages to be installed: 216

The process will require 691 MiB more space.

Proceed with this action? [y/N]: y
```



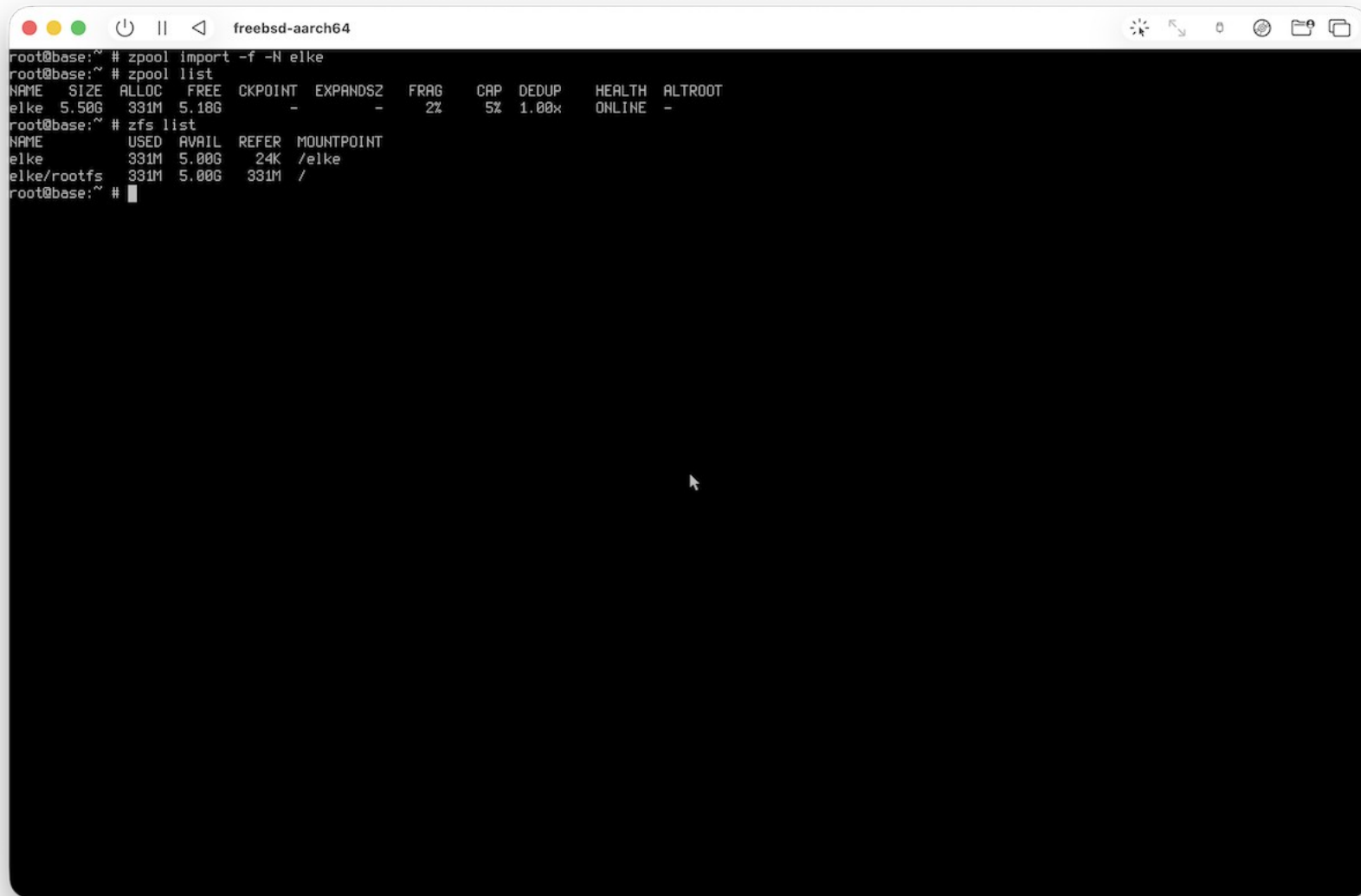
chegamos até aqui; agora é macetar o apocalipse



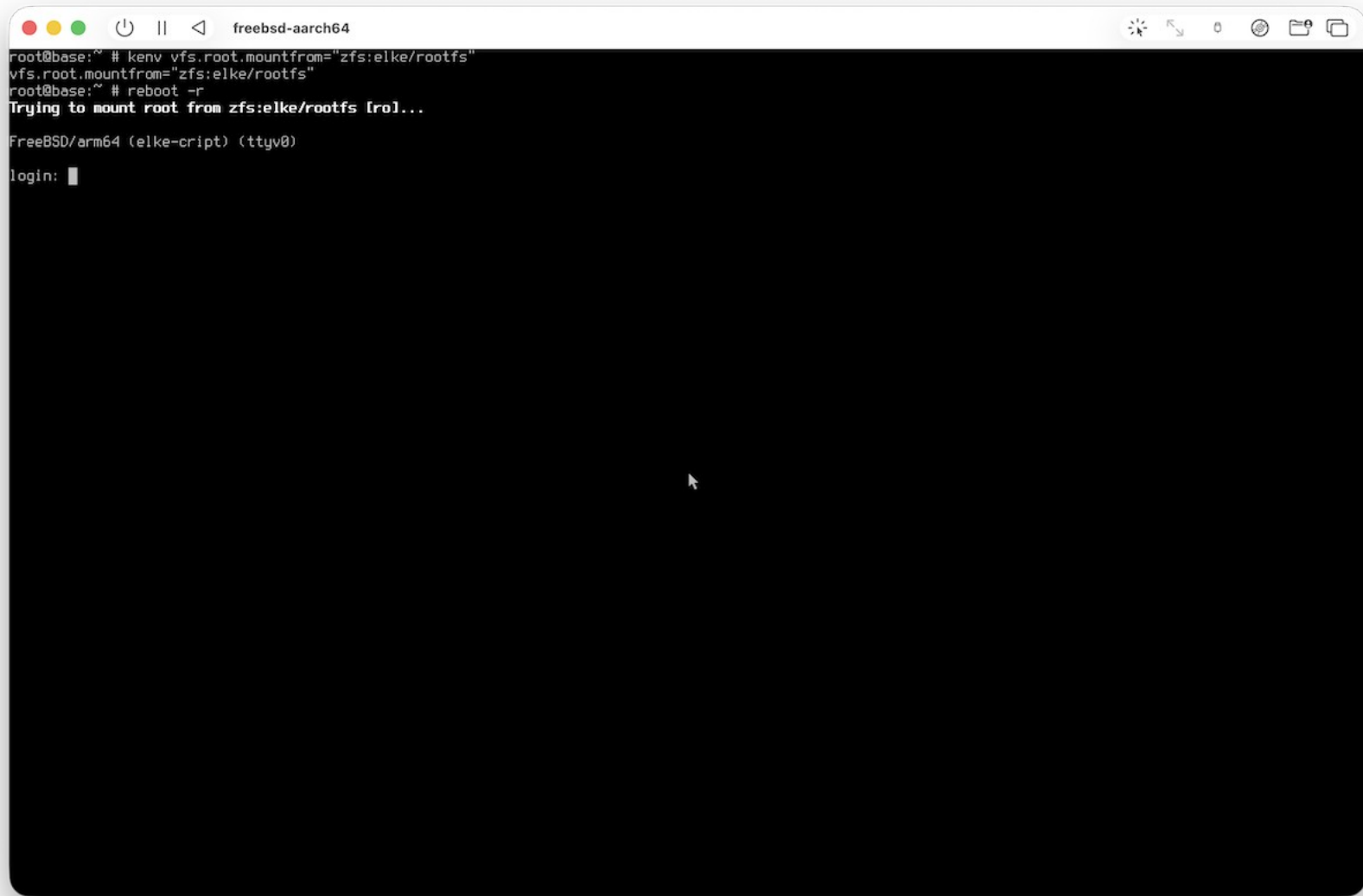
```
root@base:~ # gpart show -p nda0
=>      40  16777136   nda0  GPT  (8G)
        40    262144  nda0p1  efi  (128M)
       262184  4194384  nda0p2  freebsd-ufs  (2G)
      4456488 12320688  nda0p3  freebsd-zfs  (6G)

root@base:~ # zpool list
ZFS filesystem version: 5
ZFS storage pool version: features support (5000)
no pools available
root@base:~ # zfs list
no datasets available
root@base:~ # zpool import -f -a
no pools available to import
root@base:~ #
```

```
root@base:~ # geli attach nda0p3
Enter passphrase:
GEOM_ELI: Device nda0p3.eli created.
GEOM_ELI: Encryption: AES-XTS 128
GEOM_ELI: Crypto: accelerated software
root@base:~ # geli status
      Name Status Components
nda0p3.eli ACTIVE nda0p3
root@base:~ # geli list
Geom name: nda0p3.eli
State: ACTIVE
EncryptionAlgorithm: AES-XTS
KeyLength: 128
Crypto: accelerated software
Version: 7
UsedKey: 0
Flags: AUTORESIZE
KeysAllocated: 12
KeysTotal: 12
Providers:
1. Name: nda0p3.eli
   Mediasize: 6308191744 (5.9G)
   Sectorsize: 512
   Mode: r0u0e0
Consumers:
1. Name: nda0p3
   Mediasize: 6308192256 (5.9G)
   Sectorsize: 512
   Stripessize: 0
   Stripeoffset: 2281721856
   Mode: r1w1e1
root@base:~ # █
```



```
root@base:~ # zpool import -f -N elke
root@base:~ # zpool list
NAME      SIZE  ALLOC   FREE CKPOINT  EXPANDSZ   FRAG    CAP  DEDUP    HEALTH  ALTROOT
elke  5.50G   331M  5.18G        -         -     2%    5%   1.00x    ONLINE  -
root@base:~ # zfs list
NAME                USED  AVAIL  REFER  MOUNTPOINT
elke                 331M  5.00G   24K    /elke
elke/rootfs          331M  5.00G   331M    /
root@base:~ #
```




```
freebsd-aarch64
root@base:~ # kenv vfs.root.mountfrom="zfs:elke/rootfs"
vfs.root.mountfrom="zfs:elke/rootfs"
root@base:~ # reboot -r
Trying to mount root from zfs:elke/rootfs [ro]...

FreeBSD/arm64 (elke-crypt) (ttyv0)

login: root
FreeBSD 16.0-CURRENT (GENERIC) #0 main-n285005-e9fc0c538264: Mon Apr 13 12:48:35 UTC 2026

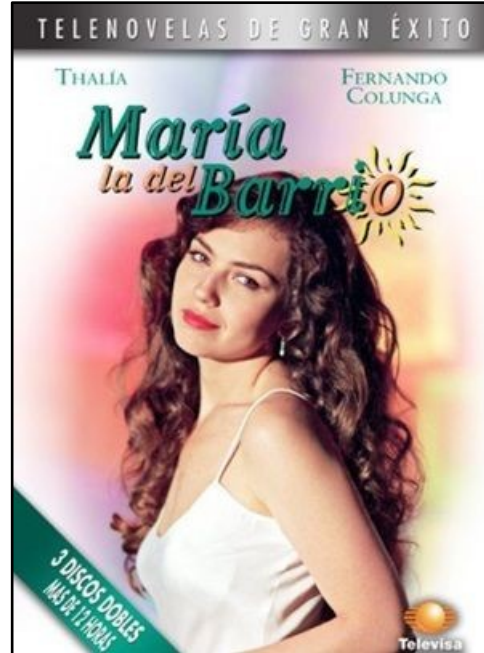
Welcome to FreeBSD!

Release Notes, Errata: https://www.FreeBSD.org/releases/
Security Advisories:  https://www.FreeBSD.org/security/
FreeBSD Handbook:     https://www.FreeBSD.org/handbook/
FreeBSD FAQ:          https://www.FreeBSD.org/faq/
Questions List:       https://www.FreeBSD.org/lists/questions/
FreeBSD Forums:       https://forums.FreeBSD.org/

Documents installed with the system are in the /usr/local/share/doc/freebsd/
directory, or can be installed later with:  pkg install en-freebsd-doc
For other languages, replace "en" with a language code like de or fr.

Show the version of FreeBSD installed:  freebsd-version ; uname -a
Please include that output and any error messages when posting questions.
Introduction to manual pages:  man man
FreeBSD directory layout:      man hier

To change this login announcement, see motd(5).
root@elke-crypt:~ # touch aaa
root@elke-crypt:~ # ls -l
total 6
-rw-r--r--  1 root wheel 1011 Apr 13 04:38 .cshrc
-rw-r--r--  1 root wheel   66 Apr 13 04:38 .k5login
-rw-r--r--  1 root wheel  316 Apr 13 04:38 .login
-rw-r--r--  1 root wheel  472 Apr 13 04:38 .profile
-rw-r--r--  1 root wheel 1213 Apr 13 04:38 .shrc
-rw-r--r--  1 root wheel    0 May  6 17:06 aaa
root@elke-crypt:~ #
```



a mesma base gera várias novelas



CRYPTORAVE
2026

evolução & refinamento

pendrive bootável (memstick)

- sistema base
- armazenamento persistente
 - particionamentos extra
 - criptografado (?)
- finalização remota do setup
 - configuração assistida

ssh(1), ssh_config(5), sshd(8) & sshd_config(5)

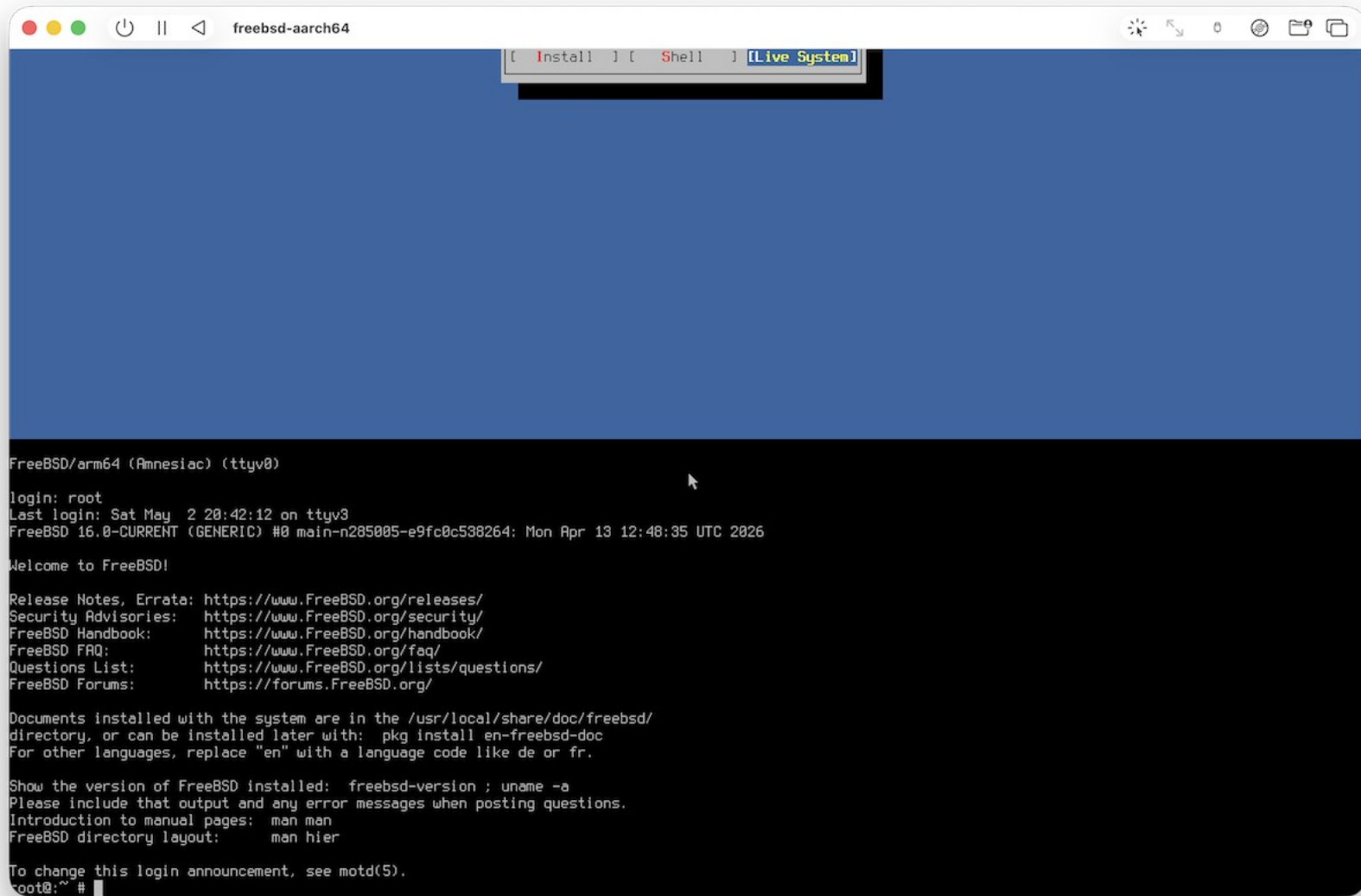
- allowusers, permitrootlogin, ...
- algoritmos, ciphers, hashes, mac (msg. auth. code), ...
- hostkeyalias, stricthostkeycheck, verifyhostdns, ...
- proxycommand & proxyjump
- userknownhostsfile

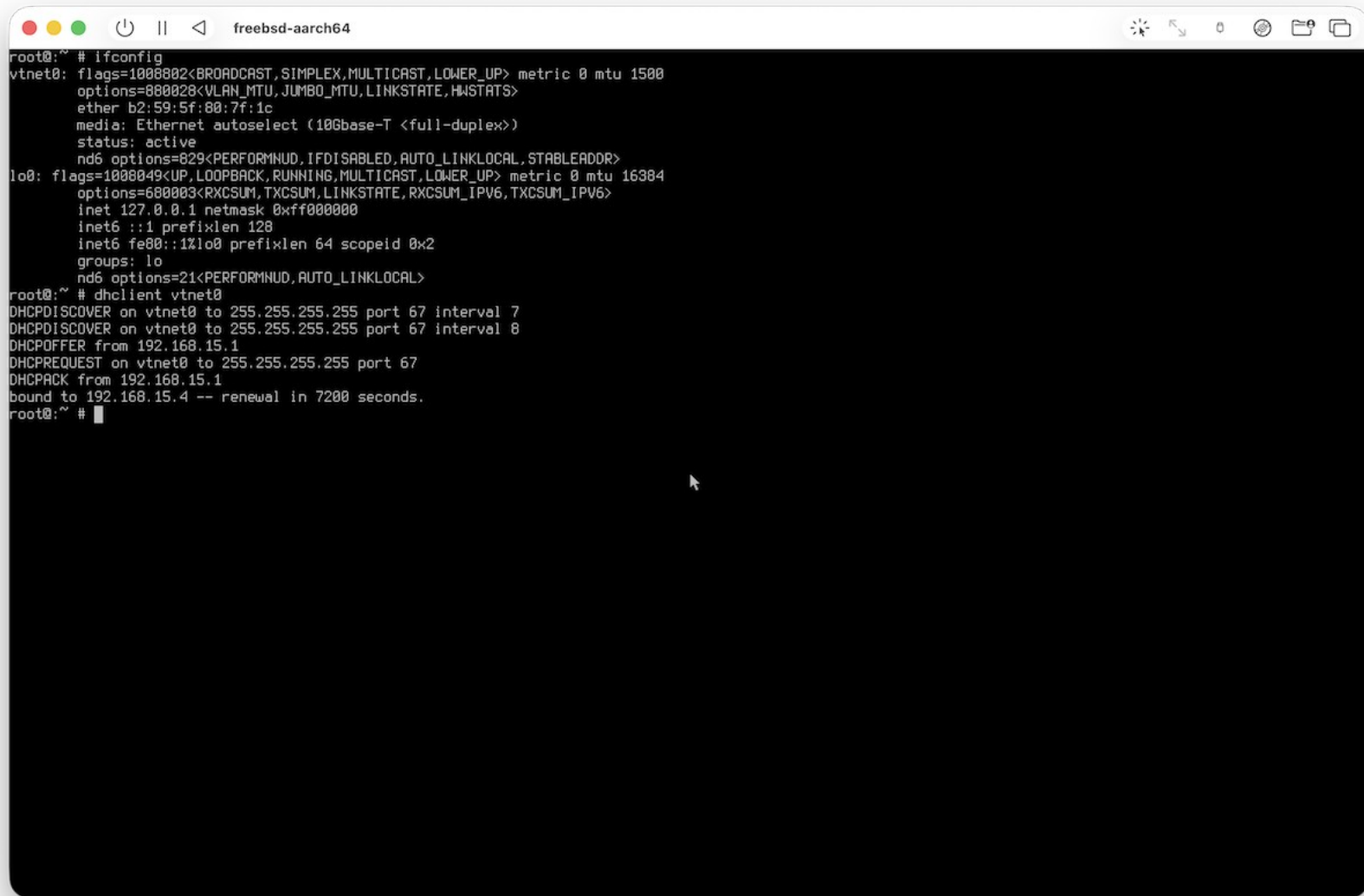
md/mfs & initramfs/initrd

- baseado no truenas (R.I.P.) & mfsbsd
- otimização de tamanho
- replicação em diferentes ambientes

chroot "base"

- /tmp/base, ufs (sem criptografia)
 - módulos do kernel
 - console "inseguro"
 - loader.conf & rc.conf
- pacotes extras (pkg)
 - tor? ;3
- customizações na configuração de rede
- serviços adicionais
 - monitoramento
 - túneis & vpn





```
root@:~ # ifconfig
vtnet0: flags=1008802<BROADCAST,SIMPLEX,MULTICAST,LOWER_UP> metric 0 mtu 1500
options=880028<VLAN_MTU,JUMBO_MTU,LINKSTATE,HMSTATS>
ether b2:59:5f:80:7f:1c
media: Ethernet autoselect (10Gbase-T <full-duplex>)
status: active
nd6 options=829<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL,STABLEADDR>
lo0: flags=1008049<UP,LOOPBACK,RUNNING,MULTICAST,LOWER_UP> metric 0 mtu 16384
options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_IPV6>
inet 127.0.0.1 netmask 0xff000000
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x2
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
root@:~ # dhclient vtnet0
DHCPDISCOVER on vtnet0 to 255.255.255.255 port 67 interval 7
DHCPDISCOVER on vtnet0 to 255.255.255.255 port 67 interval 8
DHCPOFFER from 192.168.15.1
DHCPREQUEST on vtnet0 to 255.255.255.255 port 67
DHCPACK from 192.168.15.1
bound to 192.168.15.4 -- renewal in 7200 seconds.
root@:~ #
```




```
freebsd-aarch64
root@:~ # gpart show
=>      9  590112  cd0  MBR (16)
      9  590112      - free - (16)

=>      9  590112  iso9660/16_0_CURRENT_AARCH64_CD  MBR (16)
      9  590112      - free - (16)

root@:~ # gpart create -s gpt nda0
nda0 created
root@:~ # gpart add -t efi -s 128m nda0
nda0p1 added
root@:~ # gpart add -t freebsd-ufs -s 2g nda0
nda0p2 added
root@:~ # gpart show -p nda0
=>      40  16777136  nda0  GPT (8G)
      40    262144  nda0p1  efi (128M)
      262184  4194304  nda0p2  freebsd-ufs (2G)
      4456488 12320688      - free - (6G)

root@:~ # █
```

```
freebsd-aarch64
root@:~ # gpart show
=>      9  590112  cd0  MBR (16)
      9  590112      - free - (16)

=>      9  590112  iso9660/16_0_CURRENT_AARCH64_CD  MBR (16)
      9  590112      - free - (16)

root@:~ # gpart create -s gpt nda0
nda0 created
root@:~ # gpart add -t efi -s 128m nda0
nda0p1 added
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nda0p2 added
root@:~ # gpart show -p nda0
=>      40  16777136  nda0  GPT (8G)
      40    262144  nda0p1  efi (128M)
      262184  4194304  nda0p2  freebsd-ufs (2G)
      4456488 12320688      - free - (6G)

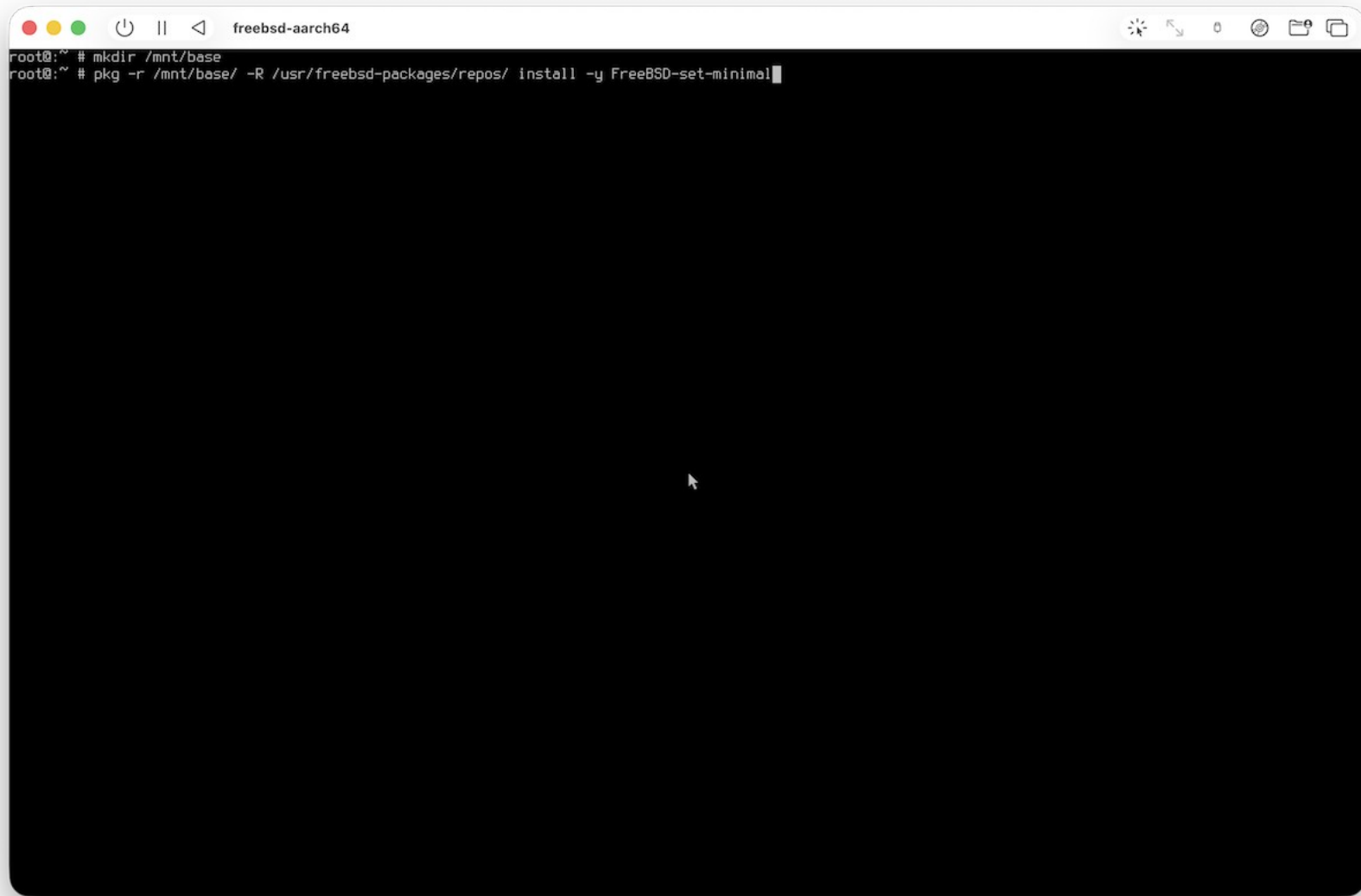
root@:~ # newfs_msdos -c 1 -F 32 nda0p1
/dev/nda0p1: 258078 sectors in 258078 FAT32 clusters (512 bytes/cluster)
BytesPerSec=512 SecPerClust=1 ResSectors=32 FATs=2 Media=0xf0 SecPerTrack=63 Heads=255 HiddenSecs=0 HugeSectors=262144 FATsecs=2017 RootCluster=2 FSInfo=1 Backup=2
root@:~ # newfs nda0p2
/dev/nda0p2: 2048.0MB (4194304 sectors) block size 32768, fragment size 4096
        using 4 cylinder groups of 512.03MB, 16385 blks, 65664 inodes.
        with soft updates
super-block backups (for fsck_ffs -b #) at:
 192, 1048832, 2097472, 3146112
root@:~ # █
```

```
freebsd-aarch64
root@:~ # gpart show -p nda0
=>  40  16777136  nda0  GPT  (8G)
    40    262144  nda0p1  efi  (128M)
   262184  4194304  nda0p2  freebsd-ufs  (2G)
  4456488 12320688  - free -  (6G)

root@:~ # mount /dev/nda0p2 /mnt
root@:~ # mkdir -p /mnt/boot/efi
root@:~ # mount_msdosfs /dev/nda0p1 /mnt/boot/efi
root@:~ # df -Th
Filesystem                                Type      Size    Used    Avail Capacity  Mounted on
/dev/iso9660/16_0_CURRENT_AARCH64_CD      cd9660    1.1G    1.1G      0B    100%          /
devfs                                       devfs     1.0K      0B    1.0K      0%          /dev
tmpfs                                       tmpfs     32M    128K    32M      0%          /var
tmpfs                                       tmpfs     20M     12K    20M      0%          /tmp
/dev/nda0p2                                ufs       1.9G    16M    1.8G      1%          /mnt
/dev/nda0p1                                msdosfs   126M    512B   126M      0%          /mnt/boot/efi
root@:~ # mkdir /mnt/base
root@:~ #
```

```
root@:~ # df -Th
Filesystem                                Type      Size  Used Avail Capacity  Mounted on
/dev/iso9660/16_0_CURRENT_AARCH64_CD     cd9660     1.1G   1.1G    0B    100%      /
devfs                                     devfs       1.0K     0B   1.0K     0%     /dev
tmpfs                                     tmpfs       32M    132K   32M     0%     /var
tmpfs                                     tmpfs       20M    4.0K   20M     0%     /tmp
/dev/nda0p2                               ufs        1.9G    16K   1.8G     0%     /mnt
/dev/nda0p1                               msdosfs    126M   855K   125M     1%     /mnt/boot/efi
root@:~ # find /mnt/
/mnt/
/mnt/.snap
/mnt/boot
/mnt/boot/efi
/mnt/boot/efi/efi
/mnt/boot/efi/efi/freebsd
/mnt/boot/efi/efi/freebsd/loader.efi
root@:~ #
```

```
freebsd-aarch64
root@:~ # efibootmgr -a -c -B 1 -l /mnt/boot/efi/efi/freebsd/loader.efi -L freebsd
Boot to FW : false
BootCurrent: 0001
Timeout : 3 seconds
BootOrder : 0005, 0001, 0003, 0000, 0002, 0004
Boot0005* freebsd
+Boot0001* UEFI QEMU QEMU USB HARDDRIVE 1-0000:00:03.0-4.1
Boot0003* UEFI QEMU NVMe Ctrl 2EEFCEBC-79C2-4B6A-A 1
Boot0000* UIApp
Boot0002* UEFI Misc Device
Boot0004* EFI Internal Shell
root@:~ #
```



```
freebsd-aarch64
[45/69] Extracting FreeBSD-rescue-16.snap20260413132855: 100%
[46/69] Installing FreeBSD-resolvconf-16.snap20260413132855...
[46/69] Extracting FreeBSD-resolvconf-16.snap20260413132855: 100%
[47/69] Installing FreeBSD-dhclient-16.snap20260413132855...
[47/69] Extracting FreeBSD-dhclient-16.snap20260413132855: 100%
[48/69] Installing FreeBSD-runtime-16.snap20260413132855...
[48/69] Extracting FreeBSD-runtime-16.snap20260413132855: 100%
[49/69] Installing FreeBSD-at-16.snap20260413132855...
[49/69] Extracting FreeBSD-at-16.snap20260413132855: 100%
[50/69] Installing FreeBSD-devd-16.snap20260413132855...
[50/69] Extracting FreeBSD-devd-16.snap20260413132855: 100%
[51/69] Installing FreeBSD-rc-16.snap20260413132855...
[51/69] Extracting FreeBSD-rc-16.snap20260413132855: 100%
[52/69] Installing FreeBSD-syslogd-16.snap20260413132855...
[52/69] Extracting FreeBSD-syslogd-16.snap20260413132855: 100%
[53/69] Installing FreeBSD-tcpd-16.snap20260413132855...
[53/69] Extracting FreeBSD-tcpd-16.snap20260413132855: 100%
[54/69] Installing FreeBSD-ufs-16.snap20260413132855...
[54/69] Extracting FreeBSD-ufs-16.snap20260413132855: 100%
[55/69] Installing FreeBSD-ufs-lib-16.snap20260413132855...
[55/69] Extracting FreeBSD-ufs-lib-16.snap20260413132855: 100%
[56/69] Installing FreeBSD-utilities-16.snap20260413132855...
[56/69] Extracting FreeBSD-utilities-16.snap20260413132855: 100%
[57/69] Installing FreeBSD-vi-16.snap20260413132855...
[57/69] Extracting FreeBSD-vi-16.snap20260413132855: 100%
[58/69] Installing FreeBSD-vt-data-16.snap20260413132855...
[58/69] Extracting FreeBSD-vt-data-16.snap20260413132855: 100%
[59/69] Installing FreeBSD-wpa-16.snap20260413132855...
[59/69] Extracting FreeBSD-wpa-16.snap20260413132855: 100%
[60/69] Installing FreeBSD-xz-16.snap20260413132855...
[60/69] Extracting FreeBSD-xz-16.snap20260413132855: 100%
[61/69] Installing FreeBSD-xz-lib-16.snap20260413132855...
[61/69] Extracting FreeBSD-xz-lib-16.snap20260413132855: 100%
[62/69] Installing FreeBSD-yp-lib-16.snap20260413132855...
[62/69] Extracting FreeBSD-yp-lib-16.snap20260413132855: 100%
[63/69] Installing FreeBSD-zfs-16.snap20260413132855...
[63/69] Extracting FreeBSD-zfs-16.snap20260413132855: 100%
[64/69] Installing FreeBSD-zfs-lib-16.snap20260413132855...
[64/69] Extracting FreeBSD-zfs-lib-16.snap20260413132855: 100%
[65/69] Installing FreeBSD-zlib-16.snap20260413132855...
[65/69] Extracting FreeBSD-zlib-16.snap20260413132855: 100%
[66/69] Installing FreeBSD-zoneinfo-16.snap20260413132855...
[66/69] Extracting FreeBSD-zoneinfo-16.snap20260413132855: 100%
[67/69] Installing FreeBSD-zstd-16.snap20260413132855...
[67/69] Extracting FreeBSD-zstd-16.snap20260413132855: 100%
[68/69] Installing FreeBSD-zstd-lib-16.snap20260413132855...
[68/69] Extracting FreeBSD-zstd-lib-16.snap20260413132855: 100%
[69/69] Installing FreeBSD-set-minimal-16.snap20260413132855...
==> Running trigger: mandoc.ucl
root@:~ #
```

```
freebsd-aarch64
root@:~ # pkg -r /mnt/ -R /usr/freebsd-packages/repos/ install -y FreeBSD-bootloader FreeBSD-kernel-generic
Updating FreeBSD-base repository catalogue...
Fetching meta.conf: 100% 179 B 0.2 kB/s 00:01
Fetching data: 100% 61 KiB 62.0 kB/s 00:01
Processing entries: 100%
FreeBSD-base repository update completed. 336 packages processed.
All repositories are up to date.
Checking integrity... done (0 conflicting)
The following 2 package(s) will be affected (of 0 checked):

New packages to be INSTALLED:
  FreeBSD-bootloader: 16.snap20260413132855
  FreeBSD-kernel-generic: 16.snap20260413132855

Number of packages to be installed: 2

The process will require 87 MiB more space.
[1/2] Installing FreeBSD-bootloader-16.snap20260413132855...
[1/2] Extracting FreeBSD-bootloader-16.snap20260413132855: 100%
[2/2] Installing FreeBSD-kernel-generic-16.snap20260413132855...
[2/2] Extracting FreeBSD-kernel-generic-16.snap20260413132855: 100%
root@:~ #
```



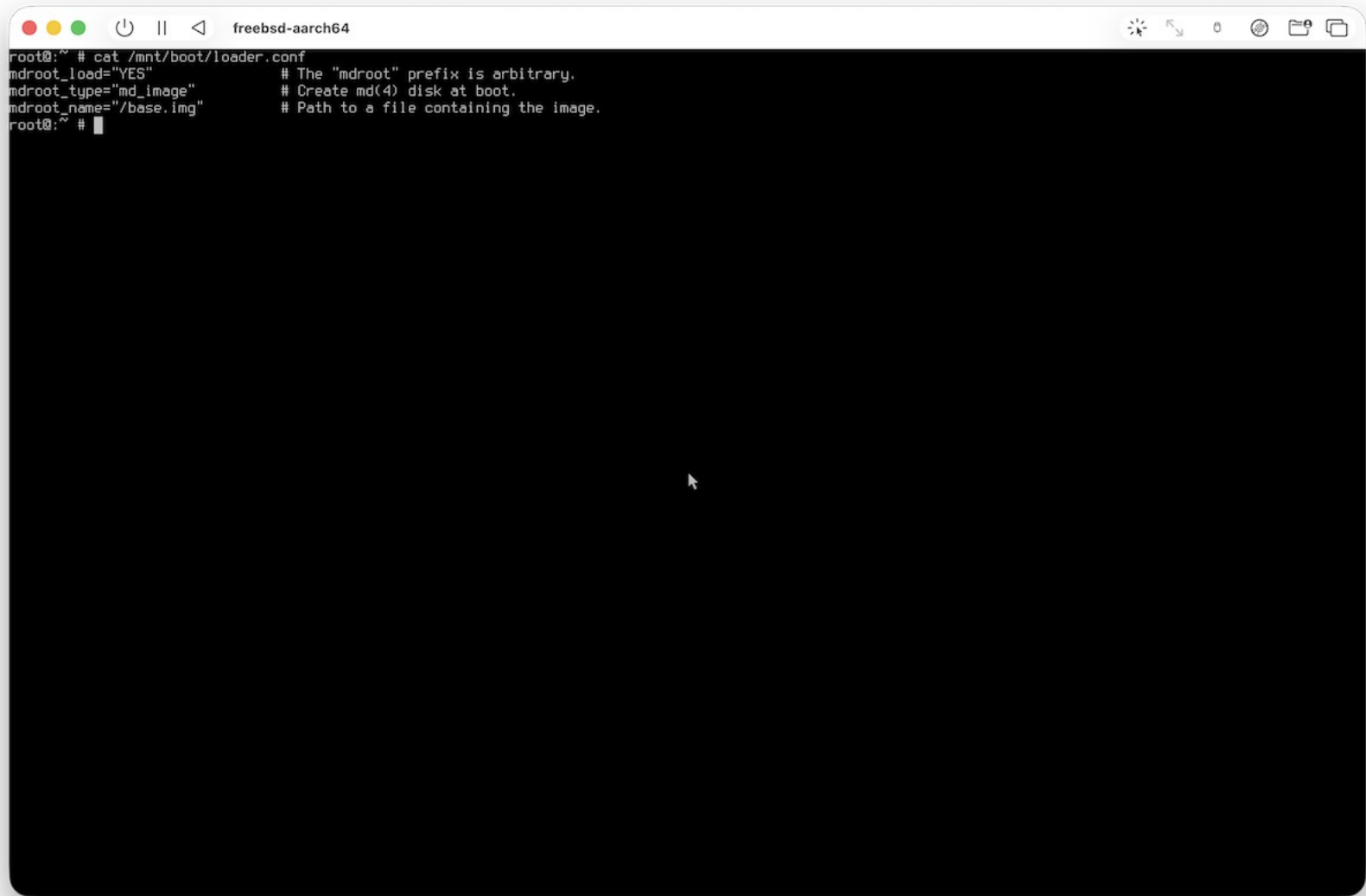
```
freebsd-aarch64
root@:~ # pkg -r /mnt/ -R /usr/freebsd-packages/repos/ install -y FreeBSD-bootloader FreeBSD-kernel-generic
Updating FreeBSD-base repository catalogue...
Fetching meta.conf: 100% 179 B 0.2 kB/s 00:01
Fetching data: 100% 61 KiB 62.0 kB/s 00:01
Processing entries: 100%
FreeBSD-base repository update completed. 336 packages processed.
All repositories are up to date.
Checking integrity... done (0 conflicting)
The following 2 package(s) will be affected (of 0 checked):

New packages to be INSTALLED:
    FreeBSD-bootloader: 16.snap20260413132855
    FreeBSD-kernel-generic: 16.snap20260413132855

Number of packages to be installed: 2

The process will require 87 MiB more space.
[1/2] Installing FreeBSD-bootloader-16.snap20260413132855...
[1/2] Extracting FreeBSD-bootloader-16.snap20260413132855: 100%
[2/2] Installing FreeBSD-kernel-generic-16.snap20260413132855...
[2/2] Extracting FreeBSD-kernel-generic-16.snap20260413132855: 100%
root@:~ # grep mdroot /boot/defaults/loader.conf >>/mnt/boot/loader.conf
root@:~ # cat /mnt/boot/loader.conf
#mdroot_load="YES"           # The "mdroot" prefix is arbitrary.
#mdroot_type="md_image"      # Create md(4) disk at boot.
#mdroot_name="/boot/root.img" # Path to a file containing the image.
root@:~ #
```





```
root@:~ # cat /mnt/boot/loader.conf
mdroot_load="YES"           # The "mdroot" prefix is arbitrary.
mdroot_type="md_image"      # Create md(4) disk at boot.
mdroot_name="/base.img"    # Path to a file containing the image.
root@:~ #
```

```
freebsd-aarch64

New packages to be INSTALLED:
  libevent: 2.1.12 [FreeBSD-ports]
  tor: 0.4.9.6 [FreeBSD-ports]
  zstd: 1.5.7_1 [FreeBSD-ports]

Number of packages to be installed: 3

The process will require 33 MiB more space.
5 MiB to be downloaded.
[1/3] Fetching zstd-1.5.7_1: 100% 430 KiB 440.4 kB/s 00:01
[2/3] Fetching tor-0.4.9.6: 100% 3873 KiB 4.0 MB/s 00:01
[3/3] Fetching libevent-2.1.12: 100% 338 KiB 346.4 kB/s 00:01
Checking integrity... done (0 conflicting)
[1/3] Installing libevent-2.1.12...
[1/3] Extracting libevent-2.1.12: 100%
[2/3] Installing zstd-1.5.7_1...
[2/3] Extracting zstd-1.5.7_1: 100%
[3/3] Installing tor-0.4.9.6...
==> Creating groups
Creating group '_tor' with gid '256'
==> Creating users
Creating user '_tor' with uid '256'
==> Creating homedir(s)
[3/3] Extracting tor-0.4.9.6: 100%
=====
Message from tor-0.4.9.6:

--
To enable the tor server, set tor_enable="YES" in your /etc/rc.conf
and edit /usr/local/etc/tor/torrc as desired. (However, note that the
/usr/local/etc/rc.d/tor rc.subr script can override some torrc
options: see that script for details.) To use the torify script, install
the net/torsocks port.

Tor users are strongly advised to prevent traffic analysis that
exploits sequential IP IDs by setting:

sysctl net.inet.ip.random_id=1

(see sysctl.conf(5)).

In order to run additional, independent instances of tor on the same machine
set tor_instances="inst1 inst2 ..." in your /etc/rc.conf, and create the
corresponding additional configuration files /usr/local/etc/tor/torrc@inst1, ...

Alternatively, you can use the extended instance definition to specify all
instance parameters explicitly:
inst_name(:inst_conf:inst_user:inst_group:inst_pidfile:inst_data_dir)
root@:/ #
```



the onion router - Tor

- <https://github.com/egypcio/criptobaiao>
- <https://torproject.org/pt-BR>
- <https://youtube.com/watch?v=mM07ERWb6Zw>



```
root@:~ # pkg -r /mnt/base/ update -f
Updating FreeBSD-ports repository catalogue...
Fetching meta.conf: 100% 179 B 0.2 kB/s 00:01
Fetching data: 100% 10 MiB 2.1 MB/s 00:05
pkg: Error extracting the archive: 'Write error'
pkg: No signature found
FreeBSD-ports repository is up to date.
Updating FreeBSD-ports-kmods repository catalogue...
Fetching meta.conf: 100% 179 B 0.2 kB/s 00:01
Fetching data: 100% 16 KiB 16.7 kB/s 00:01
Processing entries: 100%
FreeBSD-ports-kmods repository update completed. 78 packages processed.
All repositories are up to date.
root@:~ # pkg -r /mnt/base/ install rsync
Updating FreeBSD-ports repository catalogue...
FreeBSD-ports repository is up to date.
Updating FreeBSD-ports-kmods repository catalogue...
FreeBSD-ports-kmods repository is up to date.
All repositories are up to date.
pkg: warning: database version 39 is newer than libpkg(3) version 38, but still compatible
The following 4 package(s) will be affected (of 0 checked):

New packages to be INSTALLED:
  libiconv: 1.18_1 [FreeBSD-ports]
  liblz4: 1.10.0_2,1 [FreeBSD-ports]
  rsync: 3.4.1_7 [FreeBSD-ports]
  xxhash: 0.8.3 [FreeBSD-ports]

Number of packages to be installed: 4

The process will require 5 MiB more space.
1 MiB to be downloaded.

Proceed with this action? [y/N]:
```

```
freebsd-aarch64
FreeBSD-ports repository is up to date.
Updating FreeBSD-ports-knobs repository catalogue...
FreeBSD-ports-knobs repository is up to date.
All repositories are up to date.
pkg: warning: database version 39 is newer than libpkg(3) version 38, but still compatible
The following 4 package(s) will be affected (of 0 checked):

New packages to be INSTALLED:
  libiconv: 1.18_1 [FreeBSD-ports]
  liblz4: 1.10.0_2,1 [FreeBSD-ports]
  rsync: 3.4.1_7 [FreeBSD-ports]
  xxhash: 0.8.3 [FreeBSD-ports]

Number of packages to be installed: 4

The process will require 5 MiB more space.
1 MiB to be downloaded.

Proceed with this action? [y/N]: y
[1/4] Fetching rsync-3.4.1_7: 100% 399 KiB 408.2 kB/s 00:01
[2/4] Fetching liblz4-1.10.0_2,1: 100% 150 KiB 154.1 kB/s 00:01
[3/4] Fetching libiconv-1.18_1: 100% 722 KiB 739.3 kB/s 00:01
[4/4] Fetching xxhash-0.8.3: 100% 94 KiB 96.2 kB/s 00:01
Checking integrity... done (0 conflicting)
[1/4] Installing libiconv-1.18_1...
[1/4] Extracting libiconv-1.18_1: 100%
[2/4] Installing liblz4-1.10.0_2,1...
[2/4] Extracting liblz4-1.10.0_2,1: 100%
[3/4] Installing xxhash-0.8.3...
[3/4] Extracting xxhash-0.8.3: 100%
[4/4] Installing rsync-3.4.1_7...
[4/4] Extracting rsync-3.4.1_7: 100%
=====
Message from rsync-3.4.1_7:

--
Some scripts provided by rsync, such as rrsync,
require Python, which is not installed by default.
root@:~ # umount /mnt/base/dev
root@:~ # cd /mnt/base
root@:/mnt/base # makefs /mnt/base.img .
Calculated size of '/mnt/base.img': 383713280 bytes, 7318 inodes
Extent size set to 32768
/mnt/base.img: 365.9MiB (749440 sectors) block size 32768, fragment size 4096
using 1 cylinder groups of 365.94MiB, 11710 blks, 7936 inodes.
super-block backups (for fsck -b #) at:
 64
Populating '/mnt/base.img'
Image '/mnt/base.img' complete
root@:/mnt/base #
```



```
freebsd-aarch64

All repositories are up to date.
pkg: warning: database version 39 is newer than libpkg(3) version 38, but still compatible
The following 4 package(s) will be affected (of 0 checked):

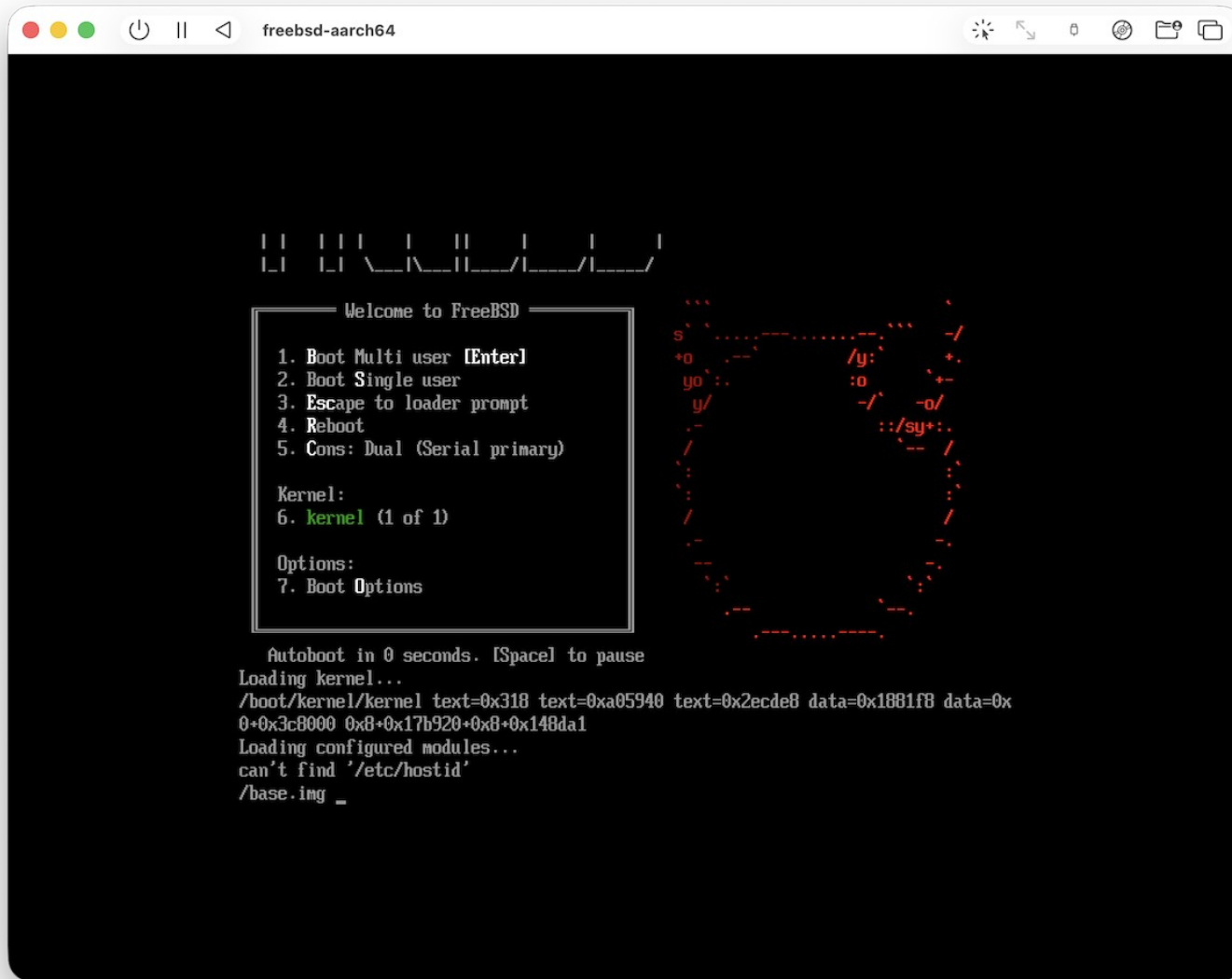
New packages to be INSTALLED:
  libiconv: 1.18_1 [FreeBSD-ports]
  liblz4: 1.10.0_2.1 [FreeBSD-ports]
  rsync: 3.4.1_7 [FreeBSD-ports]
  xxhash: 0.8.3 [FreeBSD-ports]

Number of packages to be installed: 4

The process will require 5 MiB more space.
1 MiB to be downloaded.

Proceed with this action? [y/N]: y
[1/4] Fetching rsync-3.4.1_7: 100% 399 KiB 408.2 kB/s 00:01
[2/4] Fetching liblz4-1.10.0_2.1: 100% 150 KiB 154.1 kB/s 00:01
[3/4] Fetching libiconv-1.18_1: 100% 722 KiB 739.3 kB/s 00:01
[4/4] Fetching xxhash-0.8.3: 100% 94 KiB 96.2 kB/s 00:01
Checking integrity... done (0 conflicting)
[1/4] Installing libiconv-1.18_1...
[1/4] Extracting libiconv-1.18_1: 100%
[2/4] Installing liblz4-1.10.0_2.1...
[2/4] Extracting liblz4-1.10.0_2.1: 100%
[3/4] Installing xxhash-0.8.3...
[3/4] Extracting xxhash-0.8.3: 100%
[4/4] Installing rsync-3.4.1_7...
[4/4] Extracting rsync-3.4.1_7: 100%
=====
Message from rsync-3.4.1_7:

--
Some scripts provided by rsync, such as rrsync,
require Python, which is not installed by default.
root@:~ # umount /mnt/base/dev
root@:~ # cd /mnt/base
root@:/mnt/base # makefs /mnt/base.img .
Calculated size of '/mnt/base.img': 383713280 bytes, 7318 inodes
Extent size set to 32768
/mnt/base.img: 365.9MiB (749440 sectors) block size 32768, fragment size 4096
        using 1 cylinder groups of 365.9MiB, 11710 blks, 7936 inodes.
super-block backups (for fsck -b #) at:
 64
Populating '/mnt/base.img'
Image '/mnt/base.img' complete
root@:/mnt/base # reboot
Waiting (max 60 seconds) for system process 'vnlr' to stop... done
Waiting (max 60 seconds) for system process 'syncer' to stop...
Syncing disks, vnodes remaining... 0
```

```
freebsd-aarch64
Auxiliary Features 1 = <>
AArch32 Instruction Set Attributes 5 = <>
AArch32 Media and VFP Features 0 = <>
AArch32 Media and VFP Features 1 = <>
TCP_ratelimit: Is now initialized
WARNING: WITNESS option enabled, expect reduced performance.
Trying to mount root from ufs:/dev/md0 [1...]
Dual Console: Serial Primary, Video Secondary
nda0 at nvme0 bus 0 scbus0 target 0 lun 1
nda0: <QEMU NVMe Ctrl 10.0.2 2EEFCEBC-79C2-4B6A-A>
nda0: Serial Number 2EEFCEBC-79C2-4B6A-A
nda0: nvme version 1.4
nda0: 8192MB (16777216 512 byte sectors)
uhub0: 8 ports with 8 removable, self powered
uhub1: 8 ports with 8 removable, self powered
ugen0.2: <QEMU QEMU USB Tablet> at usb0
usbhid0 on uhub1
usbhid0: <QEMU QEMU USB Tablet, class 0/0, rev 2.00/0.00, addr 1> on usb0
hidbus0: <HID bus> on usbhid0
ugen0.3: <QEMU QEMU USB Mouse> at usb0
usbhid1 on uhub1
usbhid1: <QEMU QEMU USB Mouse, class 0/0, rev 2.00/0.00, addr 2> on usb0
hidbus1: <HID bus> on usbhid1
ugen0.4: <QEMU QEMU USB Keyboard> at usb0
usbhid2 on uhub1
usbhid2: <QEMU QEMU USB Keyboard, class 0/0, rev 2.00/0.00, addr 3> on usb0
hidbus2: <HID bus> on usbhid2
hkbd0: <QEMU QEMU USB Keyboard> on hidbus2
kbd1 at hkbd0
ugen0.5: <QEMU QEMU USB Hub> at usb0
uhub2 on uhub1
uhub2: <QEMU QEMU USB Hub, class 9/0, rev 1.10/1.01, addr 4> on usb0
uhub2: 8 ports with 8 removable, self powered
ugen0.6: <QEMU QEMU USB HARDDRIVE> at usb0
umass0 on uhub2
umass0: <QEMU QEMU USB HARDDRIVE, class 0/0, rev 2.00/0.00, addr 5> on usb0
umass0: SCSI over Bulk-Only; quirks = 0x0
umass0:1:0: Attached to scbus1
cd0 at umass-sim0 bus 0 scbus1 target 0 lun 0
cd0: <QEMU QEMU CD-ROM 2.5+> Removable CD-ROM SPC-3 SCSI device
cd0: Serial Number 1-0000:00:03.0-4.1
cd0: 1.000MB/s transfers
cd0: 1152MB (590121 2048 byte sectors)
cd0: quirks=0x10<10_BYTE_ONLY>
uhub_explore: illegal enable change, port 1
lo0: link state changed to UP

FreeBSD/arm64 (base-md) (ttyv0)
login: 
```

```
root@base-md:~ # pkg info | grep -v FreeBSD
libevent-2.1.12      API for executing callback functions on events or timeouts
libcconv-1.18_1     Character set conversion library
liblz4-1.10.0_2,1   LZ4 compression library, lossless and very fast
pkg-2.7.4           Package manager
rsync-3.4.1_7       Network file distribution/synchronization utility
tor-0.4.9.6         Anonymizing overlay network for TCP
xxhash-0.8.3        Extremely fast non-cryptographic hash algorithm
zstd-1.5.7_1        Fast real-time compression algorithm
root@base-md:~ #
```

```
freebsd-aarch64
root@base-md:~ # ps ax
PID TT STAT TIME COMMAND
0 - DLS 0:00.01 [kernel]
1 - ILs 0:00.00 /sbin/init
2 - WL 0:00.02 [clock]
3 - DL 0:00.00 [crypto]
4 - DL 0:00.00 [cam]
5 - DL 0:00.00 [busdma]
6 - DL 0:00.00 [rand_harvestq]
7 - DL 0:00.00 [md0]
8 - DL 0:00.00 [pagedaemon]
9 - DL 0:00.00 [vmdaemon]
10 - DL 0:00.00 [audit]
11 - RNL 0:44.75 [idle]
12 - WL 0:00.05 [intr]
13 - DL 0:00.00 [geom]
14 - DL 0:00.03 [usb]
15 - DL 0:00.00 [bufdaemon]
16 - DL 0:00.00 [vnlr]
17 - DL 0:00.00 [syncer]
2334 - Is 0:00.00 /sbin/devd
2559 - SCs 0:00.00 /usr/sbin/syslogd -s
2562 - I 0:00.00 syslogd: syslogd.casper (syslogd)
2563 - Is 0:00.00 syslogd: system.net (syslogd)
2626 - Ss 0:00.00 /usr/sbin/cron -s
2674 - Is 0:00.00 logger: system.syslog (logger)
2667 u0- I 0:00.00 /bin/sh -o verify /etc/rc autoboot
2668 u0- IC 0:00.00 logger -p daemon.notice -t fsck
2670 u0- IC 0:00.00 sleep 60
2686 u0 Is+ 0:00.00 /usr/libexec/getty 3wire ttyu0
2678 v0 Is 0:00.00 login [pam] (login)
2687 v0 S 0:00.00 -sh (sh)
2696 v0 R+ 0:00.00 ps ax
2679 v1 Is+ 0:00.00 /usr/libexec/getty Pc ttyv1
2680 v2 Is+ 0:00.00 /usr/libexec/getty Pc ttyv2
2681 v3 Is+ 0:00.00 /usr/libexec/getty Pc ttyv3
2682 v4 Is+ 0:00.00 /usr/libexec/getty Pc ttyv4
2683 v5 Is+ 0:00.00 /usr/libexec/getty Pc ttyv5
2684 v6 Is+ 0:00.00 /usr/libexec/getty Pc ttyv6
2685 v7 Is+ 0:00.00 /usr/libexec/getty Pc ttyv7
root@base-md:~ # cat /etc/rc.conf
hostname=base-md
root@base-md:~ #
```

```
freebsd-aarch64
root@base-md:~ # pkg search FreeBSD-ssh
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/latest/meta.txz: Non-recoverable resolver failure
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/latest/data.pkg: Non-recoverable resolver failure
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/latest/data.tzst: Non-recoverable resolver failure
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/latest/packagesite.pkg: Non-recoverable resolver failure
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/latest/packagesite.tzst: Non-recoverable resolver failure
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/kmods_latest/meta.txz: Non-recoverable resolver failure
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/kmods_latest/data.pkg: Non-recoverable resolver failure
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/kmods_latest/data.tzst: Non-recoverable resolver failure
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/kmods_latest/packagesite.pkg: Non-recoverable resolver failure
pkg: https://pkg.freebsd.org/FreeBSD:16:aarch64/kmods_latest/packagesite.tzst: Non-recoverable resolver failure
root@base-md:~ # cat /etc/resolv.conf
nameserver 1.1.1.1
root@base-md:~ # ifconfig
vtnet0: flags=1000002<BROADCAST,SIMPLEX,MULTICAST,LOWER_UP> metric 0 mtu 1500
    options=800020<VLAN_MTU,JUMBO_MTU,LINKSTATE,HWSTATS>
    ether b2:59:5f:80:7f:1c
    media: Ethernet autoselect (10Gbase-T <full-duplex>)
    status: active
    nd6 options=829<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL,STABLEADDR>
lo0: flags=1000049<UP,LOOPBACK,RUNNING,MULTICAST,LOWER_UP> metric 0 mtu 16384
    options=600003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_IPV6>
    inet 127.0.0.1 netmask 0xff000000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x2
    groups: lo
    nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
root@base-md:~ #
```



customizações extras

- repetir o que já foi feito, acrescentando modificações personalizadas
- definir uma senha `pr_usuári_root`
 - que a gente não fez, até agora
- gerar uma nova imagem com `makefs(8)`
- utilizar ambiente de testes
 - virtualização pode não cobrir peculiaridades
 - alguma arquitetura específica
- lembre-se que a partição base está exposta
 - console "inseguro" via `ttys(5)`

relembrando, que ...

- saúde mental é importante; quando possível, fazer uma pausa
- se nada der certo? paciência, leveza & convicção



firmwares de código aberto

- <https://libreboot.org/docs/bsd>
 - <https://minifree.org>
- <https://www.coreboot.org>
 - <https://system76.com/servers>

uefi secureboot & freebsd

- <https://freebsdoundation.org/freebsd-uefi-secure-boot>
- <https://wiki.freebsd.org/SecureBoot>
- tpm^{*}
 - <https://github.com/paigeadelethompson/freebsd-secureboot>

suporte ao hardware

- <https://www.freebsd.org/releases/15.0R/hardware>

geli (partição criptografada)

- chave extra
- diferentes credenciais

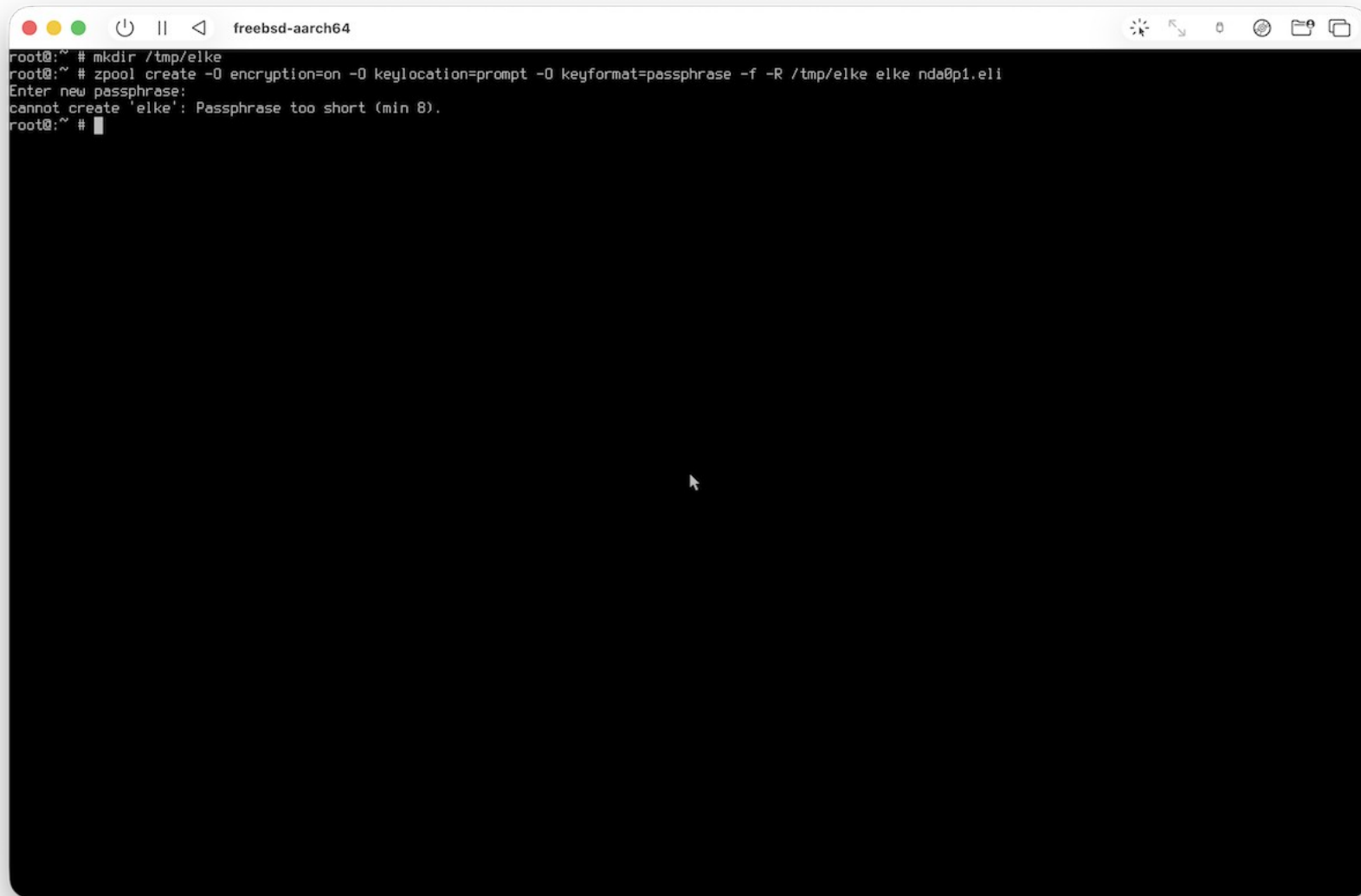
dataset criptografado (zfs)

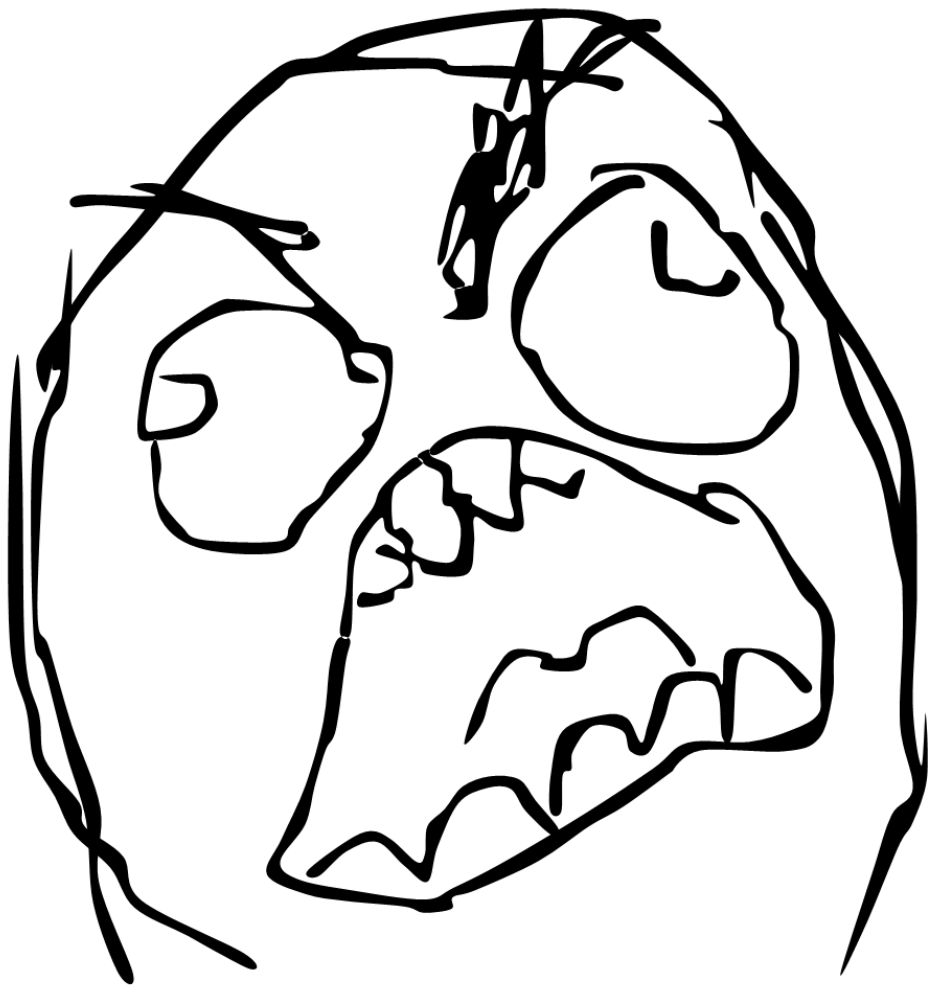
- raíz com senha única
- dataset adicional
- zfs be (boot environment)
 - diferentes ambientes
 - senhas distintas
- chave extra

```
freebsd-aarch64
root@:~ # sysctl kern.disks
kern.disks: cd0 nda0
root@:~ # gpart show nda0
gpart: No such geom: nda0.
root@:~ # gpart create -s gpt nda0
nda0 created
root@:~ # gpart add -t freebsd-zfs nda0
nda0p1 added
root@:~ # kldstat
Id Refs Address          Size Name
  1   19 0xffff000000000000 15126d8 kernel
  2    1 0xffff000096200000   22000 hms.ko
  3    1 0xffff000096222000   24000 hidmap.ko
  4    1 0xffff000096246000   25000 virtio_console.ko
  5    1 0xffff00009626b000   23000 virtio_random.ko
root@:~ # kldload zfs geom_eli
ZFS filesystem version: 5
ZFS storage pool version: features support (5000)
root@:~ # kldstat
Id Refs Address          Size Name
  1   31 0xffff000000000000 15126d8 kernel
  2    1 0xffff000096200000   22000 hms.ko
  3    1 0xffff000096222000   24000 hidmap.ko
  4    1 0xffff000096246000   25000 virtio_console.ko
  5    1 0xffff00009626b000   23000 virtio_random.ko
  7    1 0xffff000096400000   46e000 zfs.ko
  8    1 0xffff00009686e000   33000 geom_eli.ko
root@:~ #
```

```
root@:~ # df -Th
Filesystem                                Type      Size  Used Avail Capacity  Mounted on
/dev/iso9660/16_0_CURRENT_AARCH64_CD    cd9660    1.1G   1.1G    0B   100%      /
devfs                                     devfs      1.0K    0B   1.0K    0%      /dev
tmpfs                                     tmpfs      32M   156K   32M    0%      /var
tmpfs                                     tmpfs      20M    4.0K   20M    0%      /tmp
root@:~ # dd if=/dev/random of=/tmp/fulano.chave bs=4096 count=1
1+0 records in
1+0 records out
4096 bytes transferred in 0.000042 secs (96469535 bytes/sec)
root@:~ # dd if=/dev/random of=/tmp/beltrana.chave bs=4096 count=1
1+0 records in
1+0 records out
4096 bytes transferred in 0.000040 secs (102829312 bytes/sec)
root@:~ # cat /tmp/*.chave | geli init -P -B none -K - nda0p1
root@:~ # echo $?
0
root@:~ # #
```

```
freebsd-aarch64
root@:~ # cat /tmp/*.chave | geli attach -p -k - nda0p1
GEOM_ELI: Device nda0p1.eli created.
GEOM_ELI: Encryption: AES-XTS 128
GEOM_ELI: Crypto: accelerated software
root@:~ # geli list
Geom name: nda0p1.eli
State: ACTIVE
EncryptionAlgorithm: AES-XTS
KeyLength: 128
Crypto: accelerated software
Version: 7
UsedKey: 0
Flags: AUTORESIZE
KeysAllocated: 16
KeysTotal: 16
Providers:
1. Name: nda0p1.eli
   Mediasize: 8589893120 (8.0G)
   Sectorsize: 512
   Mode: r0w0e0
Consumers:
1. Name: nda0p1
   Mediasize: 8589893632 (8.0G)
   Sectorsize: 512
   Stripsize: 0
   Stripeoffset: 20480
   Mode: r1w1e1
root@:~ #
```





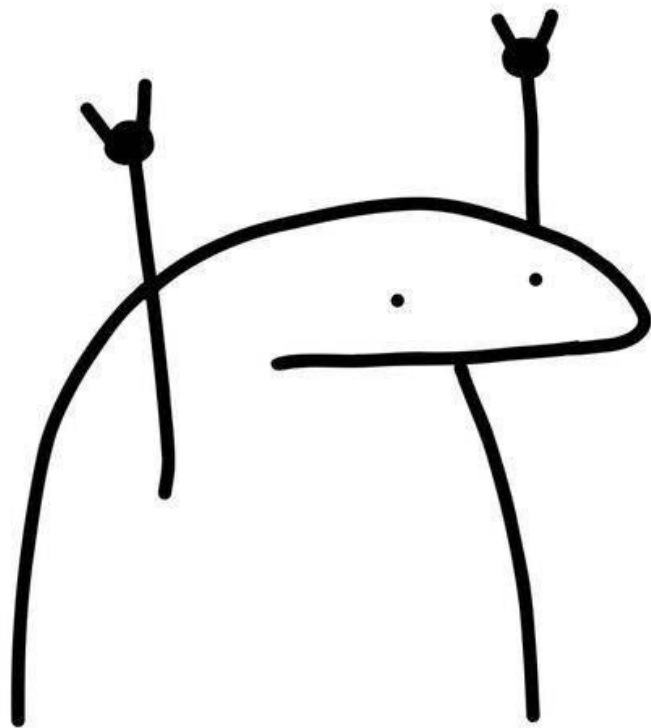
ffffff

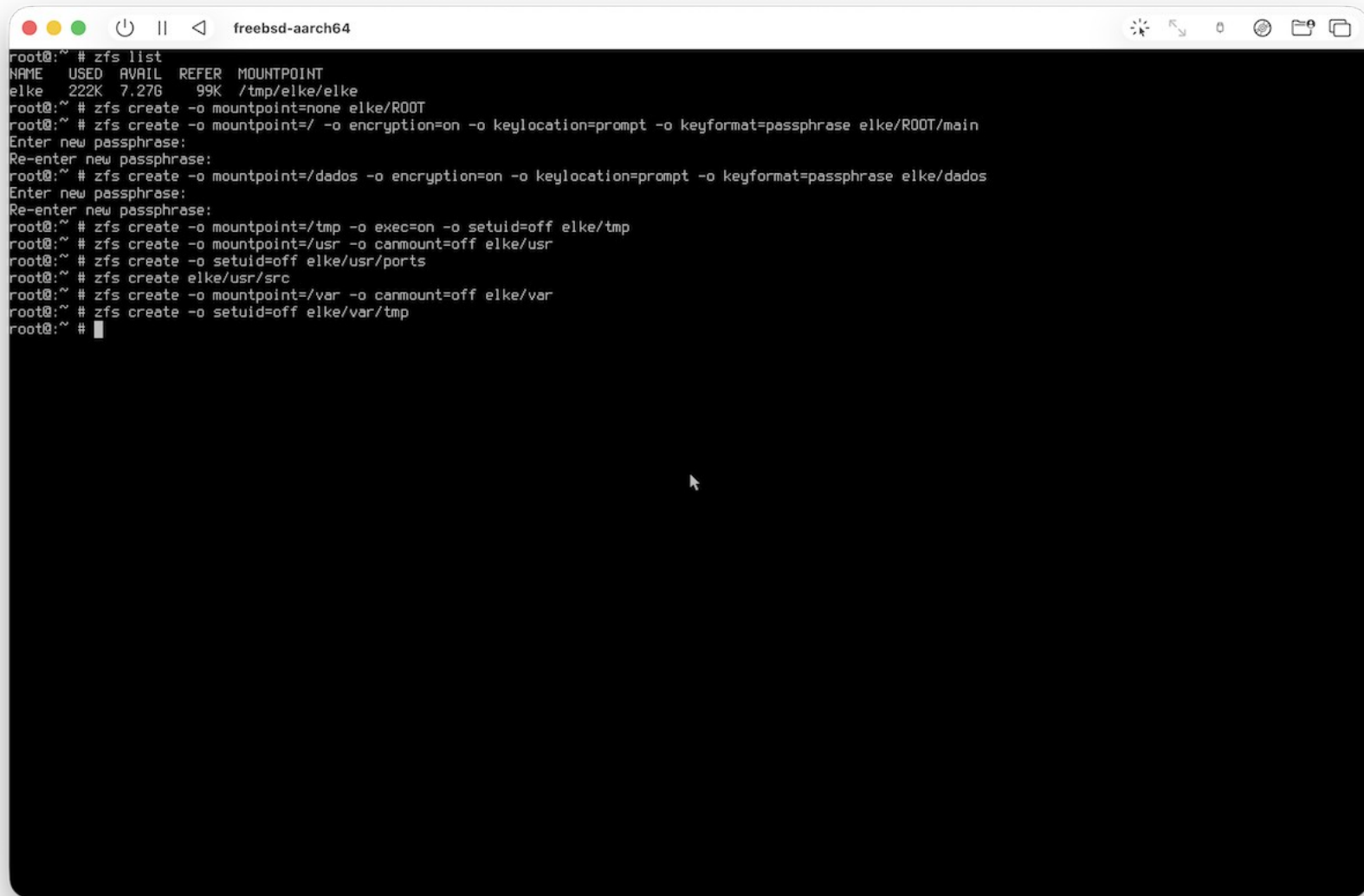
fffuuuu

uuuuuuuu

uuuu-

```
freebsd-aarch64
root@:~ # mkdir /tmp/elke
root@:~ # zpool create -O encryption=on -O keylocation=prompt -O keyformat=passphrase -f -R /tmp/elke elke nda0p1.eli
Enter new passphrase:
cannot create 'elke': Passphrase too short (min 8).
root@:~ # zpool create -O encryption=on -O keylocation=prompt -O keyformat=passphrase -f -R /tmp/elke elke nda0p1.eli
Enter new passphrase:
Re-enter new passphrase:
root@:~ # zpool list
NAME      SIZE  ALLOC   FREE CKPOINT  EXPANDSZ   FRAG    CAP  DEDUP    HEALTH  ALTROOT
elke  7.50G   250K  7.50G      -         -       0%    0%   1.00x    ONLINE  /tmp/elke
root@:~ # zfs list
NAME      USED  AVAIL  REFER  MOUNTPOINT
elke    222K  7.27G   99K    /tmp/elke/elke
root@:~ #
```



```
root@:~ # zfs list
NAME      USED  AVAIL  REFER  MOUNTPOINT
elke      222K  7.27G   99K    /tmp/elke/elke
root@:~ # zfs create -o mountpoint=none elke/ROOT
root@:~ # zfs create -o mountpoint=/ -o encryption=on -o keylocation=prompt -o keyformat=passphrase elke/ROOT/main
Enter new passphrase:
Re-enter new passphrase:
root@:~ # zfs create -o mountpoint=/dados -o encryption=on -o keylocation=prompt -o keyformat=passphrase elke/dados
Enter new passphrase:
Re-enter new passphrase:
root@:~ # zfs create -o mountpoint=/tmp -o exec=on -o setuid=off elke/tmp
root@:~ # zfs create -o mountpoint=/usr -o canmount=off elke/usr
root@:~ # zfs create -o setuid=off elke/usr/ports
root@:~ # zfs create elke/usr/src
root@:~ # zfs create -o mountpoint=/var -o canmount=off elke/var
root@:~ # zfs create -o setuid=off elke/var/tmp
root@:~ #
```

```
freebsd-aarch64
root@:~ # zfs list
NAME      USED  AVAIL  REFER  MOUNTPOINT
elke      222K  7.27G   99K    /tmp/elke/elke
root@:~ # zfs create -o mountpoint=none elke/ROOT
root@:~ # zfs create -o mountpoint=/ -o encryption=on -o keylocation=prompt -o keyformat=passphrase elke/ROOT/main
Enter new passphrase:
Re-enter new passphrase:
root@:~ # zfs create -o mountpoint=/dados -o encryption=on -o keylocation=prompt -o keyformat=passphrase elke/dados
Enter new passphrase:
Re-enter new passphrase:
root@:~ # zfs create -o mountpoint=/tmp -o exec=on -o setuid=off elke/tmp
root@:~ # zfs create -o mountpoint=/usr -o canmount=off elke/usr
root@:~ # zfs create -o setuid=off elke/usr/ports
root@:~ # zfs create elke/usr/src
root@:~ # zfs create -o mountpoint=/var -o canmount=off elke/var
root@:~ # zfs create -o setuid=off elke/var/tmp
root@:~ # zfs list
NAME      USED  AVAIL  REFER  MOUNTPOINT
elke      1.33M  7.26G   99K    /tmp/elke/elke
elke/ROOT  205K   7.26G   99K    none
elke/ROOT/main  106K   7.26G  106K    /tmp/elke
elke/dados   99K   7.26G   99K    /tmp/elke/dados
elke/tmp     99K   7.26G   99K    /tmp/elke/tmp
elke/usr    297K   7.26G   99K    /tmp/elke/usr
elke/usr/ports  99K   7.26G   99K    /tmp/elke/usr/ports
elke/usr/src  99K   7.26G   99K    /tmp/elke/usr/src
elke/var    198K   7.26G   99K    /tmp/elke/var
elke/var/tmp  99K   7.26G   99K    /tmp/elke/var/tmp
root@:~ #
```

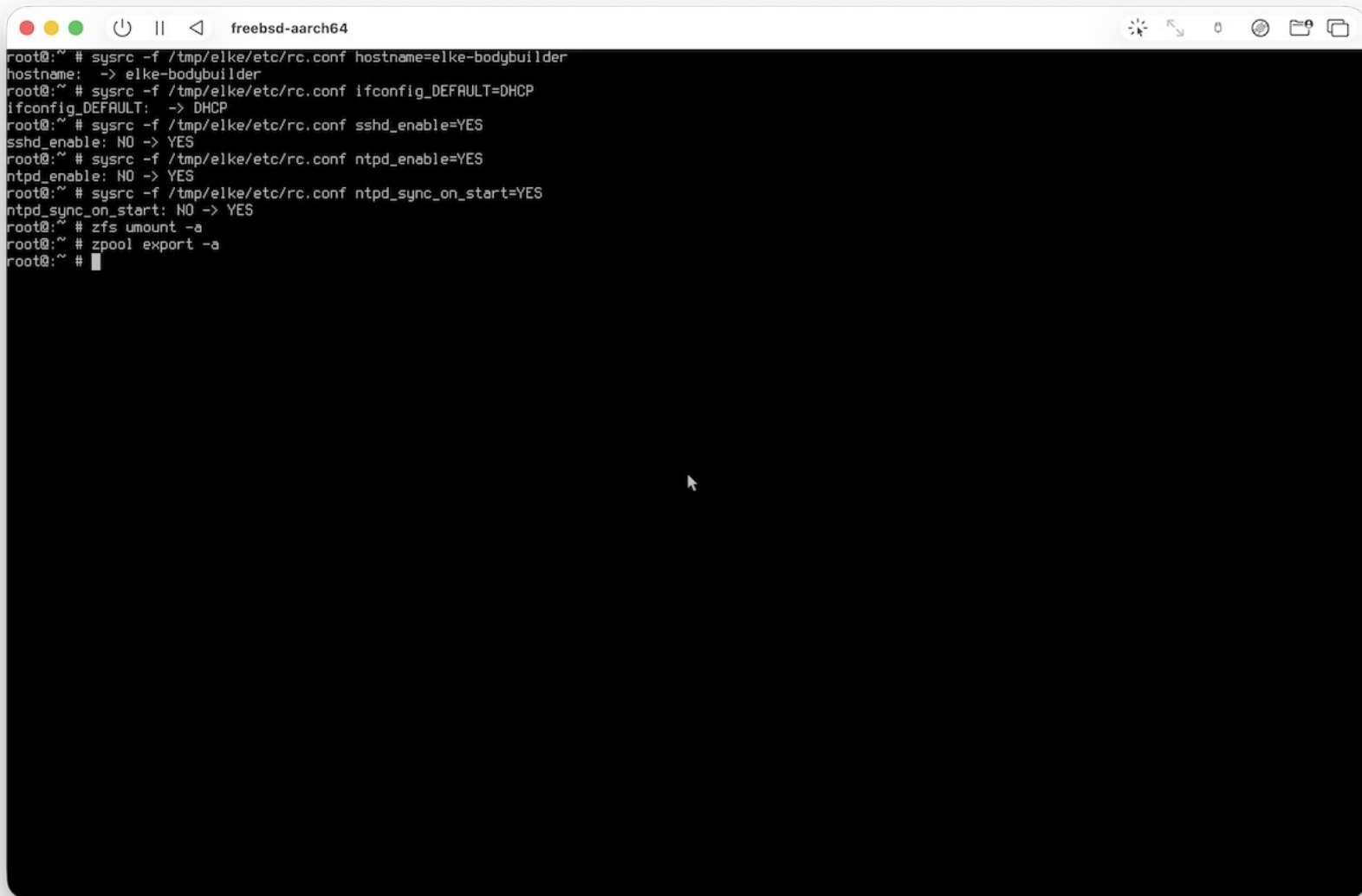
```
freebsd-aarch64
[195/216] Installing FreeBSD-vt-data-16.snap20260413132855...
[195/216] Extracting FreeBSD-vt-data-16.snap20260413132855: 100%
[196/216] Installing FreeBSD-wpa-16.snap20260413132855...
[196/216] Extracting FreeBSD-wpa-16.snap20260413132855: 100%
[197/216] Installing FreeBSD-xz-16.snap20260413132855...
[197/216] Extracting FreeBSD-xz-16.snap20260413132855: 100%
[198/216] Installing FreeBSD-xz-dev-16.snap20260413132855...
[198/216] Extracting FreeBSD-xz-dev-16.snap20260413132855: 100%
[199/216] Installing FreeBSD-xz-lib-16.snap20260413132855...
[199/216] Extracting FreeBSD-xz-lib-16.snap20260413132855: 100%
[200/216] Installing FreeBSD-yp-16.snap20260413132855...
[200/216] Extracting FreeBSD-yp-16.snap20260413132855: 100%
[201/216] Installing FreeBSD-yp-dev-16.snap20260413132855...
[201/216] Extracting FreeBSD-yp-dev-16.snap20260413132855: 100%
[202/216] Installing FreeBSD-yp-lib-16.snap20260413132855...
[202/216] Extracting FreeBSD-yp-lib-16.snap20260413132855: 100%
[203/216] Installing FreeBSD-zfs-16.snap20260413132855...
[203/216] Extracting FreeBSD-zfs-16.snap20260413132855: 100%
[204/216] Installing FreeBSD-zfs-dev-16.snap20260413132855...
[204/216] Extracting FreeBSD-zfs-dev-16.snap20260413132855: 100%
[205/216] Installing FreeBSD-zfs-lib-16.snap20260413132855...
[205/216] Extracting FreeBSD-zfs-lib-16.snap20260413132855: 100%
[206/216] Installing FreeBSD-zlib-16.snap20260413132855...
[206/216] Extracting FreeBSD-zlib-16.snap20260413132855: 100%
[207/216] Installing FreeBSD-zlib-dev-16.snap20260413132855...
[207/216] Extracting FreeBSD-zlib-dev-16.snap20260413132855: 100%
[208/216] Installing FreeBSD-zoneinfo-16.snap20260413132855...
[208/216] Extracting FreeBSD-zoneinfo-16.snap20260413132855: 100%
[209/216] Installing FreeBSD-zstd-16.snap20260413132855...
[209/216] Extracting FreeBSD-zstd-16.snap20260413132855: 100%
[210/216] Installing FreeBSD-zstd-dev-16.snap20260413132855...
[210/216] Extracting FreeBSD-zstd-dev-16.snap20260413132855: 100%
[211/216] Installing FreeBSD-set-devel-16.snap20260413132855...
[212/216] Installing FreeBSD-zstd-lib-16.snap20260413132855...
[212/216] Extracting FreeBSD-zstd-lib-16.snap20260413132855: 100%
[213/216] Installing FreeBSD-set-minimal-16.snap20260413132855...
[214/216] Installing FreeBSD-set-optional-16.snap20260413132855...
[215/216] Installing FreeBSD-set-base-16.snap20260413132855...
[216/216] Installing pkg-2.6.2_1...
[216/216] Extracting pkg-2.6.2_1: 100%
==> Running trigger: mandoc.ucl
root@:~ # sysrc -f /tmp/elke/etc/rc.conf hostname=elke-bodybuilder
hostname: -> elke-bodybuilder
root@:~ # sysrc -f /tmp/elke/etc/rc.conf ifconfig_DEFAULT=DHCP
ifconfig_DEFAULT: -> DHCP
root@:~ # sysrc -f /tmp/elke/etc/rc.conf ntpd_enable=YES
ntpd_enable: NO -> YES
root@:~ # sysrc -f /tmp/elke/etc/rc.conf sshd_enable=YES
sshd_enable: NO -> YES
root@:~ #
```

```
freebsd-aarch64
root@:~ # dhclient vtnet0
DHCPDISCOVER on vtnet0 to 255.255.255.255 port 67 interval 3
DHCPDISCOVER on vtnet0 to 255.255.255.255 port 67 interval 4
DHCPOFFER from 192.168.15.1
DHCPOFFER from 192.168.15.1
DHCPOFFER already seen.
DHCPOFFER from 192.168.15.1
DHCPOFFER already seen.
DHCPOFFER from 192.168.15.1
DHCPOFFER already seen.
DHCPREQUEST on vtnet0 to 255.255.255.255 port 67
DHCPACK from 192.168.15.1
bound to 192.168.15.7 -- renewal in 7200 seconds.
root@:~ # cp /etc/resolv.conf /tmp/elke/etc/.
root@:~ # pkg -c /tmp/elke/ install -y rsync tor openvpn nginx
Updating FreeBSD-ports repository catalogue...
Fetching meta.conf: 100% 179 B 0.2 kB/s 00:01
Fetching data: 100% 10 MiB 2.1 MB/s 00:05
Processing entries: 0%
Processing entries: 100%
FreeBSD-ports repository update completed. 34235 packages processed.
Updating FreeBSD-ports-kmods repository catalogue...
Fetching meta.conf: 100% 179 B 0.2 kB/s 00:01
Fetching data: 100% 16 KiB 16.7 kB/s 00:01
Processing entries: 100%
FreeBSD-ports-kmods repository update completed. 70 packages processed.
All repositories are up to date.
New version of pkg detected; it needs to be installed first.
The following 1 package(s) will be affected (of 0 checked):

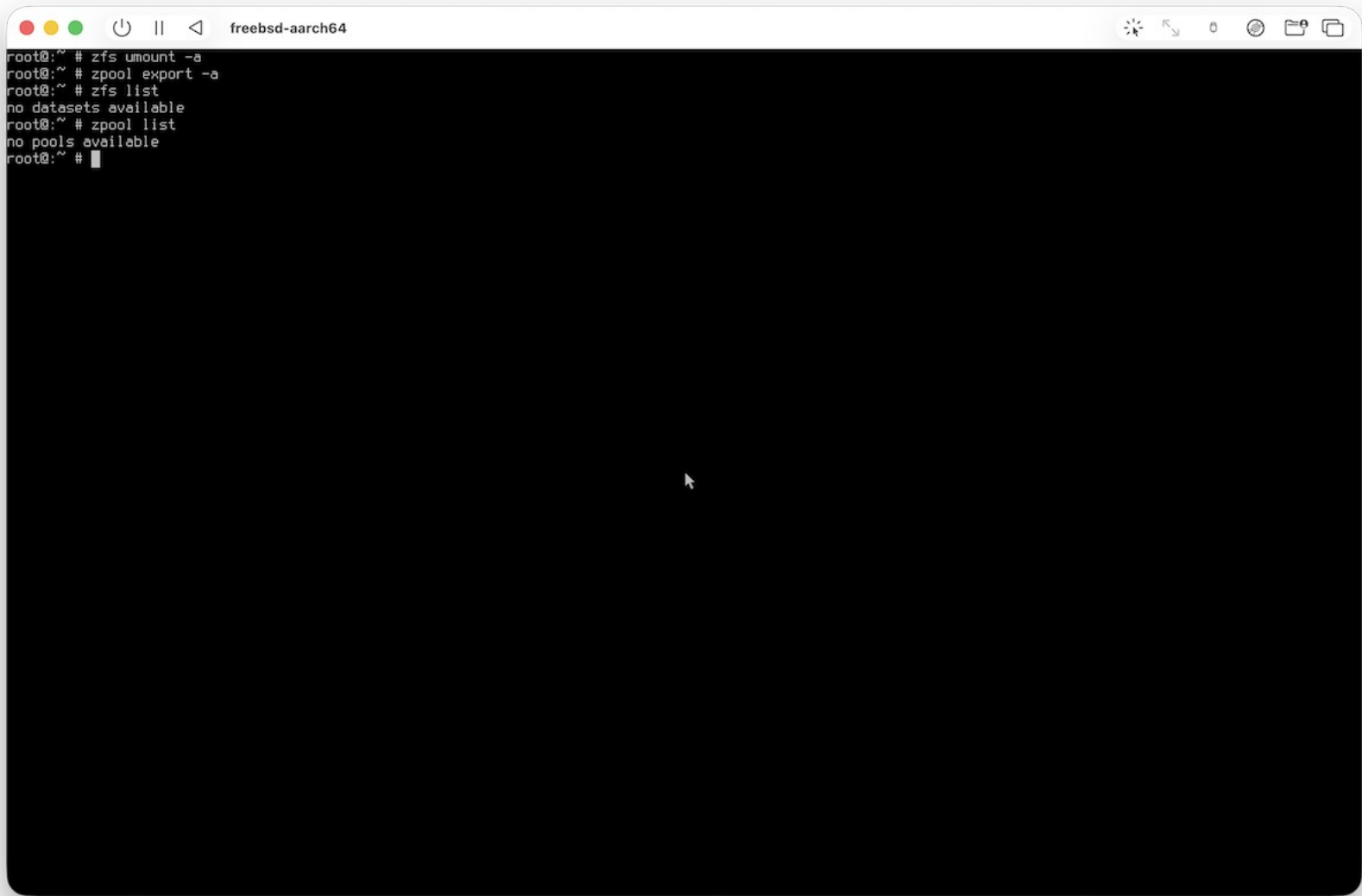
Installed packages to be UPGRADED:
  pkg: 2.6.2_1 -> 2.7.4 [FreeBSD-ports]

Number of packages to be upgraded: 1

The operation will free 3 MiB.
5 MiB to be downloaded.
[1/1] Fetching pkg-2.7.4: 100% 5403 KiB 5.5 MB/s 00:01
Checking integrity... done (0 conflicting)
[1/1] Upgrading pkg from 2.6.2_1 to 2.7.4...
[1/1] Extracting pkg-2.7.4: 100%
Updating FreeBSD-ports repository catalogue...
FreeBSD-ports repository is up to date.
Updating FreeBSD-ports-kmods repository catalogue...
```



```
root@:~ # sysrc -f /tmp/elke/etc/rc.conf hostname=elke-bodybuilder
hostname: -> elke-bodybuilder
root@:~ # sysrc -f /tmp/elke/etc/rc.conf ifconfig_DEFAULT=DHCP
ifconfig_DEFAULT: -> DHCP
root@:~ # sysrc -f /tmp/elke/etc/rc.conf sshd_enable=YES
sshd_enable: NO -> YES
root@:~ # sysrc -f /tmp/elke/etc/rc.conf ntpd_enable=YES
ntpd_enable: NO -> YES
root@:~ # sysrc -f /tmp/elke/etc/rc.conf ntpd_sync_on_start=YES
ntpd_sync_on_start: NO -> YES
root@:~ # zfs umount -a
root@:~ # zpool export -a
root@:~ #
```



```
freebsd-aarch64
root@:~ # zfs umount -a
root@:~ # zpool export -a
root@:~ # zfs list
no datasets available
root@:~ # zpool list
no pools available
root@:~ # zpool import -f -N elke
root@:~ # zfs list
NAME                USED  AVAIL  REFER  MOUNTPOINT
elke                 440M  6.84G   99K    /elke
elke/ROOT            439M  6.84G   99K    none
elke/ROOT/main       439M  6.84G  439M    /
elke/dados           99K    6.84G   99K    /dados
elke/tmp             99K    6.84G   99K    /tmp
elke/usr            297K    6.84G   99K    /usr
elke/usr/ports       99K    6.84G   99K    /usr/ports
elke/usr/src         99K    6.84G   99K    /usr/src
elke/var            200K    6.84G   99K    /var
elke/var/tmp        101K    6.84G  101K    /var/tmp
root@:~ # zfs load-key -a
Enter passphrase for 'elke':
Enter passphrase for 'elke/ROOT/main':
Enter passphrase for 'elke/dados':
3 / 3 key(s) successfully loaded
root@:~ #
```



```
freebsd-aarch64
root@:~ # zfs umount -a
root@:~ # zpool export -a
root@:~ # zfs list
no datasets available
root@:~ # zpool list
no pools available
root@:~ # zpool import -f -N elke
root@:~ # zfs list
NAME                USED  AVAIL  REFER  MOUNTPOINT
elke                 440M  6.84G   99K    /elke
elke/ROOT            439M  6.84G   99K    none
elke/ROOT/main       439M  6.84G  439M    /
elke/dados           99K    6.84G   99K    /dados
elke/tmp             99K    6.84G   99K    /tmp
elke/usr             297K    6.84G   99K    /usr
elke/usr/ports       99K    6.84G   99K    /usr/ports
elke/usr/src         99K    6.84G   99K    /usr/src
elke/var             200K    6.84G   99K    /var
elke/var/tmp        101K    6.84G  101K    /var/tmp
root@:~ # zfs load-key -a
Enter passphrase for 'elke':
Enter passphrase for 'elke/ROOT/main':
Enter passphrase for 'elke/dados':
3 / 3 key(s) successfully loaded
root@:~ # kenv vfs.root.mountfrom="zfs:elke/ROOT/main"
vfs.root.mountfrom="zfs:elke/ROOT/main"
root@:~ # reboot -r
Trying to mount root from zfs:elke/ROOT/main [ro]...

FreeBSD/arm64 (elke-bodybuilder) (ttyv0)

login: █
```

```
freebsd-aarch64
root@:~ # zpool list
no pools available
root@:~ # zpool import -f -N elke
root@:~ # zfs list
NAME                USED  AVAIL  REFER  MOUNTPOINT
elke                 440M  6.84G   99K    /elke
elke/ROOT            439M  6.84G   99K    none
elke/ROOT/main       439M  6.84G  439M    /
elke/dados           99K    6.84G   99K    /dados
elke/tmp             99K    6.84G   99K    /tmp
elke/usr            297K    6.84G   99K    /usr
elke/usr/ports       99K    6.84G   99K    /usr/ports
elke/usr/src         99K    6.84G   99K    /usr/src
elke/var            200K    6.84G   99K    /var
elke/var/tmp        101K    6.84G  101K    /var/tmp
root@:~ # zfs load-key -a
Enter passphrase for 'elke':
Enter passphrase for 'elke/ROOT/main':
Enter passphrase for 'elke/dados':
3 / 3 key(s) successfully loaded
root@:~ # kenv vfs.root.mountfrom="zfs:elke/ROOT/main"
vfs.root.mountfrom="zfs:elke/ROOT/main"
root@:~ # reboot -r
Trying to mount root from zfs:elke/ROOT/main [ro]...

FreeBSD/arm64 (elke-bodybuilder) (ttyv0)

login: root
FreeBSD 16.0-CURRENT (GENERIC) #0 main-n285005-e9fc0c538264: Mon Apr 13 12:48:35 UTC 2026

Welcome to FreeBSD!

Release Notes, Errata: https://www.FreeBSD.org/releases/
Security Advisories:  https://www.FreeBSD.org/security/
FreeBSD Handbook:     https://www.FreeBSD.org/handbook/
FreeBSD FAQ:          https://www.FreeBSD.org/faq/
Questions List:        https://www.FreeBSD.org/lists/questions/
FreeBSD Forums:        https://forums.FreeBSD.org/

Documents installed with the system are in the /usr/local/share/doc/freebsd/
directory, or can be installed later with: pkg install en-freebsd-doc
For other languages, replace "en" with a language code like de or fr.

Show the version of FreeBSD installed: freebsd-version ; uname -a
Please include that output and any error messages when posting questions.
Introduction to manual pages: man man
FreeBSD directory layout:      man hier

To change this login announcement, see motd(5).
root@elke-bodybuilder:~ #
```

acesso remoto restrito/restritivo

- openssh
 - autenticações extras
 - otp/mfa
 - certificados
- túnel de vpn da sua escolha
 - wireguard, openvpn, ipsec, ...
- tor (onion service)
 - <https://community.torproject.org/onion-services/advanced/client-auth>

```
freebsd-aarch64

root@elke-bodybuilder:~ # uname -a
FreeBSD elke-bodybuilder 16.0-CURRENT FreeBSD 16.0-CURRENT #0 main-n285085-e9fc0c538264: Mon Apr 13 12:48:35 UTC 2026    root@releng3.nyi.freebsd.org:/usr/obj/
usr/src/arm64.aarch64/sys/GENERIC arm64
root@elke-bodybuilder:~ # tor --version
Tor version 0.4.9.6.
This build of Tor is covered by the GNU General Public License (https://www.gnu.org/licenses/gpl-3.0.en.html)
Tor is running on FreeBSD with Libevent 2.1.12-stable, OpenSSL 3.5.6, Zlib 1.3.2, Liblzma 5.8.3, Libzstd 1.5.7 and BSD 1600016 as libc.
Tor compiled with clang version 19.1.7
root@elke-bodybuilder:~ # nginx -V
nginx version: nginx/1.28.3
built with OpenSSL 3.5.6 7 Apr 2026
TLS SNI support enabled
configure arguments: --prefix=/usr/local/etc/nginx --with-cc-opt='-I /usr/local/include' --conf-path=/usr/local/etc/nginx/nginx.conf --sbin-path=/usr/local/sbin
/nginx --pid-path=/var/run/nginx.pid --error-log-path=/var/log/nginx/error.log --user=www --group=www --with-compat --with-pcre --modules-path=/usr/local/libex
c/nginx --with-file-aio --http-client-body-temp-path=/var/tmp/nginx/client_body_temp --http-fastcgi-temp-path=/var/tmp/nginx/fastcgi_temp --http-proxy-temp-path
=/var/tmp/nginx/proxy_temp --http-scgi-temp-path=/var/tmp/nginx/scgi_temp --http-uwsgi-temp-path=/var/tmp/nginx/uwsgi_temp --http-log-path=/var/log/nginx/access
.log --with-http_v2_module --with-http_v3_module --with-http_addition_module --with-http_auth_request_module --with-http_dav_module --with-http_flv_module --wit
h-http_gunzip_module --with-http_gzip_static_module --with-http_mp4_module --with-http_random_index_module --with-http_realip_module --with-http_secure_link_mod
ule --with-http_slice_module --with-http_ssl_module --with-http_stub_status_module --with-http_sub_module --without-mail_imap_module --without-mail_pop3_module
--without-mail_smtp_module --with-mail_ssl_module --with-stream --with-stream_realip_module --with-stream_ssl_module --with-stream_ssl_preread_module --with-thr
eads --with-mail=dynamic --with-stream=dynamic --with-ld-opt='-L /usr/local/lib'
root@elke-bodybuilder:~ # rsync --version
rsync version 3.4.1 protocol version 32
Copyright (C) 1996-2025 by Andrew Tridgell, Wayne Davison, and others.
Web site: https://rsync.samba.org/
Capabilities:
  64-bit files, 64-bit inums, 64-bit timestamps, 64-bit long ints,
  socketpairs, symlinks, symtimes, hardlinks, hardlink-specials,
  hardlink-symlinks, IPv6, atimes, batchfiles, inplace, append, ACLs,
  xattrs, optional secluded-args, iconv, no prealloc, stop-at, no ctimes,
  file-flags
Optimizations:
  no SIMD-roll, no asm-roll, openssl-crypto, no asm-MD5
Checksum list:
  xxh128 xxh3 xxh64 (xxhash) md5 md4 sha1 none
Compress list:
  zstd lz4 zlibx zlib none
Daemon auth list:
  sha512 sha256 sha1 md5 md4

rsync comes with ABSOLUTELY NO WARRANTY. This is free software, and you
are welcome to redistribute it under certain conditions. See the GNU
General Public Licence for details.
root@elke-bodybuilder:~ #
```



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automatização

atualização (automática & regular) do ecossistema

- pacotes
 - ports
 - base
- kernel & módulos
- mdroot

cron & agendamentos corriqueiros

- <https://community.torproject.org/relay/setup/guard/freebsd/updates>

apresentações na eurobsdcon

- <https://slideshare.net/>...

- [slideshow/eurobsdcon-2021-autoinstalling-bsd-systems/250243872](https://slideshare.net/slideshow/eurobsdcon-2021-autoinstalling-bsd-systems/250243872)
- [slideshow/eurobsdcon-2023-autoinstalling-bsd-systems-cases-using-pfsense-truenas-and-more/261357863](https://slideshare.net/slideshow/eurobsdcon-2023-autoinstalling-bsd-systems-cases-using-pfsense-truenas-and-more/261357863)



boot via rede usando freebsd

- <https://docs.freebsd.org/pt-br/books/handbook/advanced-networking/#network-diskless>
 - exports(5)
 - nfsd(8)
 - pxeboot(8)
 - tftpd(8)

netboot

- <https://netboot.xyz>
- <https://github.com/netbootxyz/netboot.xyz>

ipxe

- autoridade certificadora própria
- verificação de integridade
- tlsv1.2 (https)
- scripting & customização automatizada
 - vlan
 - endereço mac
 - hostname

dnsmasq

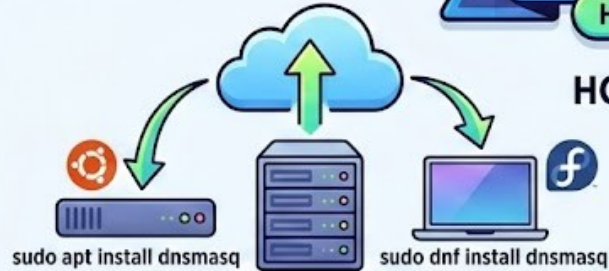
- boot remoto (bootp/tftp)
 - uefi
- configuração da rede
 - domínio, dns, ntp, ...

dnsmasq

How to Use dnsmasq on Linux

HOW TO USE DNSMASQ ON LINUX (minimal texts)

DNS CACHING & FORWARDING



QUICK INSTALL



DHCP SERVER



LOCAL HOSTS FILE

CONFIG FILE (OPTIONAL)



`dnsmasq --version`

`dnsmasq [options]`

`dnsmasq -C /etc/custom.conf`

TIP
`dnsmasq --test`
to check config

<https://commandlinux.com/arch-linux/dnsmasq/>

“gitops”

- pipeline, testes, integração, ...
- infraestrutura em código

ansible

- <https://duckduckgo.com/?q=ansible+dropbear+luks>
- playbook customizado
 - não tem python3 na base do freebsd
- role de setup & instalação
 - tag específica
 - variáveis de ambiente
 - chaves & senhas
- tower (novo nome: automation platform controller?)
 - semaphore (código aberto)

puppet

- agente
- r10k
 - servidor_ & repositório (git)
 - ambientes dinâmicos (branches)
- eyaml & hiera
 - "vault"

```
---  
my-base-packages:  
  - bind-utils  
  - curl  
  
my-plain-password: vetinhoR0ch3d@  
  
my-encrypted-property: >  
ENC[PKCS7,Y22exl+OvjDe+drmik2XEeD3VQt1luZJXFFF2NnrMXDWx0csyqLB/2NOWefv  
  NBTZfOlPvMlAesyr4bUY4I5XeVbVvk38XKxeriH69EFAD4CahIZlC8lkE/uDh  
  jJGQfh052eonkungHIcuGKY/5sEbbZl/qufjAtp/ufor15VBJtsXt17tXP4y  
  15ZP119Fwq8xiREGOL0lVvFYJz2hZclppPCNG5lwuLnTekXN/OazNYpf4CMd  
  /HjZFXwcXRtTlzewJLc+/gox2IfByQRhsI/AgogRfYQKocZgFb/DOZoXR7wm  
  IZGeunzwhqfmEtGiqpvJJQ5wVRdzJVpTnANBA5qxeA==]
```



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referências extras

setup de boot de sistemas via rede

- <https://aniket.foo/posts/20260505-netboot>
- <https://www.tunbury.org/2025/05/06/freebsd-uefi>

configuração de um onion service na rede tor

- <https://community.torproject.org/pt-BR/onion-services/setup>

partição de swap criptografada no freebsd

- <https://docs.freebsd.org/en/books/handbook/disks/#swap-encrypting>

autenticação de múltiplos fatores com openssh

- <https://duo.com/docs/loginduo>
- <https://www.simplified.guide/ssh/use-otp-2fa>

criptografia de datasets zfs

- <https://freebsdoundation.org/wp-content/uploads/2024/09/angel.pdf>
- https://openzfs.org/wiki/ZFS-Native_Encryption

wiki do freebsd sobre suporte a placas gráficas

- <https://wiki.freebsd.org/Graphics>

um pouco mais de geli e criptografia de partições com freebsd

- <https://www.siberoloji.com/how-to-enable-full-disk-encryption-with-geli-during-freebsd-installation/>

instalações automatizadas de pfsense & truenas

- <https://youtu.be/kMwnRBy6vkQ> (pfsense)
- <https://youtu.be/J1txpe4YcNU> (truenas)

hardening para setups com freebsd

- <https://freebsdsoftware.org/blog/freebsd-security-handbook.html>
- <https://computingforgeeks.com/freebsd-15-ssh-key-auth-hardening>

instalar freebsd de maneira automatizada

- <https://github.com/pasztor/bsdai>
- <https://siberoloji.com/how-to-use-bsdinstall-for-automated-deployments-on-freebsd>

utilizando "reroot" no freebsd

- <https://lobste.rs/s/zyj97x>





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ELKE

encrypted and lovely kage environment

UMA MARAVILHA DE AMBIENTE CRIPTOGRAFADO COM FREEBSD



vinícius zavam

<https://keybase.io/egypcio>

<https://slideshare.net/egypcio>